Fourier Series

Type I: Complex form of Fourier series

1. Obtain complex form of Fourier series for $f(x) = e^{ax}$ in $(-l, l)$
[N14/Chem/6M][N15/ElexExtcElectBiomInst/6M][N15/Chem/7M][M18/Chem/6M][N19/Extc/8M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad \left[\because e^{\pm in\pi} = (-1)^n\right]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi}\times\frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n \sinh al\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,
$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

2. Obtain complex form of Fourier series for $f(x) = e^{ax}$ in $(-\pi, \pi)$
[N15/Biot/6M][M17/AutoMechCivil/6M][N17/Inst/6M][M19/Extc/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n \sinh al\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Put $l = \pi$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a\pi\ (a\pi+in\pi)}{a^2\pi^2+n^2\pi^2} e^{inx}$$

3. Obtain complex form of Fourier series for $f(x) = e^{ax}$ in $(-\pi,\pi)$ where a is a real constant. Hence deduce that $\frac{\pi}{a\sin h a\pi} = \sum_{-\infty}^{\infty}\frac{(-1)^n}{(a^2+n^2)}$

[N17/Extc/6M][M18/Extc/6M]

Solution:

Let $f(x) = e^{ax}$ in $(-l,l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right] = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n \sinh al \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Put $l = \pi$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a\pi\ (a\pi+in\pi)}{a^2\pi^2+n^2\pi^2} e^{inx}$$

Now, putting $x = 0$, we get

$$e^0 = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a\pi\ (a\pi+in\pi)}{a^2\pi^2+n^2\pi^2} e^0$$

$$1 = \frac{\pi \sinh a\pi}{\pi^2} \sum_{-\infty}^{\infty} \frac{(-1)^n\ (a+in)}{a^2+n^2}$$

$$\frac{\pi}{\sinh a\pi} = \sum_{-\infty}^{\infty} \frac{(-1)^n\ a}{a^2+n^2} + i\sum_{-\infty}^{\infty} \frac{(-1)^n\ n}{a^2+n^2}$$

Comparing real and imaginary, we get

$$\frac{\pi}{\sinh a\pi} = \sum_{-\infty}^{\infty} \frac{(-1)^n\ a}{a^2+n^2} \qquad \& \qquad 0 = \sum_{-\infty}^{\infty} \frac{(-1)^n\ n}{a^2+n^2}$$

4. Obtain complex form of Fourier series for $f(x) = e^{-ax}$ in $(-\pi, \pi)$ **[M14/AutoMechCivil/6M]**

Solution:

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al - in\pi} \times \frac{al + in\pi}{al + in\pi}$$
$$C_n = (-1)^n \sinh al \frac{al + in\pi}{a^2 l^2 + n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al + in\pi)}{a^2 l^2 + n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$ and putting $l = \pi$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(-a\pi)(-a\pi + in\pi)}{a^2\pi^2 + n^2\pi^2} e^{\frac{in\pi x}{\pi}}$$
$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(a\pi)(a\pi - in\pi)}{a^2\pi^2 + n^2\pi^2} e^{inx}$$

5. Obtain complex form of Fourier series for $f(x) = e^{ax}$ in $(-1,1)$
[M14/ElexExtcElectBiomInst/5M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a - \frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a - \frac{in\pi}{l}\right)x}}{a - \frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a - \frac{in\pi}{l}\right)}\left[e^{\left(a - \frac{in\pi}{l}\right)l} - e^{-\left(a - \frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al - in\pi} - e^{-al + in\pi}\right]$$
$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al - in\pi)}\left[\frac{e^{al} - e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al - in\pi} \times \frac{al + in\pi}{al + in\pi}$$
$$C_n = (-1)^n \sinh al \frac{al + in\pi}{a^2 l^2 + n^2\pi^2}$$

Now, $f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al + in\pi)}{a^2 l^2 + n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = 1$, we get

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a.(a + in\pi)}{a^2 + n^2\pi^2} e^{in\pi x}$$

6. Obtain complex form of Fourier series for $f(x) = e^{ax}$ in $(-a, a)$
[M19/Elex/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n sinhal\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,
$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\,(al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = a$, we get
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a^2.(a^2+in\pi)}{a^4+n^2\pi^2} e^{\frac{in\pi x}{a}}$$

7. Find complex form of Fourier series of $f(x) = e^{3x}$ in $(-\pi, \pi)$
[M18/N18/Biom/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right] = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sin hal}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin hal\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Put $a = 3, l = \pi$

$$e^{3x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin h3\pi\ (3\pi+in\pi)}{9\pi^2+n^2\pi^2} e^{inx}$$

8. If $f(x) = e^{-3x}, -1 < x < 1$. Obtain complex form of $f(x)$ in $(-1,1)$ **[N17/MTRX/6M]**

Solution:

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sin hal}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal \frac{al + in\pi}{a^2 l^2 + n^2 \pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin hal\ (al + in\pi)}{a^2 l^2 + n^2 \pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = 1$, we get

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin ha.(a + in\pi)}{a^2 + n^2 \pi^2} e^{in\pi x}$$

Putting $a = -3$, we get

$$e^{-3x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh (-3).(-3 + in\pi)}{(-3)^2 + n^2 \pi^2} e^{in\pi x}$$

$$e^{-3x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 3.(3 - in\pi)}{9 + n^2 \pi^2} e^{in\pi x}$$

9. Obtain complex form of Fourier series for $f(x) = e^x$ in $(-\pi, \pi)$

[M18/Comp/6M]

Solution:

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l} \int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l} \int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l} \int_{-l}^{l} e^{\left(a - \frac{in\pi}{l}\right)x} dx$$

$$C_n = \frac{1}{2l} \left[\frac{e^{\left(a - \frac{in\pi}{l}\right)x}}{a - \frac{in\pi}{l}} \right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a - \frac{in\pi}{l}\right)} \left[e^{\left(a - \frac{in\pi}{l}\right)l} - e^{-\left(a - \frac{in\pi}{l}\right)l} \right]$$

$$C_n = \frac{1}{2(al - in\pi)} \left[e^{al - in\pi} - e^{-al + in\pi} \right]$$

$$C_n = \frac{1}{2(al - in\pi)} \left[e^{al} . e^{-in\pi} - e^{-al} . e^{in\pi} \right]$$

$$C_n = \frac{1}{2(al - in\pi)} \left[e^{al} (-1)^n - e^{-al} (-1)^n \right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al - in\pi)} \left[\frac{e^{al} - e^{-al}}{2} \right]$$

$$C_n = \frac{(-1)^n \sin hal}{al - in\pi} \times \frac{al + in\pi}{al + in\pi}$$

$$C_n = (-1)^n sinhal \frac{al + in\pi}{a^2 l^2 + n^2 \pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin hal\ (al + in\pi)}{a^2 l^2 + n^2 \pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 1, l = \pi$, we get

$$e^x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh\pi.(\pi + in\pi)}{\pi^2 + n^2\pi^2} e^{inx}$$

10. Obtain complex form of Fourier series for $f(x) = e^x$ in $(-1,1)$
[N17/AutoMechCivil/5M][M18/AutoMechCivil/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a - \frac{in\pi}{l}\right)x} dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a - \frac{in\pi}{l}\right)x}}{a - \frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a - \frac{in\pi}{l}\right)}\left[e^{\left(a - \frac{in\pi}{l}\right)l} - e^{-\left(a - \frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al - in\pi} - e^{-al + in\pi}\right]$$

$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al - in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al - in\pi)}\left[\frac{e^{al} - e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al - in\pi} \times \frac{al + in\pi}{al + in\pi}$$

$$C_n = (-1)^n \sinh al \frac{al + in\pi}{a^2 l^2 + n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al + in\pi)}{a^2 l^2 + n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 1, l = 1$, we get

$$e^x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 1.(1 + in\pi)}{1 + n^2\pi^2} e^{in\pi x}$$

11. Obtain complex form of Fourier series for $f(x) = e^{-x}$ in $(-1,1)$
[M18/MTRX/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad \left[\because e^{\pm in\pi} = (-1)^n\right]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n \sinh al \, \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al \,(al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = 1$, we get

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a.(a+in\pi)}{a^2+n^2\pi^2} e^{in\pi x}$$

Putting $a = -1$, we get

$$e^{-x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(-1).(-1+in\pi)}{(-3)^2+n^2\pi^2} e^{in\pi x}$$
$$e^{-x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 1.(1-in\pi)}{1+n^2\pi^2} e^{in\pi x}$$

12. Obtain complex form of Fourier series for $f(x) = coshax \ in \ (-l, l)$
[M14/Biot/6M][N14/Biot/6M][N18/Biot/6M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n \sinh al \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(-al)\,(-al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that,

$$\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$$
$$\cosh ax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\,[(al+in\pi)+(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$\cosh ax = al\sinh al \sum \frac{(-1)^n e^{\frac{in\pi x}{l}}}{a^2l^2+n^2\pi^2}$$

13. Obtain complex form of Fourier series for $f(x) = \sinh ax$ in $(-l, l)$ **[N14/CompIT/6M]**

Solution:

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n \sinh al \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(-al)\ (-al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that,

$$\sinh ax = \frac{e^{ax}-e^{-ax}}{2}$$
$$\sinh ax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al[(al+in\pi)-(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$\sinh ax = \sinh al \sum \frac{(-1)^n in\pi}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

14. Find complex form of Fourier series for $f(x) = \cosh ax$ in $(-a, a)$ **[M18/Elex/6M]**

Solution:

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$

$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right]$ $\quad [\because e^{\pm in\pi} = (-1)^n]$

$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$

$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi} = (-1)^n \sinh al \frac{al+in\pi}{a^2l^2+n^2\pi^2}$

Now, $\quad f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$

Replacing a by $-a$, we get

$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$

We know that, $\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$

$\cosh ax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al[(al+in\pi)+(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$

$\cosh ax = al \sinh al \sum \frac{(-1)^n e^{\frac{in\pi x}{l}}}{a^2l^2+n^2\pi^2}$

Putting $l = a$, we get $\cosh ax = a^2 \sinh a^2 \sum \frac{(-1)^n e^{\frac{in\pi x}{a}}}{a^4+n^2\pi^2}$

15. Obtain complex form of Fourier series for $f(x) = \cosh 2x$ in $(-3,3)$ **[M19/MTRX/6M]**

Solution:

Let $f(x) = e^{ax}$ in $(-l,l)$

$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$

$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$

$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$

$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$

$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$

$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$

$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$

$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right]$ $\quad [\because e^{\pm in\pi} = (-1)^n]$

$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$

$$C_n=\frac{(-1)^n \sinh al}{al-in\pi}\times\frac{al+in\pi}{al+in\pi}=(-1)^n \sinh al\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x)=\sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax}=\sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\,(al+in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax}=\sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\,(al-in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

We know that, $\cosh ax=\frac{e^{ax}+e^{-ax}}{2}$

$$\cosh ax=\frac{1}{2}\sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\,[(al+in\pi)+(al-in\pi)]}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

$$\cosh ax=al\sinh al\sum\frac{(-1)^n e^{\frac{in\pi x}{l}}}{a^2l^2+n^2\pi^2}$$

Putting $l=3, a=2$, we get $\cosh 2x=6\sinh 6\sum\frac{(-1)^n e^{\frac{in\pi x}{2}}}{36+n^2\pi^2}$

16. Obtain complex form of Fourier series for $f(x)=\cosh x$ in $(-\pi,\pi)$ **[N18/MTRX/6M]**

Solution:

Let $f(x)=e^{ax}$ in $(-l,l)$

$$C_n=\frac{1}{2l}\int_{-l}^{l}f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n=\frac{1}{2l}\int_{-l}^{l}e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n=\frac{1}{2l}\int_{-l}^{l}e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n=\frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n=\frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l}-e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n=\frac{1}{2(al-in\pi)}\left[e^{al-in\pi}-e^{-al+in\pi}\right]$$

$$C_n=\frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi}-e^{-al}.e^{in\pi}\right]$$

$$C_n=\frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n-e^{-al}(-1)^n\right]\qquad[\because e^{\pm in\pi}=(-1)^n]$$

$$C_n=\frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n=\frac{(-1)^n \sinh al}{al-in\pi}\times\frac{al+in\pi}{al+in\pi}=(-1)^n \sinh al\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x)=\sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax}=\sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\,(al+in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that, $coshax = \frac{e^{ax}+e^{-ax}}{2}$

$$coshax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al[(al+in\pi)+(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$coshax = alsinhal \sum \frac{(-1)^n e^{\frac{in\pi x}{l}}}{a^2l^2+n^2\pi^2}$$

Putting $l=\pi, a=1$, we get $coshx = \pi sinh\pi \sum \frac{(-1)^n e^{inx}}{\pi^2+n^2\pi^2}$

17. Obtain complex form of Fourier series for $f(x) = coshx$ in $(-l,l)$
[M18/Biom/8M]
Solution:
Let $f(x) = e^{ax}$ in $(-l,l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi} = (-1)^n sinhal \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $\quad f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that, $coshax = \frac{e^{ax}+e^{-ax}}{2}$

$$coshax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al[(al+in\pi)+(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$coshax = alsinhal \sum \frac{(-1)^n e^{\frac{in\pi x}{l}}}{a^2l^2+n^2\pi^2}$$

Putting $a = 1$, we get $coshx = lsinhl \sum \frac{(-1)^n e^{\frac{in\pi x}{l}}}{l^2+n^2\pi^2}$

18. Obtain complex form of Fourier series for $f(x) = sinh2x$ in $(-2,2)$ **[N19/Elex/6M]**

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh(-al)\ (-al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$e^{-ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that, $sinhax = \frac{e^{ax}-e^{-ax}}{2}$

$$sinhax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al[(al+in\pi)-(al-in\pi)]}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$sinhax = sinhal \sum \frac{(-1)^n in\pi}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 2, l = 2$, we get $sinh2x = sinh4 \sum \frac{(-1)^n in\pi}{16+n^2\pi^2} e^{\frac{in\pi x}{2}}$

19. Obtain complex form of Fourier series for $f(x) = sinhx\ in\ (-l, l)$
[N17/Elect/8M][N17/Biom/8M][M19/Inst/8M]
Solution:
Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}\, e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi}\times\frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now, $f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$

$$e^{ax} = \sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-ax} = \sum_{-\infty}^{\infty}\frac{(-1)^n \sinh(-al)\,(-al+in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

$$e^{-ax} = \sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al\ (al-in\pi)}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

We know that,

$$sinhax = \frac{e^{ax}-e^{-ax}}{2}$$

$$sinhax = \frac{1}{2}\sum_{-\infty}^{\infty}\frac{(-1)^n \sinh al[(al+in\pi)-(al-in\pi)]}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

$$sinhax = sinhal\sum\frac{(-1)^n in\pi}{a^2l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$$

Putting $a = 1$, we get $sinhx = sinhl\sum\frac{(-1)^n in\pi}{l^2+n^2\pi^2}e^{\frac{in\pi x}{l}}$

20. Obtain complex form of Fourier series for $f(x) = coshax + sinhax$ in $(-l, l)$ Ans. $f(x) = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al.(al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$

21. Find complex form of Fourier series for $f(x) = cosh2x + sinh2x \ in \ (-2, 2)$
[M15/AutoMechCivil/6M][N15/AutoMechCivil/6M][M16/Biot/6M][N19/Elect/8M]

Solution:

We have,

$$cosh2x + sinh2x = \frac{e^{2x}+e^{-2x}+e^{2x}-e^{-2x}}{2} = \frac{2e^{2x}}{2} = e^{2x}$$

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax} e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x} dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad \left[\because e^{\pm in\pi} = (-1)^n\right]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al \ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 2$ & $l = 2$, we get

$$e^{2x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 4.(4+in\pi)}{16+n^2\pi^2} e^{\frac{in\pi x}{2}}$$

$$cosh2x + sinh2x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 4.(4+in\pi)}{16+n^2\pi^2} e^{\frac{in\pi x}{2}}$$

22. Obtain complex form of Fourier series for $f(x) = coshax + sinhax$ in $(-a, a)$

[N17/Elex/6M]

Solution:

We have,

$$coshax + sinhax = \frac{e^{ax} + e^{-ax}}{2} + \frac{e^{ax} - e^{-ax}}{2} = \frac{e^{ax} + e^{-ax} + e^{ax} - e^{-ax}}{2}$$
$$= \frac{2e^{ax}}{2}$$
$$= e^{ax}$$

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad \left[\because e^{\pm in\pi} = (-1)^n\right]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n sinhal\, \frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = a$, we get

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a^2\ (a^2+in\pi)}{a^4+n^2\pi^2} e^{\frac{in\pi x}{a}}$$
$$coshax + sinhax = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh a^2.(a^2+in\pi)}{a^4+n^2\pi^2} e^{\frac{in\pi x}{a}}$$

23. Find complex form of Fourier series for $f(x) = cosh2x + sinh2x$ in $(-5, 5)$
[M19/Elect/5M]
Solution:
We have,
$$cosh2x + sinh2x = \frac{e^{2x}+e^{-2x}+e^{2x}-e^{-2x}}{2} = \frac{2e^{2x}}{2} = e^{2x}$$
Let $f(x) = e^{ax}$ in $(-l, l)$
$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad \left[\because e^{\pm in\pi} = (-1)^n\right]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n sinhal\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$
Now,
$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$
Putting $a = 2$ & $l = 5$, we get
$$e^{2x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 10.(10+in\pi)}{100+n^2\pi^2} e^{\frac{in\pi x}{5}}$$
$$cosh2x + sinh2x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 10.(10+in\pi)}{100+n^2\pi^2} e^{\frac{in\pi x}{5}}$$

24. Find complex form of Fourier series for
$f(x) = cosh2x + sinh2x$ in $(-1, 1)$
[M19/Biom/6M]
Solution:
We have,

$$cosh2x + sinh2x = \frac{e^{2x}+e^{-2x}+e^{2x}-e^{-2x}}{2} = \frac{2e^{2x}}{2} = e^{2x}$$

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$

$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$

$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$

$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$

$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$

$$C_n = \frac{(-1)^n\sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$

$$C_n = (-1)^n sinhal\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 2$ & $l = 1$, we get

$$e^{2x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 2.(2+in\pi)}{4+n^2\pi^2} e^{in\pi x}$$

$$cosh2x + sinh2x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 2.(2+in\pi)}{4+n^2\pi^2} e^{in\pi x}$$

25. Find complex form of Fourier series of $cosh3x + sinh3x\ in\ (-3,3)$
[M14/CompIT/6M][M16/Chem/7M][N16/Chem/6M][N18/Inst/8M]
Solution:
We have,

$$cosh3x + sinh3x = \frac{e^{3x}+e^{-3x}}{2} + \frac{e^{3x}-e^{-3x}}{2}$$
$$= \frac{e^{3x}+e^{-3x}+e^{3x}-e^{-3x}}{2}$$
$$= \frac{2e^{3x}}{2}$$
$$= e^{3x}$$

Let $f(x) = e^{ax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{ax}\, e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(a-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(a-\frac{in\pi}{l}\right)x}}{a-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(a-\frac{in\pi}{l}\right)}\left[e^{\left(a-\frac{in\pi}{l}\right)l} - e^{-\left(a-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al-in\pi} - e^{-al+in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}.e^{-in\pi} - e^{-al}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2(al-in\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-in\pi)}\left[\frac{e^{al}-e^{-al}}{2}\right]$$
$$C_n = \frac{(-1)^n \sinh al}{al-in\pi} \times \frac{al+in\pi}{al+in\pi}$$
$$C_n = (-1)^n \sinh al\,\frac{al+in\pi}{a^2l^2+n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{ax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh al\ (al+in\pi)}{a^2l^2+n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $a = 3$ & $l = 3$, we get

$$e^{3x} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 9.(9+in\pi)}{81+n^2\pi^2} e^{\frac{in\pi x}{3}}$$
$$\cosh 3x + \sinh 3x = \sum_{-\infty}^{\infty} \frac{(-1)^n \sinh 9.(9+in\pi)}{81+n^2\pi^2} e^{\frac{in\pi x}{3}}$$

26. Find complex form of Fourier series for $\cos ax$, where 'a' is not an integer in $(-\pi, \pi)$

[M15/Biot/6M][N19/Comp/6M]

Solution:

Here, $f(x) = \cos ax$ in $(-\pi, \pi)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos ax\, e^{-inx}dx$$
$$C_n = \frac{1}{2\pi}\left[\frac{e^{-inx}}{(-in)^2+a^2}(-in\cos ax + a\sin ax)\right]_{-\pi}^{\pi}$$
$$C_n = \frac{1}{2\pi}\left[\frac{e^{-in\pi}}{i^2n^2+a^2}(-in\cos a\pi + a\sin a\pi) - \frac{e^{in\pi}}{i^2n^2+a^2}(-in\cos(-a\pi) + a\sin(-a\pi)\right]_{-\pi}^{\pi}$$

$$C_n = \frac{1}{2\pi}.\frac{1}{a^2-n^2}\left[in\ cosa\pi\ (-e^{-in\pi} + e^{in\pi}) + asina\pi(e^{-in\pi} + e^{in\pi})\right]$$
$$C_n = \frac{1}{\pi(a^2-n^2)}\left[a\ sina\pi\left(\frac{e^{in\pi}+e^{-in\pi}}{2}\right) + in\ cosa\pi\ \left(\frac{e^{in\pi}-e^{-in\pi}}{2}\right)\right]$$
$$C_n = \frac{1}{\pi(a^2-n^2)}\left[a\ sina\pi\left(\frac{e^{in\pi}+e^{-in\pi}}{2}\right) + i^2 n\ cosa\pi\ \left(\frac{e^{in\pi}-e^{-in\pi}}{2i}\right)\right]$$
$$C_n = \frac{1}{\pi(a^2-n^2)}[a\ sina\pi\ cosn\pi - n\ cosa\pi\ sinn\pi]$$
$$C_n = \frac{1}{\pi(a^2-n^2)}[a\ sina\pi\ (-1)^n - n\ cosa\pi\ (0)]$$
$$C_n = \frac{(-1)^n a\ sina\pi}{\pi(a^2-n^2)}$$
$$C_n = \frac{(-1)^n a\ sina\pi}{\pi(a^2-n^2)}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$cosax = \frac{sina\,\pi}{\pi}\sum\frac{(-1)^n.a.e^{inx}}{a^2-n^2}$$

OR

Let $f(x) = e^{iax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{iax}\, e^{-\frac{in\pi x}{l}}dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(ia-\frac{in\pi}{l}\right)x}dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(ia-\frac{in\pi}{l}\right)x}}{ia-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(ia-\frac{in\pi}{l}\right)}\left[e^{\left(ia-\frac{in\pi}{l}\right)l} - e^{-\left(ia-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}\left[e^{ial-in\pi} - e^{-ial+in\pi}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}\left[e^{ial}.e^{-in\pi} - e^{-ial}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}[e^{al}(-1)^n - e^{-al}(-1)^n] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-n\pi)}\left[\frac{e^{al}-e^{-al}}{2i}\right]$$
$$C_n = \frac{(-1)^n sinal}{al-n\pi}\times\frac{al+n\pi}{al+n\pi}$$
$$C_n = (-1)^n sinal\frac{al+n\pi}{a^2l^2-n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{iax} = \sum_{-\infty}^{\infty}\frac{(-1)^n sinal\ (al+n\pi)}{a^2l^2-n^2\pi^2}e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-iax} = \sum_{-\infty}^{\infty} \frac{(-1)^n \sin(-al)\,(-al+n\pi)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$e^{-iax} = \sum_{-\infty}^{\infty} \frac{(-1)^n sinal\ (al-n\pi)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that,

$$cosax = \frac{e^{iax}+e^{-iax}}{2}$$

$$cosax = \frac{1}{2}\sum_{-\infty}^{\infty} \frac{(-1)^n sinal\,[(al+n\pi)+(al-n\pi)]}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

$$cosax = \sum \frac{(-1)^n sinal\,(al)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = \pi$, we get

$$cosax = sina\pi \sum \frac{(-1)^n . a\pi . e^{inx}}{a^2\pi^2-n^2\pi^2}$$

27. Obtain complex form of Fourier series for $f(x) = sinax, x \in (-\pi, \pi)$
[N14/ElexExtcElectBiomInst/6M][M15/Chem/6M][M17/CompIT/6M]
Solution:
Here, $f(x) = sinax$ in $(-\pi, \pi)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} sinax\ e^{-inx} dx$$

$$C_n = \frac{1}{2\pi}\left[\frac{e^{-inx}}{(-in)^2+a^2}(-in\ sinax - a\ cosax)\right]_{-\pi}^{\pi}$$

$$C_n = \frac{1}{2\pi}\left[\frac{e^{-in\pi}}{i^2n^2+a^2}(-insina\pi - acosa\pi) - \frac{e^{in\pi}}{i^2n^2+a^2}(-in\sin(-a\pi) - a\cos(-a\pi)\right]_{-\pi}^{\pi}$$

$$C_n = \frac{1}{2\pi}\cdot\frac{1}{a^2-n^2}\left[in\ sina\pi\left(-e^{-in\pi} - e^{in\pi}\right) - acosa\pi\left(e^{-in\pi} - e^{in\pi}\right)\right]$$

$$C_n = \frac{1}{\pi(a^2-n^2)}\left[a\ cosa\pi\left(\frac{e^{in\pi}-e^{-in\pi}}{2}\right) - in\ sina\pi\ \left(\frac{e^{in\pi}+e^{-in\pi}}{2}\right)\right]$$

$$C_n = \frac{i}{\pi(a^2-n^2)}\left[a\ cosa\pi\left(\frac{e^{in\pi}-e^{-in\pi}}{2i}\right) - n\ sina\pi\ \left(\frac{e^{in\pi}+e^{-in\pi}}{2}\right)\right]$$

$$C_n = \frac{i}{\pi(a^2-n^2)}[a\ cosa\pi\ sinn\pi - n\ sina\pi\ cosn\pi]$$

$$C_n = \frac{i}{\pi(a^2-n^2)}[a\ cosa\pi\ (0) - n\ sina\pi\ (-1)^n]$$

$$C_n = \frac{-(-1)^n i\ n\ sina\pi}{\pi(a^2-n^2)}$$

$$C_n = \frac{(-1)^n n\ sina\pi}{i\pi(a^2-n^2)}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$sinax = \frac{sina\pi}{\pi i}\sum \frac{(-1)^n . n . e^{inx}}{a^2-n^2}$$

OR

Let $f(x) = e^{iax}$ in $(-l, l)$

$$C_n = \frac{1}{2l}\int_{-l}^{l} f(x) e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{iax}\, e^{-\frac{in\pi x}{l}} dx$$
$$C_n = \frac{1}{2l}\int_{-l}^{l} e^{\left(ia-\frac{in\pi}{l}\right)x} dx$$
$$C_n = \frac{1}{2l}\left[\frac{e^{\left(ia-\frac{in\pi}{l}\right)x}}{ia-\frac{in\pi}{l}}\right]_{-l}^{l}$$
$$C_n = \frac{1}{2l\left(ia-\frac{in\pi}{l}\right)}\left[e^{\left(ia-\frac{in\pi}{l}\right)l} - e^{-\left(ia-\frac{in\pi}{l}\right)l}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}\left[e^{ial-in\pi} - e^{-ial+in\pi}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}\left[e^{ial}.e^{-in\pi} - e^{-ial}.e^{in\pi}\right]$$
$$C_n = \frac{1}{2i(al-n\pi)}\left[e^{al}(-1)^n - e^{-al}(-1)^n\right] \qquad [\because e^{\pm in\pi} = (-1)^n]$$
$$C_n = \frac{(-1)^n}{(al-n\pi)}\left[\frac{e^{al}-e^{-al}}{2i}\right]$$
$$C_n = \frac{(-1)^n sinal}{al-n\pi}\times\frac{al+n\pi}{al+n\pi}$$
$$C_n = (-1)^n sinal\,\frac{al+n\pi}{a^2l^2-n^2\pi^2}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$e^{iax} = \sum_{-\infty}^{\infty}\frac{(-1)^n sinal\ (al+n\pi)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Replacing a by $-a$, we get

$$e^{-iax} = \sum_{-\infty}^{\infty}\frac{(-1)^n \sin(-al)\,(-al+n\pi)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$e^{-iax} = \sum_{-\infty}^{\infty}\frac{(-1)^n sinal\ (al-n\pi)}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

We know that,

$$sinax = \frac{e^{iax}-e^{-iax}}{2i}$$
$$sinax = \frac{1}{2i}\sum_{-\infty}^{\infty}\frac{(-1)^n sinal\,[(al+n\pi)-(al-n\pi)]}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$
$$sinax = \frac{sinal}{i}\sum\frac{(-1)^n n\pi}{a^2l^2-n^2\pi^2} e^{\frac{in\pi x}{l}}$$

Putting $l = \pi$, we get

$$sinax = \frac{sina\pi}{i}\sum\frac{(-1)^n.n\pi.e^{inx}}{a^2\pi^2-n^2\pi^2}$$

28. Find complex form of fourier series $f(x) = e^{3x}$ in $0 < x < 3$
[N13/AutoMechCivil/6M][N14/AutoMechCivil/6M]
[M16/AutoMechCivil/5M][N18/Elex/6M]

Solution:

We have, $f(x) = e^{3x}$ in $(0,3)$

$$C_n = \frac{1}{2l}\int_0^{2l} f(x)e^{-\frac{in\pi x}{l}}dx$$

Here, $2l = 3$ i.e. $l = \frac{3}{2}$

$$C_n = \frac{1}{3}\int_0^3 e^{3x}e^{-\frac{i2n\pi x}{3}}dx$$

$$C_n = \frac{1}{3}\int_0^3 e^{\left(3-\frac{i2n\pi}{3}\right)x}dx$$

$$C_n = \frac{1}{3}\left[\frac{e^{\left(3-\frac{i2n\pi}{3}\right)x}}{3-\frac{i2n\pi}{3}}\right]_0^3$$

$$C_n = \frac{1}{3\left(3-\frac{i2n\pi}{3}\right)}\left[e^{\left(3-\frac{i2n\pi}{3}\right)3} - e^0\right]$$

$$C_n = \frac{1}{(9-i2n\pi)}\left[e^{9-2in\pi} - 1\right]$$

$$C_n = \frac{1}{(9-2in\pi)}\left[e^9.e^{-2in\pi} - 1\right]$$

$$C_n = \frac{1}{(9-2in\pi)}\left[e^9 - 1\right] \qquad \because e^{-i2n\pi} = cos2n\pi - isin2n\pi = 1$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{3x} = \sum_{-\infty}^{\infty} \frac{e^9-1}{9-2in\pi} e^{\frac{i2n\pi x}{3}}$$

29. Obtain complex form of Fourier series for $f(x) = 2x$ in $(0, 2\pi)$
[N17/Comp/6M]

Solution:

Let $f(x) = 2x$ in $(0,2\pi)$

$$C_n = \frac{1}{2l}\int_0^{2l} f(x)e^{-\frac{in\pi x}{l}}dx$$

$$C_n = \frac{1}{2\pi}\int_0^{2\pi} 2x\, e^{-inx}\,dx$$

$$C_n = \frac{1}{2\pi}\left[(2x)\left(\frac{e^{-inx}}{-in}\right) - (2)\left(\frac{e^{-inx}}{i^2n^2}\right)\right]_0^{2\pi}$$

$$C_n = \frac{1}{2\pi}\left[4\pi\left(\frac{e^{-in\,2\pi}}{-in}\right) - 2\frac{e^{-in\,2\pi}}{i^2n^2} - 0 + 2\frac{e^0}{i^2n^2}\right]$$

$$C_n = \frac{1}{2\pi}\left[-\frac{4\pi}{in}(cos2n\pi - isin2n\pi) + \frac{2}{n^2}(cos2n\pi - isin2n\pi) - \frac{2}{n^2}\right]$$

$$C_n = \frac{1}{2\pi}\left[-\frac{4\pi}{in} + \frac{2}{n^2} - \frac{2}{n^2}\right]$$

$$C_n = -\frac{2}{in}$$

$$C_n = \frac{2i}{n}, n \neq 0$$

Thus,

$$C_0 = \frac{1}{2\pi}\int_0^{2\pi} 2x\,dx = \frac{1}{2\pi}[x^2]_0^{2\pi} = \frac{1}{2\pi}(4\pi^2) = 2\pi$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$f(x) = \sum_{-\infty}^{-1} C_n e^{\frac{in\pi x}{l}} + C_0 + \sum_{1}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$2x = \sum_{-\infty}^{-1} \frac{2i}{n}\, e^{inx} + 2\pi + \sum_{1}^{\infty} \frac{2i}{n} e^{inx}$$

30. Find complex form of fourier series for $e^{2x}; 0 < x < 2$
[M15/CompIT/6M][N16/AutoMechCivil/6M][N18/Comp/5M]
[N18/Chem/6M]
Solution:
We have, $f(x) = e^{2x}$ in $(0,2)$

$$C_n = \frac{1}{2l}\int_0^{2l} f(x) e^{-\frac{in\pi x}{l}} dx$$

Here, $2l = 2$ i.e. $l = 1$

$$C_n = \frac{1}{2}\int_0^{2} e^{2x} e^{-in\pi x}\,dx$$

$$C_n = \frac{1}{2}\int_0^{2} e^{(2-in\pi)x} dx$$

$$C_n = \frac{1}{2}\left[\frac{e^{(2-in\pi)x}}{2-in\pi}\right]_0^2$$

$$C_n = \frac{1}{2(2-in\pi)}\left[e^{(2-in\pi)2} - e^0\right]$$

$$C_n = \frac{1}{2(2-in\pi)}\left[e^{4-2in\pi} - 1\right]$$

$$C_n = \frac{1}{2(2-in\pi)}\left[e^{4}.e^{-2in\pi} - 1\right]$$

$$C_n = \frac{1}{2(2-in\pi)}\left[e^{4} - 1\right] \qquad \because e^{-i2n\pi} = cos2n\pi - isin2n\pi = 1$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$

$$e^{2x} = \sum_{-\infty}^{\infty} \frac{e^4-1}{4-i2n\pi} e^{in\pi x}$$

31. Obtain complex form of Fourier series for $f(x) = 3x$ in $(0, 2\pi)$
[M19/Comp/6M]
Solution:
Let $f(x) = 3x$ in $(0,2\pi)$

$$C_n = \frac{1}{2l}\int_0^{2l} f(x) e^{-\frac{in\pi x}{l}} dx$$

$$C_n = \frac{1}{2\pi}\int_0^{2\pi} 3x\, e^{-inx}\,dx$$

$$C_n = \frac{1}{2\pi}\left[(3x)\left(\frac{e^{-inx}}{-in}\right) - (3)\left(\frac{e^{-inx}}{i^2n^2}\right)\right]_0^{2\pi}$$

$$C_n = \frac{1}{2\pi}\left[6\pi\left(\frac{e^{-in\,2\pi}}{-in}\right) - 3\,\frac{e^{-in\,2\pi}}{i^2n^2} - 0 + 3\,\frac{e^0}{i^2n^2}\right]$$
$$C_n = \frac{1}{2\pi}\left[-\frac{6\pi}{in}(cos2n\pi - isin2n\pi) + \frac{3}{n^2}(cos2n\pi - isin2n\pi) - \frac{3}{n^2}\right]$$
$$C_n = \frac{1}{2\pi}\left[-\frac{6\pi}{in} + \frac{3}{n^2} - \frac{3}{n^2}\right]$$
$$C_n = -\frac{3}{in}$$
$$C_n = \frac{3i}{n}, n \neq 0$$

Thus,

$$C_0 = \frac{1}{2\pi}\int_0^{2\pi} 3x\,dx = \frac{1}{2\pi}\left[\frac{3x^2}{2}\right]_0^{2\pi} = \frac{1}{2\pi}(6\pi^2) = 3\pi$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$f(x) = \sum_{-\infty}^{-1} C_n e^{\frac{in\pi x}{l}} + C_0 + \sum_1^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$3x = \sum_{-\infty}^{-1} \frac{3i}{n}\,e^{inx} + 3\pi + \sum_1^{\infty} \frac{3i}{n} e^{inx}$$

32. Obtain the complex form of Fourier series for $f(x) = 2x - x^2$ in $(0,2)$
[M19/Chem/8M]
Solution:
We have, $f(x) = 2x - x^2$ in $(0,2)$

$$C_n = \frac{1}{2l}\int_0^{2l} f(x)e^{-\frac{in\pi x}{l}}dx$$

Here, $2l = 2$ i.e. $l = 1$

$$C_n = \frac{1}{2}\int_0^2 (2x - x^2)e^{-in\pi x}\,dx$$
$$C_n = \frac{1}{2}\left[(2x - x^2)\frac{e^{-in\pi x}}{-in\pi} - (2 - 2x)\frac{e^{-in\pi x}}{i^2n^2\pi^2} + (-2)\frac{e^{-in\pi x}}{-i^3n^3\pi^3}\right]_0^2$$
$$C_n = \frac{1}{2}\left[0 - (-2)\frac{e^{-i2n\pi}}{i^2n^2\pi^2} + (-2)\frac{e^{-i2n\pi}}{-i^3n^3\pi^3} - 0 + (2)\frac{1}{i^2n^2\pi^2} - (-2)\left(\frac{1}{-i^3n^3\pi^3}\right)\right]$$
$$C_n = \frac{1}{2}\left[\frac{4}{i^2n^2\pi^2}\right] \qquad \because e^{-i2n\pi} = cos2n\pi - isin2n\pi = 1$$
$$C_n = -\frac{2}{n^2\pi^2}, n \neq 0$$

Thus,

$$C_0 = \frac{1}{2}\int_0^2 (2x - x^2)\,dx = \frac{1}{2}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^2 = \frac{1}{2}\left[4 - \frac{8}{3}\right] = \frac{2}{3}$$

Now,

$$f(x) = \sum_{-\infty}^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$f(x) = \sum_{-\infty}^{-1} C_n e^{\frac{in\pi x}{l}} + C_0 + \sum_1^{\infty} C_n e^{\frac{in\pi x}{l}}$$
$$2x - x^2 = \sum_{-\infty}^{-1} -\frac{2}{n^2\pi^2}\,e^{in\pi x} + \frac{2}{3} + \sum_1^{\infty} -\frac{2}{n^2\pi^2} e^{in\pi x}$$

Type II: Orthogonality, Orthonormality

1. Show that the functions $cosnx, n = 1,2,3, \dots\dots$ is orthogonal on$(0,2\pi)$
[N14/AutoMechCivil/5M][M18/AutoMechCivil/5M]
[N18/AutoMechCivil/5M]
Hence, construct orthonormal set of functions
[N19/Comp/6M]
Solution:
Let $f_n(x) = cosnx$
Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_0^{2\pi} cosmx.cosnx\ dx$$
$$= \frac{1}{2}\int_0^{2\pi} 2cosmx.cosnx\ dx$$
$$= \frac{1}{2}\int_0^{2\pi} \cos(m+n)\,x + \cos(m-n)\,x\ \ dx$$
$$= \frac{1}{2}\left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n}\right]_0^{2\pi}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^{2\pi} \cos^2 nx\, dx$$
$$= \int_0^{2\pi} \frac{[1+cos\,2nx\ ]}{2} dx$$
$$= \frac{1}{2}\int_0^{2\pi} [1 + cos2nx] dx$$
$$= \frac{1}{2}\left[x + \frac{sin\,2nx}{2n}\right]_0^{2\pi}$$
$$= \frac{1}{2}[2\pi] = \pi \neq 0$$

Therefore, $cosnx$ is orthogonal over $(0,2\pi)$
We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here,

$$\int_a^b [f_n(x)]^2 dx = \pi$$
$$\frac{1}{\pi}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{1}{\sqrt{\pi}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{1}{\sqrt{\pi}} cosnx$ becomes orthonormal over $(0,2\pi)$

2. Show that the set of functions $cosx, cos2x, cos3x$, ….a set of orthogonal functions over$[-\pi,\pi]$Hence, construct a set of orthonormal functions.
[M14/Biot/5M][M15/Biot/5M[M16/ElexExtcElectBiomInst/6M]
[N17/Comp/6M][N17/MTRX/5M][M18/N18/Biom/5M]
Solution:

Let $f_n(x) = cosnx$

Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_{-\pi}^{\pi} cosmx.cosnx\, dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi} 2cosmx.cosnx\, dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi} \cos(m+n)\,x + \cos(m-n)\,x\;\; dx$$
$$= \frac{1}{2}\left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n}\right]_{-\pi}^{\pi}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_{-\pi}^{\pi} \cos^2 nx\, dx$$
$$= \int_{-\pi}^{\pi} \frac{[1+cos\,2nx\;]}{2} dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi} [1 + cos2nx] dx$$
$$= \frac{1}{2}\left[x + \frac{sin\,2nx}{2n}\right]_{-\pi}^{\pi}$$
$$= \frac{1}{2}[\pi - (-\pi)] = \pi \neq 0$$

Therefore, $cosnx$ is orthogonal over $(-\pi, \pi)$

We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here,

$$\int_a^b [f_n(x)]^2 dx = \pi$$
$$\frac{1}{\pi}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{1}{\sqrt{\pi}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{1}{\sqrt{\pi}} cosnx$ becomes orthonormal over $(-\pi, \pi)$

3. Show that the set of functions $cosx, cos3x, cos5x, \ldots \ldots$ is orthogonal over $\left(0, \frac{\pi}{2}\right)$. Hence construct orthonormal set of functions

[N19/Elex/6M]

Solution:

Let $f_n(x) = \cos(2n+1)\,x$

Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_0^{\frac{\pi}{2}} \cos(2m+1)\,x\,.\cos(2n+1)\,x\,dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} 2\cos(2m+1)\,x\,.\cos(2n+1)\,x\,dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos(2m+2n+2)\,x + \cos(2m-2n)\,x\;\; dx$$

$$= \frac{1}{2}\left[\frac{\sin(2m+2n+2)x}{2m+2n+2} + \frac{\sin(2m-2n)x}{2m-2n}\right]_0^{\frac{\pi}{2}}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^{\frac{\pi}{2}} \cos^2(2n+1)x \ dx$$
$$= \int_0^{\frac{\pi}{2}} \frac{[1+\cos 2(2n+1)x\ ]}{2} dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} [1 + \cos 2(2n+1)x] dx$$
$$= \frac{1}{2}\left[x + \frac{\sin 2(2n+1)x}{2(2n+1)}\right]_0^{\frac{\pi}{2}}$$
$$= \frac{1}{2}\left[\frac{\pi}{2}\right] = \frac{\pi}{4} \neq 0$$

Therefore, $\cos(2n+1)\,x$ is orthogonal over $\left(0, \frac{\pi}{2}\right)$

We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here,

$$\int_a^b [f_n(x)]^2 dx = \frac{\pi}{4}$$
$$\frac{4}{\pi}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{2}{\sqrt{\pi}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{2}{\sqrt{\pi}}\cos(2n+1)\,x$ becomes orthonormal over $\left(0, \frac{\pi}{2}\right)$

4. Show that the functions $\sin(2n+1)\,x, n = 0,1,2,..$ is orthogonal over $\left[0, \frac{\pi}{2}\right]$

Hence, Construct orthonormal set of functions.

[N13/ElexExtcElectBiomInst/6M][N14/Chem/7M][N14/Biot/5M] [M15/Chem/5M][M16/CompIT/6M][N16/ElexExtcElectBiomInst/6M] [M18/Elect/6M][M18/Inst/6M][M19/Comp/6M]

Solution:

Let $f_n(x) = \sin(2n+1)\,x$

Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_0^{\frac{\pi}{2}} \sin(2m+1)\,x \,.\, \sin(2n+1)\,x\ dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} 2\sin(2m+1)\,x \,.\, \sin(2n+1)\,x\ dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos(2m-2n)\,x - \cos(2m+2n+2)\,x\ \ dx$$
$$= \frac{1}{2}\left[\frac{\sin(2m-2n)x}{2m-2n} - \frac{\sin(2m+2n+2)x}{2m+2n+2}\right]_0^{\frac{\pi}{2}} = 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^{\frac{\pi}{2}} \sin^2(2n+1)x \; dx$$
$$= \int_0^{\frac{\pi}{2}} \frac{[1-cos\,2(2n+1)x\,]}{2} dx$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} [1 - cos2(2n+1)x]dx$$
$$= \frac{1}{2}\left[x - \frac{sin\,2(2n+1)x}{2(2n+1)}\right]_0^{\frac{\pi}{2}}$$
$$= \frac{1}{2}\left[\frac{\pi}{2}\right] = \frac{\pi}{4} \neq 0$$

Therefore, $\sin(2n+1)\,x$ is orthogonal over $\left(0, \frac{\pi}{2}\right)$

We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here,

$$\int_a^b [f_n(x)]^2 dx = \frac{\pi}{4}$$
$$\frac{4}{\pi}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{2}{\sqrt{\pi}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{2}{\sqrt{\pi}}\sin(2n+1)\,x$ becomes orthonormal over $\left(0, \frac{\pi}{2}\right)$

5. Show that the set of functions $\sin\left[\frac{\pi x}{2l}\right], \sin\left[\frac{3\pi x}{2l}\right], \sin\left[\frac{5\pi x}{2l}\right]$ ….is orthogonal over$(0, l)$.

[N13/AutoMechCivil/5M][M14/Chem/6M][N15/Biot/6M]
[N15/Chem/7M][M16/Chem/6M]

Hence construct orthonormal set of functions.

[N16/AutoMechCivil/8M][N18/Elect/6M][M19/Elect/6M]

Solution:

Let $f_n(x) = \sin\left[\frac{(2n+1)\pi x}{2l}\right]$

Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_0^l \sin\left[\frac{(2m+1)\pi x}{2l}\right] \sin\left[\frac{(2n+1)\pi x}{2l}\right] dx$$
$$= \frac{1}{2}\int_0^l 2\sin\left[\frac{(2m+1)\pi x}{2l}\right] \sin\left[\frac{(2n+1)\pi x}{2l}\right] dx$$
$$= \frac{1}{2}\int_0^l cos\left[\frac{(2m-2n)\pi x}{2l}\right] - cos\left[\frac{(2m+2n+2)\pi x}{2l}\right] dx$$
$$= \frac{1}{2}\left[\frac{\sin\left[\frac{(2m-2n)\pi x}{2l}\right]}{\frac{(2m-2n)\pi}{2l}} - \frac{sin\left[\frac{(2m+2n+2)\pi x}{2l}\right]}{\frac{(2m+2n+2)\pi}{2l}}\right]_0^l$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^l \sin^2\left[\frac{(2n+1)\pi x}{2l}\right] dx$$
$$= \int_0^l \frac{\left[1-\cos\left[\frac{2(2n+1)\pi x}{2l}\right]\right]}{2} dx$$
$$= \frac{1}{2}\int_0^l \left[1-\cos\left[\frac{2(2n+1)\pi x}{2l}\right]\right] dx$$
$$= \frac{1}{2}\left[x - \frac{\sin\left[\frac{2(2n+1)\pi x}{2l}\right]}{\frac{2(2n+1)\pi}{2l}}\right]_0^l$$
$$= \frac{1}{2}[l] = \frac{l}{2} \neq 0$$

Therefore, $\sin\left[\frac{(2n+1)\pi x}{2l}\right]$ is orthogonal over $(0, l)$

We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here, $\int_a^b [f_n(x)]^2 dx = \frac{l}{2}$

$$\frac{2}{l}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{\sqrt{2}}{\sqrt{l}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{\sqrt{2}}{\sqrt{l}}\sin\left[\frac{(2n+1)\pi x}{2l}\right]$ becomes orthonormal over $(0, l)$

6. Show that the set of functions $\frac{sinx}{\sqrt{\pi}}, \frac{sin\,2x}{\sqrt{\pi}}, \frac{sin\,3x}{\sqrt{\pi}}$, …. form a normal set in the interval $[-\pi, \pi]$

[N18/Extc/6M]

Solution:

Let $f_n(x) = \frac{sinnx}{\sqrt{\pi}}$

Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_{-\pi}^{\pi} \frac{sinmx}{\sqrt{\pi}}.\frac{sinnx}{\sqrt{\pi}}\ dx$$
$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} 2sinmx.sinnx\ dx$$
$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(m-n)\,x - \cos(m+n)\,x\ \ dx$$
$$= \frac{1}{2\pi}\left[\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n}\right]_{-\pi}^{\pi}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_{-\pi}^{\pi} \frac{\sin^2 nx}{\pi} dx$$
$$= \int_{-\pi}^{\pi} \frac{[1-cos\,2nx\,]}{2\pi} dx$$
$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} [1 - cos2nx] dx$$

$$= \frac{1}{2\pi}\left[x - \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi}$$
$$= \frac{1}{2\pi}[\pi - (-\pi)] = 1$$

Therefore, $\frac{sinnx}{\sqrt{\pi}}$ is orthonormal over $(-\pi, \pi)$

7. Show that the set of functions $\{sinx, sin3x, sin5x, \ldots.\}$ is orthogonal over $(0,2\pi)$. Construct orthonormal set of functions
[M19/Biom/6M]
Solution:
Let $f_n(x) = \sin(2n+1)\,x$
Consider, $m \neq n$

$$\int_a^b f_m(x) f_n(x) dx = \int_0^{2\pi} \sin(2m+1)\,x\,.\sin(2n+1)\,x\,dx$$
$$= \frac{1}{2}\int_0^{2\pi} 2\sin(2m+1)\,x\,.\sin(2n+1)\,x\,dx$$
$$= \frac{1}{2}\int_0^{2\pi} \cos(2m-2n)\,x - \cos(2m+2n+2)\,x\;\,dx$$
$$= \frac{1}{2}\left[\frac{\sin(2m-2n)x}{2m-2n} - \frac{\sin(2m+2n+2)x}{2m+2n+2}\right]_0^{2\pi} = 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^{2\pi} \sin^2(2n+1)x\;\,dx$$
$$= \int_0^{2\pi} \frac{[1 - cos\,2(2n+1)x\,]}{2} dx$$
$$= \frac{1}{2}\int_0^{2\pi} [1 - cos2(2n+1)x] dx$$
$$= \frac{1}{2}\left[x - \frac{sin\,2(2n+1)x}{2(2n+1)}\right]_0^{2\pi}$$
$$= \frac{1}{2}[2\pi] = \pi \neq 0$$

Therefore, $\sin(2n+1)\,x$ is orthogonal over $(0,2\pi)$
We know that, for orthonormality

$$\int_a^b [f_n(x)]^2 dx = 1$$

But here,

$$\int_a^b [f_n(x)]^2 dx = \pi$$
$$\frac{1}{\pi}\int_a^b [f_n(x)]^2 dx = 1$$
$$\int_a^b \left[\frac{1}{\sqrt{\pi}} f_n(x)\right]^2 dx = 1$$

Thus, $\frac{1}{\sqrt{\pi}}\sin(2n+1)\,x$ becomes orthonormal over $(0,2\pi)$

8. Show that $\{sinnx, n = 1,2,3, \ldots\}$ is a set of orthogonal function over an interval $(-\pi, \pi)$
[M19/MTRX/5M]

Solution:
Let $f_n(x) = sinnx$
Consider, $m \neq n$

$$\int_a^b f_m(x)f_n(x)dx = \int_{-\pi}^{\pi} sinmx.sinnx\ dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi} 2sinmx.sinnx\ dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi} \cos(m-n)\,x - \cos(m+n)\,x\ \ dx$$
$$= \frac{1}{2}\left[\frac{\sin(m-n)x}{m+n} - \frac{\sin(m+n)x}{m-n}\right]_{-\pi}^{\pi}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_{-\pi}^{\pi} \sin^2 nx\ dx$$
$$= \int_{-\pi}^{\pi} \frac{[1-cos\,2nx\ ]}{2} dx$$
$$= \frac{1}{2}\int_{-\pi}^{\pi}[1-cos2nx]dx$$
$$= \frac{1}{2}\left[x - \frac{sin\,2nx}{2n}\right]_{-\pi}^{\pi}$$
$$= \frac{1}{2}[2\pi] = \pi \neq 0$$

Therefore, $sinnx$ is orthogonal over $(-\pi, \pi)$

9. Prove that $sinx, sin2x, sin3x, \ldots \ldots$ is orthogonal on $[0,2\pi]$
[M17/AutoMechCivil/5M][M19/AutoMechCivil/5M]
Solution:
Let $f_n(x) = sinnx$
Consider, $m \neq n$

$$\int_a^b f_m(x)f_n(x)dx = \int_0^{2\pi} sinmx.sinnx\ dx$$
$$= \frac{1}{2}\int_0^{2\pi} 2sinmx.sinnx\ dx$$
$$= \frac{1}{2}\int_0^{2\pi} \cos(m-n)\,x - \cos(m+n)\,x\ \ dx$$
$$= \frac{1}{2}\left[\frac{\sin(m-n)x}{m+n} - \frac{\sin(m+n)x}{m-n}\right]_0^{2\pi} = 0$$

Consider, $m = n$

$$\int_a^b [f_n(x)]^2 dx = \int_0^{2\pi} \sin^2 nx\ dx$$
$$= \int_0^{2\pi} \frac{[1-cos\,2nx\ ]}{2} dx$$
$$= \frac{1}{2}\int_0^{2\pi}[1-cos2nx]dx$$
$$= \frac{1}{2}\left[x - \frac{sin\,2nx}{2n}\right]_0^{2\pi}$$
$$= \frac{1}{2}[2\pi] = \pi \neq 0$$

Therefore, $sinnx$ is orthogonal over $(0,2\pi)$

10. Show that the set of functions $1, \sin\frac{\pi x}{l}, \cos\frac{\pi x}{l}, \sin\frac{2\pi x}{l}, \cos\frac{2\pi x}{l}, \dots\dots$ form an orthogonal set in$(-l, l)$ and construct an orthonormal set.
[M16/AutoMechCivil/8M][M16/Biot/6M][N16/CompIT/6M] [N16/Chem/6M]
Solution:
Let $f_n(x) = sin\frac{n\pi x}{l}, g_n(x) = \cos\frac{n\pi x}{l}$
Consider, $m \neq n$

$$\begin{aligned}
\int_a^b f_m(x)f_n(x)dx &= \int_{-l}^{l} sin\frac{m\pi x}{l} sin\frac{n\pi x}{l}\ dx \\
&= \frac{1}{2}\int_{-l}^{l} 2\ sin\frac{m\pi x}{l} sin\frac{n\pi x}{l} dx \\
&= \frac{1}{2}\int_{-l}^{l} \cos\frac{(m-n)\pi x}{l} - \cos\frac{(m+n)\pi x}{l}\ dx \\
&= \frac{1}{2}\left[\frac{\sin\frac{(m-n)\pi x}{l}}{\frac{(m-n)\pi x}{l}} - \frac{\sin\frac{(m+n)\pi x}{l}}{\frac{(m+n)\pi x}{l}}\right]_{-l}^{l} \\
&= 0
\end{aligned}$$

Consider, $m = n$

$$\begin{aligned}
\int_a^b [f_n(x)]^2 dx &= \int_{-l}^{l} \sin^2\frac{n\pi x}{l}\ dx \\
&= \int_{-l}^{l} \frac{\left[1-\cos\frac{2n\pi x}{l}\right]}{2} dx \\
&= \frac{1}{2}\int_{-l}^{l} \left[1 - \cos\frac{2n\pi x}{l}\right] dx \\
&= \frac{1}{2}\left[x - \frac{\sin\frac{2n\pi x}{l}}{\frac{2n\pi}{l}}\right]_{-l}^{l} \\
&= \frac{1}{2}[l - (-l)] = l \neq 0
\end{aligned}$$

Therefore, $sin\frac{n\pi x}{l}$ is orthogonal over $(-l, l)$
Consider, $m \neq n$

$$\begin{aligned}
\int_a^b g_m(x)g_n(x)dx &= \int_{-l}^{l} \cos\frac{m\pi x}{l} \cos\frac{n\pi x}{l}\ dx \\
&= \frac{1}{2}\int_{-l}^{l} 2\ \cos\frac{m\pi x}{l} \cos\frac{n\pi x}{l} dx \\
&= \frac{1}{2}\int_{-l}^{l} \cos\frac{(m+n)\pi x}{l} + \cos\frac{(m-n)\pi x}{l}\ dx \\
&= \frac{1}{2}\left[\frac{\sin\frac{(m+n)\pi x}{l}}{\frac{(m+n)\pi x}{l}} + \frac{\sin\frac{(m-n)\pi x}{l}}{\frac{(m-n)\pi x}{l}}\right]_{-l}^{l} \\
&= 0
\end{aligned}$$

Consider, $m = n$

$$\begin{aligned}
\int_a^b [g_n(x)]^2 dx &= \int_{-l}^{l} cos^2\frac{n\pi x}{l}\ dx \\
&= \int_{-l}^{l} \frac{\left[1+\cos\frac{2n\pi x}{l}\right]}{2} dx
\end{aligned}$$

$$= \frac{1}{2}\int_{-l}^{l}\left[1+\cos\frac{2n\pi x}{l}\right]dx$$
$$= \frac{1}{2}\left[x+\frac{\sin\frac{2n\pi x}{l}}{\frac{2n\pi}{l}}\right]_{-l}^{l}$$
$$= \frac{1}{2}[l-(-l)] = l \neq 0$$

Therefore, $cos\frac{n\pi x}{l}$ is orthogonal over $(-l,l)$

Consider, $m \neq n$

$$\int_a^b f_m(x)g_n(x)dx = \int_{-l}^{l}\sin\frac{m\pi x}{l}\cos\frac{n\pi x}{l}\ dx$$
$$= \frac{1}{2}\int_{-l}^{l} 2\ \sin\frac{m\pi x}{l}\cos\frac{n\pi x}{l}dx$$
$$= \frac{1}{2}\int_{-l}^{l}\sin\frac{(m+n)\pi x}{l}+\sin\frac{(m-n)\pi x}{l}\ dx$$
$$= \frac{1}{2}\left[-\frac{\cos\frac{(m+n)\pi x}{l}}{\frac{(m+n)\pi x}{l}}-\frac{\cos\frac{(m-n)\pi x}{l}}{\frac{(m-n)\pi x}{l}}\right]_{-l}^{l}$$
$$= 0$$

Consider, $m = n$

$$\int_a^b[f_n(x)]^2dx = \int_a^b[g_n(x)]^2dx = l \neq 0$$

Therefore, $\sin\frac{n\pi x}{l}$ is orthogonal to $cos\frac{n\pi x}{l}$ over $(-l,l)$

We know that, for orthonormality

$$\int_a^b[f_n(x)]^2dx = 1$$

But here,

$$\int_a^b[f_n(x)]^2dx = l$$
$$\frac{1}{l}\int_a^b[f_n(x)]^2dx = 1$$
$$\int_a^b\left[\frac{1}{\sqrt{l}}f_n(x)\right]^2dx = 1$$

Thus, $\frac{1}{\sqrt{l}},\frac{1}{\sqrt{l}}\sin\frac{\pi x}{l},\frac{1}{\sqrt{l}}\cos\frac{\pi x}{l},\frac{1}{\sqrt{l}}\sin\frac{2\pi x}{l},\frac{1}{\sqrt{l}}\cos\frac{2\pi x}{l},\ldots\ldots$ becomes orthonormal over $(-l,l)$

11. Prove that $f_1(x)=1, f_2(x)=x, f_3(x)=\frac{3x^2-1}{2}$ are orthogonal over $(-1,1)$

[M14/AutoMechCivil/5M][M14/CompIT/5M][M15/AutoMechCivil/5M][N15/CompIT/6M][M16/Chem/5M][N16/Chem/5M][M18/Comp/6M][N18/Biot/6M][M19/Extc/6M]

Solution:

Consider,

$$\int_{-1}^{1}f_1(x)f_2(x)dx = \int_{-1}^{1}1.x\,dx = \left[\frac{x^2}{2}\right]_{-1}^{1} = 0$$

$$\int_{-1}^{1} f_2(x)f_3(x)dx = \int_{-1}^{1} x.\frac{3x^2-1}{2}dx = \int_{-1}^{1} \frac{3x^3-x}{2}dx = \frac{1}{2}\left[3\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{1} = 0$$

$$\int_{-1}^{1} f_1(x)f_3(x)dx = \int_{-1}^{1} 1.\frac{3x^2-1}{2}dx = \frac{1}{2}\left[3\frac{x^3}{3} - x\right]_{-1}^{1} = 0$$

Now, consider

$$\int_{-1}^{1}[f_1(x)]^2dx = \int_{-1}^{1}[1]^2dx = [x]_{-1}^{1} = 2 \neq 0$$

$$\int_{-1}^{1}[f_2(x)]^2dx = \int_{-1}^{1}[x]^2dx = \left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{2}{3} \neq 0$$

$$\int_{-1}^{1}[f_3(x)]^2dx = \int_{-1}^{1}\left[\frac{3x^2-1}{2}\right]^2 dx = \frac{1}{4}\int_{-1}^{1}(9x^4 - 6x^2 + 1)dx$$

$$= \frac{1}{4}\left[9\frac{x^5}{5} - 6\frac{x^3}{3} + x\right]_{-1}^{1} = \frac{2}{5} \neq 0$$

Therefore, $f_1(x) = 1, f_2(x) = x, f_3(x) = \frac{3x^2-1}{2}$ are orthogonal over $(-1,1)$

12. Show that the functions $f_1(x) = 1, f_2(x) = x$ are orthogonal on $(-1,1)$. Determine the constant a and b such that the function $f_3(x) = -1 + ax + bx^2$ is orthogonal to both $f_1 \& f_2$ on that interval.
[M17/ElexExctElectBiomInst/6M][N18/Comp/6M]
[N19/AutoMechCivil/5M]
Solution:

$$\int_{-1}^{1} f_1(x)f_2(x)dx = \int_{-1}^{1} 1.x\,dx = \left[\frac{x^2}{2}\right]_{-1}^{1} = 0$$

Now, consider

$$\int_{-1}^{1}[f_1(x)]^2dx = \int_{-1}^{1}[1]^2dx = [x]_{-1}^{1} = 2 \neq 0$$

$$\int_{-1}^{1}[f_2(x)]^2dx = \int_{-1}^{1}[x]^2dx = \left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{2}{3} \neq 0$$

Therefore, $f_1(x) = 1, f_2(x) = x$ are orthogonal over $(-1,1)$

It is given that $f_3(x)$ is orthogonal to $f_1(x)$,

$$\int_{-1}^{1} f_1(x)f_3(x)dx = 0$$

$$\int_{-1}^{1} 1 \cdot (-1 + ax + bx^2)dx = 0$$

$$\int_{-1}^{1}(-1 + ax + bx^2)dx = 0$$

$$\left[-x + a\frac{x^2}{2} + b\frac{x^3}{3}\right]_{-1}^{1} = 0$$

$$\left[-1 + \frac{a}{2} + \frac{b}{3}\right] - \left[1 + \frac{a}{2} - \frac{b}{3}\right] = 0$$

$$-2 + \frac{2b}{3} = 0$$

$$\therefore b = 3$$

Also, $f_3(x)$ is orthogonal to $f_2(x)$,

$\int_{-1}^{1} f_2(x) f_3(x) dx = 0$
$\int_{-1}^{1} x \cdot (-1 + ax + bx^2) dx = 0$
$\int_{-1}^{1} (-x + ax^2 + bx^3) dx = 0$
$\left[-\frac{x^2}{2} + a\frac{x^3}{3} + b\frac{x^4}{4}\right]_{-1}^{1} = 0$
$\left[-\frac{1}{2} + \frac{a}{3} + \frac{b}{4}\right] - \left[-\frac{1}{2} - \frac{a}{3} + \frac{b}{4}\right] = 0$
$\frac{2a}{3} = 0$
$\therefore a = 0$

13. If $f(x) = c_1\varphi_1(x) + c_2\varphi_2(x) + c_3\varphi_3(x)$ where c_1, c_2, c_3 are constants and $\varphi_1, \varphi_2, \varphi_3$ are orthonormal sets on (a, b),
show that $\int_a^b [f(x)]^2\, dx = c_1^2 + c_2^2 + c_3^2$
[M15/ElexExtcElectBiomInst/6M]
Solution:
$f(x) = c_1\varphi_1(x) + c_2\varphi_2(x) + c_3\varphi_3(x)$
Squaring both the sides, we get
$[f(x)]^2 = [c_1\varphi_1(x) + c_2\varphi_2(x) + c_3\varphi_3(x)]^2$
$[f(x)]^2 = c_1^2[\varphi_1(x)]^2 + c_2^2[\varphi_2(x)]^2 + c_3^2[\varphi_3(x)]^2 +$
$2c_1c_2\varphi_1(x)\varphi_2(x) + 2c_2c_3\varphi_2(x)\varphi_3(x) + 2c_1c_3\varphi_1(x)\varphi_3(x)$
Integrating, we get
$\int_a^b [f(x)]^2 dx = c_1^2 \int_a^b [\varphi_1(x)]^2\, dx + c_2^2 \int_a^b [\varphi_2(x)]^2 dx +$
$c_3^2 \int_a^b [\varphi_3(x)]^2 dx + 2c_1c_2 \int_a^b \varphi_1(x)\varphi_2(x)\, dx + 2c_2c_3 \int_a^b \varphi_2(x)\varphi_3(x)\, dx$
$+2c_1c_3 \int_a^b \varphi_1(x)\varphi_3(x)\, dx$
Since, $\varphi_1, \varphi_2, \varphi_3$ are orthonormal sets on (a, b), we know that
$\int_a^b [\varphi_1(x)]^2\, dx = \int_a^b [\varphi_2(x)]^2 dx = \int_a^b [\varphi_3(x)]^2 dx = 1$
$\int_a^b \varphi_1(x)\varphi_2(x)\, dx = \int_a^b \varphi_2(x)\varphi_3(x)\, dx = \int_a^b \varphi_1(x)\varphi_3(x)\, dx = 0$
$\therefore \int_a^b [f(x)]^2\, dx = c_1^2 + c_2^2 + c_3^2$

14. Define orthogonal set of functions on (a, b). Show that the functions $f_1(x) = 1$ and $f_2(x) = 4x$ are orthogonal on $(-1,1)$. Determine the constants P and Q such that $f_3(x) = Px^2 + Qx^3 + 4$ is orthogonal to both $f_1(x)$ & $f_2(x)$ on the same interval
[N18/Inst/6M]
Solution:

A set of functions is $f_1(x), f_2(x), f_3(x), \dots\dots, f_n(x)$ is said to be orthogonal on (a,b) if $\int_a^b f_m(x).f_n(x)dx \begin{cases} = 0 & if\ m \neq n \\ \neq 0 & if\ m = n \end{cases}$

Consider,

$$\int_{-1}^{1} f_1(x)f_2(x)dx = \int_{-1}^{1} 1.4x\,dx = \left[4\frac{x^2}{2}\right]_{-1}^{1} = 0$$

Now, consider

$$\int_{-1}^{1}[f_1(x)]^2dx = \int_{-1}^{1}[1]^2dx = [x]_{-1}^{1} = 2 \neq 0$$

$$\int_{-1}^{1}[f_2(x)]^2dx = \int_{-1}^{1}[4x]^2dx = \left[16\frac{x^3}{3}\right]_{-1}^{1} = \frac{32}{3} \neq 0$$

Therefore, $f_1(x) = 1, f_2(x) = 4x$ are orthogonal over $(-1,1)$

It is given that $f_3(x)$ is orthogonal to $f_1(x)$,

$$\int_{-1}^{1} f_1(x)f_3(x)dx = 0$$

$$\int_{-1}^{1} 1 \cdot (Px^2 + Qx^3 + 4)dx = 0$$

$$\int_{-1}^{1}(Px^2 + Qx^3 + 4)dx = 0$$

$$\left[P\frac{x^3}{3} + Q\frac{x^4}{4} + 4x\right]_{-1}^{1} = 0$$

$$\left[\frac{P}{3} + \frac{Q}{4} + 4\right] - \left[-\frac{P}{3} + \frac{Q}{4} - 4\right] = 0$$

$$\frac{2P}{3} + 8 = 0$$

$$\therefore P = -12$$

Also, $f_3(x)$ is orthogonal to $f_2(x)$,

$$\int_{-1}^{1} f_2(x)f_3(x)dx = 0$$

$$\int_{-1}^{1} 4x \cdot (Px^2 + Qx^3 + 4)dx = 0$$

$$\int_{-1}^{1}(4Px^3 + 4Qx^4 + 16x)dx = 0$$

$$\left[4P\frac{x^4}{4} + 4Q\frac{x^5}{5} + 16\frac{x^2}{2}\right]_{-1}^{1} = 0$$

$$\left[P + \frac{4Q}{5} + 8\right] - \left[P - \frac{4Q}{5} + 8\right] = 0$$

$$\frac{8Q}{5} = 0$$

$$\therefore Q = 0$$

15. Define orthogonal set of functions on (a,b). Show that the functions $f_1(x) = 1$ and $f_2(x) = 3x$ are orthogonal on $(-2,2)$. Determine the constants P and Q such that $f_3(x) = Px^2 + Qx + 9$ is orthogonal to both $f_1(x)$ & $f_2(x)$ on the same interval
[N17/Elect/6M][N17/Biom/6M][M19/Inst/6M]
Solution:

A set of functions is $f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)$ is said to be orthogonal on (a,b) if $\int_a^b f_m(x).f_n(x)dx \begin{cases} = 0 & if\ m \neq n \\ \neq 0 & if\ m = n \end{cases}$

Consider,

$$\int_{-2}^{2} f_1(x)f_2(x)dx = \int_{-2}^{2} 1.3x\,dx = \left[3\frac{x^2}{2}\right]_{-2}^{2} = 0$$

Now, consider

$$\int_{-2}^{2}[f_1(x)]^2dx = \int_{-2}^{2}[1]^2dx = [x]_{-2}^{2} = 4 \neq 0$$

$$\int_{-2}^{2}[f_2(x)]^2dx = \int_{-2}^{2}[3x]^2dx = \left[9\frac{x^3}{3}\right]_{-2}^{2} = 48 \neq 0$$

Therefore, $f_1(x) = 1, f_2(x) = 3x$ are orthogonal over $(-2,2)$

It is given that $f_3(x)$ is orthogonal to $f_1(x)$,

$$\int_{-2}^{2} f_1(x)f_3(x)dx = 0$$

$$\int_{-2}^{2} 1 \cdot (Px^2 + Qx + 9)dx = 0$$

$$\int_{-2}^{2}(Px^2 + Qx + 9)dx = 0$$

$$\left[P\frac{x^3}{3} + Q\frac{x^2}{2} + 9x\right]_{-2}^{2} = 0$$

$$\left[\frac{8P}{3} + 2Q + 18\right] - \left[-\frac{8P}{3} + 2Q - 18\right] = 0$$

$$\frac{16P}{3} + 36 = 0$$

$$\therefore P = -\frac{27}{4}$$

Also, $f_3(x)$ is orthogonal to $f_2(x)$,

$$\int_{-2}^{2} f_2(x)f_3(x)dx = 0$$

$$\int_{-2}^{2} 3x \cdot (Px^2 + Qx + 9)dx = 0$$

$$\int_{-2}^{2}(3Px^3 + 3Qx^2 + 27x)dx = 0$$

$$\left[3P\frac{x^4}{4} + 3Q\frac{x^3}{3} + 27\frac{x^2}{2}\right]_{-2}^{2} = 0$$

$$[12P + 8Q + 54] - [12P - 8Q + 24] = 0$$

$$16Q = 0$$

$$\therefore Q = 0$$

16. Define orthogonal set of functions on (a,b). Show that the functions $f_1(x) = 5$ and $f_2(x) = 3x$ are orthogonal on $(-2,2)$. Determine the constants P and Q such that $f_3(x) = Px^2 + Qx + 9$ is orthogonal to both $f_1(x)$ & $f_2(x)$ on the same interval

[M18/Biom/6M]

Solution:

A set of functions is $f_1(x), f_2(x), f_3(x), \dots\dots, f_n(x)$ is said to be orthogonal on (a,b) if $\int_a^b f_m(x).f_n(x)dx \begin{cases} = 0 & if\ m \neq n \\ \neq 0 & if\ m = n \end{cases}$

Consider,

$$\int_{-2}^{2} f_1(x)f_2(x)dx = \int_{-2}^{2} 5.3x\,dx = \left[15\frac{x^2}{2}\right]_{-2}^{2} = 0$$

Now, consider

$$\int_{-2}^{2}[f_1(x)]^2dx = \int_{-2}^{2}[5]^2dx = [25x]_{-2}^{2} = 100 \neq 0$$

$$\int_{-2}^{2}[f_2(x)]^2dx = \int_{-2}^{2}[3x]^2dx = \left[9\frac{x^3}{3}\right]_{-2}^{2} = 48 \neq 0$$

Therefore, $f_1(x) = 5, f_2(x) = 3x$ are orthogonal over $(-2,2)$

It is given that $f_3(x)$ is orthogonal to $f_1(x)$,

$$\int_{-2}^{2} f_1(x)f_3(x)dx = 0$$

$$\int_{-2}^{2} 5\cdot(Px^2 + Qx + 9)dx = 0$$

$$\int_{-2}^{2}(5Px^2 + 5Qx + 45)dx = 0$$

$$\left[5P\frac{x^3}{3} + 5Q\frac{x^2}{2} + 45x\right]_{-2}^{2} = 0$$

$$\left[\frac{40P}{3} + 10Q + 90\right] - \left[-\frac{40P}{3} + 10Q - 90\right] = 0$$

$$\frac{80P}{3} + 180 = 0$$

$$\therefore P = -\frac{27}{4}$$

Also, $f_3(x)$ is orthogonal to $f_2(x)$,

$$\int_{-2}^{2} f_2(x)f_3(x)dx = 0$$

$$\int_{-2}^{2} 3x\cdot(Px^2 + Qx + 9)dx = 0$$

$$\int_{-2}^{2}(3Px^3 + 3Qx^2 + 27x)dx = 0$$

$$\left[3P\frac{x^4}{4} + 3Q\frac{x^3}{3} + 27\frac{x^2}{2}\right]_{-2}^{2} = 0$$

$$[12P + 8Q + 54] - [12P - 8Q + 24] = 0$$

$$16Q = 0$$

$$\therefore Q = 0$$

17. Show that the functions $f_1(x) = 1, f_2(x) = x$ are orthogonal on $(-1,1)$. Determine the constant a and b such that the function $f_3(x) = 1 + ax + bx^2$ is orthogonal to both $f_1 \& f_2$ on that interval.

[N17/Extc/6M][M18/Extc/6M]

Solution:

Consider,

$$\int_{-1}^{1} f_1(x)f_2(x)dx = \int_{-1}^{1} 1.x\,dx = \left[\frac{x^2}{2}\right]_{-1}^{1} = 0$$

Now, consider

$$\int_{-1}^{1}[f_1(x)]^2dx = \int_{-1}^{1}[1]^2dx = [x]_{-1}^{1} = 2 \neq 0$$

$$\int_{-1}^{1}[f_2(x)]^2dx = \int_{-1}^{1}[x]^2dx = \left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{2}{3} \neq 0$$

Therefore, $f_1(x) = 1, f_2(x) = x$ are orthogonal over $(-1,1)$

It is given that $f_3(x)$ is orthogonal to $f_1(x)$,

$$\int_{-1}^{1} f_1(x)f_3(x)dx = 0$$

$$\int_{-1}^{1} 1\cdot(1 + ax + bx^2)dx = 0$$

$$\int_{-1}^{1}(1 + ax + bx^2)dx = 0$$

$$\left[x + a\frac{x^2}{2} + b\frac{x^3}{3}\right]_{-1}^{1} = 0$$

$$\left[1 + \frac{a}{2} + \frac{b}{3}\right] - \left[-1 + \frac{a}{2} - \frac{b}{3}\right] = 0$$

$$2 + \frac{2b}{3} = 0$$

$$\therefore b = -3$$

Also, $f_3(x)$ is orthogonal to $f_2(x)$,

$$\int_{-1}^{1} f_2(x)f_3(x)dx = 0$$

$$\int_{-1}^{1} x\cdot(1 + ax + bx^2)dx = 0$$

$$\int_{-1}^{1}(x + ax^2 + bx^3)dx = 0$$

$$\left[\frac{x^2}{2} + a\frac{x^3}{3} + b\frac{x^4}{4}\right]_{-1}^{1} = 0$$

$$\left[\frac{1}{2} + \frac{a}{3} + \frac{b}{4}\right] - \left[\frac{1}{2} - \frac{a}{3} + \frac{b}{4}\right] = 0$$

$$\frac{2a}{3} = 0$$

$$\therefore a = 0$$

18. Define orthogonal set of functions on (a, b). If $f(x) = P_1f_1(x) + P_2f_2(x) + P_3f_3(x)$ where P_1, P_2, P_3 are constants and $f_1(x), f_2(x), f_3(x)$ are orthogonal sets on (a, b), then show that $\int_a^b[f(x)]^2\,dx = P_1R_1^2 + P_2R_2^2 + P_3R_3^2$ where R_i are nonzero for $i = 1,2,3$

[N19/Elect/6M]

Solution:

$$f(x) = P_1f_1(x) + P_2f_2(x) + P_3f_3(x)$$

Squaring both the sides, we get

$$[f(x)]^2 = [P_1f_1(x) + P_2f_2(x) + P_3f_3(x)]^2$$

$$[f(x)]^2 = P_1^2[f_1(x)]^2 + P_2^2[f_2(x)]^2 + P_3^2[f_3(x)]^2 + 2P_1P_2f_1(x)f_2(x) + 2P_2P_3f_2(x)f_3(x) + 2P_1P_3f_1(x)f_3(x)$$

Integrating, we get

$\int_a^b [f(x)]^2 dx = P_1^2 \int_a^b [f_1(x)]^2\, dx + P_2^2 \int_a^b [f_2(x)]^2 dx +$
$P_3^2 \int_a^b [f_3(x)]^2 dx + 2P_1P_2 \int_a^b f_1(x) f_2(x)\, dx + 2P_2P_3 \int_a^b f_2(x) f_3(x)\, dx$
$+2P_1P_3 \int_a^b f_1(x) f_3(x)\, dx$

Since, f_1, f_2, f_3 are orthogonal sets on (a, b), we know that

$\int_a^b [f_1(x)]^2\, dx = \int_a^b [f_2(x)]^2 dx = \int_a^b [f_3(x)]^2 dx = 1$

$\int_a^b f_1(x) f_2(x)\, dx = R_1, \int_a^b f_2(x) f_3(x)\, dx = R_2, \int_a^b f_1(x) f_3(x)\, dx = R_3$

$\therefore \int_a^b [f(x)]^2\, dx = P_1 R_1^2 + P_2 R_2^2 + P_3 R_3^2$

Type III: Algebraic functions (x, x^2, x^3, etc)

1. Find the Fourier series of $f(x) = x$ in$(0,2\pi)$

[N17/Comp/5M][M18/Chem/5M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2\pi}\int_0^{2\pi} xdx$$
$$= \frac{1}{2\pi}\left[\frac{x^2}{2}\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left[\frac{4\pi^2}{2}\right]$$
$$\therefore a_0 = \pi$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi} x\cos nx \ dx$$
$$= \frac{1}{\pi}\left[(x)\left(\frac{\sin nx}{n}\right) - (1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{2\pi}$$
$$= \frac{1}{\pi n^2}[cosnx]_0^{2\pi}$$
$$= \frac{1}{\pi n^2}[1-1]$$
$$\therefore a_n = 0$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi} x\sin nx \ dx$$
$$= \frac{1}{\pi}\left[(x)\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{2\pi}$$
$$= \frac{-1}{n\pi}[(x)cosnx]_0^{2\pi}$$
$$= -\frac{1}{n\pi}[2\pi \ cos2n\pi]$$
$$\therefore b_n = \frac{-2}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$x = \pi - 2\sum\frac{sinnx}{n}$$

2. Find the Fourier series of $f(x) = \frac{\pi - x}{2}, x \in (0,2\pi)$.

[M15/CompIT/6M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\frac{\pi-x}{2}\,dx$$
$$= \frac{1}{4\pi}\left[\pi x-\frac{x^2}{2}\right]_0^{2\pi}$$
$$= \frac{1}{4\pi}\left[2\pi^2-\frac{4\pi^2}{2}\right]$$
$$\therefore a_0 = 0$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}\,dx$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi}\left(\frac{\pi-x}{2}\right)\cos nx\ dx$$
$$= \frac{1}{2\pi}\left[(\pi-x)\left(\frac{\sin nx}{n}\right)-(-1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{2\pi}$$
$$= \frac{-1}{2\pi n^2}[cosnx]_0^{2\pi}$$
$$= \frac{-1}{2\pi n^2}[1-1]$$
$$\therefore a_n = 0$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi}\frac{\pi-x}{2}\sin nx\ dx$$
$$= \frac{1}{2\pi}\left[(\pi-x)\left(-\frac{\cos nx}{n}\right)-(-1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{2\pi}$$
$$= \frac{-1}{2n\pi}[(\pi-x)cosnx]_0^{2\pi}$$
$$= -\frac{1}{2n\pi}[-\pi\ cos2n\pi-\pi cos0]$$
$$\therefore b_n = \frac{1}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0+\sum a_n\cos\frac{n\pi x}{l}+\sum b_n\sin\frac{n\pi x}{l}$$
$$\frac{\pi-x}{2} = \sum_1^\infty\frac{1}{n}\sin nx$$

3. Find half range sine series for $f(x) = \frac{\pi}{4}$ in $(0,\pi)$. Hence show that $\frac{\pi}{4} = 1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots$ ….
[M18/MTRX/5M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{\pi}\int_0^{\pi}\left(\frac{\pi}{4}\right)\sin nx\ dx$$
$$= \frac{2}{\pi}.\frac{\pi}{4}\left[\left(-\frac{\cos nx}{n}\right)\right]_0^{\pi}$$
$$= \frac{1}{2}\left[-\frac{\cos n\pi}{n}+\frac{cos\,0}{n}\right]$$
$$\therefore b_n = \frac{(1-(-1)^n)}{2n}$$

Thus,

$f(x) = \sum b_n \sin\frac{n\pi x}{l}$

$\frac{\pi}{4} = \sum \frac{(1-(-1)^n)}{2n}\ sinnx$

$\frac{\pi}{4} = \left[\frac{2sinx}{2} + 0 + \frac{2sin\,3x}{6} + 0 + \frac{2sin\,5x}{10} + \cdots.\right]$

$\frac{\pi}{4} = sinx + \frac{sin\,3x}{3} + \frac{sin\,5x}{5} + \cdots ....$

Put $x = \frac{\pi}{2}$

$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots ...$

4. Find half range cosine series for $f(x) = x, 0 < x < 2$
[M14/Biot/5M][N15/Biot/5M][M16/CompIT/5M][N18/Chem/6M]
Using Parseval's identity, deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots$
[M16/Chem/7M][N17/AutoMechCivil/8M][M18/AutoMechCivil/8M]
[N18/AutoMechCivil/8M][N17/Inst/8M][N19/AutoMechCivil/8M]

Solution:

We have,

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{2}\int_0^2 xdx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^2$

$= \frac{1}{2}\left[\frac{4}{2}\right]$

$\therefore a_0 = 1$

$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$

$a_n = \frac{2}{2}\int_0^2 x\cos\frac{n\pi x}{2}\ dx$

$= \left[x\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^2$

$= \frac{\cos\,n\pi}{\frac{n^2\pi^2}{4}} - \frac{\cos\,0}{\frac{n^2\pi^2}{4}}$

$= \frac{4}{n^2\pi^2}[(-1)^n - 1]$

$\therefore a_n = \frac{4[(-1)^n-1]}{n^2\pi^2}$

Thus, the fourier series is given by

$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$

$x = 1 + \sum_1^\infty \frac{4[(-1)^n-1]}{n^2\pi^2}\cos\frac{n\pi x}{2}$

$x = 1 + \frac{4}{\pi^2}\sum_1^\infty \frac{[(-1)^n-1]}{n^2}\cos\frac{n\pi x}{2}$

By Parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2\,dx = a_0^2 + \frac{1}{2}\sum_1^\infty a_n^2$$
$$\frac{1}{2}\int_0^2 [x]^2 dx = 1 + \frac{1}{2}\sum_1^\infty \frac{16[(-1)^n-1]^2}{\pi^4 n^4}$$
$$\frac{1}{2}\int_0^2 x^2 dx = 1 + \frac{8}{\pi^4}\left[\frac{4}{1^4}+\frac{4}{3^4}+\frac{4}{5^4}+\cdots.\right]$$
$$\frac{1}{2}\left[\frac{x^3}{3}\right]_0^2 = 1 + \frac{32}{\pi^4}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{4}{3} = 1 + \frac{32}{\pi^4}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{4}{3} - 1 = \frac{32}{\pi^4}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{1}{3} = \frac{32}{\pi^4}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{\pi^4}{96} = \left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$

5. Obtain Fourier series for $f(x) = x$ in $(-\pi,\pi)$
[N17/Inst/5M][N18/MTRX/5M]
Solution:
We see that $f(x) = x$ is an odd function
Thus,
$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{\pi}\int_0^\pi (x)\sin nx\;\,dx$$
$$= \frac{2}{\pi}\left[x\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{sinnx}{n^2}\right)\right]_0^\pi$$
$$= \frac{2}{\pi}\left[-\pi\frac{\cos n\pi}{n}\right]$$
$$\therefore b_n = \frac{-2(-1)^n}{n}$$
Thus,
$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$x = \sum \frac{-2(-1)^n}{n}\,sinnx$$

6. Obtain Fourier series for $f(x) = x$ in $(-3,3)$
[N19/Elect/5M]
Solution:
We see that $f(x) = x$ is an odd function
$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{3}\int_0^3 (x)\sin\frac{n\pi x}{3}\;dx$$
$$= \frac{2}{3}\left[x\left(-\frac{\cos\frac{n\pi x}{3}}{\frac{n\pi}{3}}\right) - (1)\left(-\frac{\sin\frac{n\pi x}{3}}{\frac{n^2\pi^2}{9}}\right)\right]_0^3$$

$$= \frac{2}{3}\left[-3\,\frac{\cos n\pi}{\frac{n\pi}{3}}\right]$$

$$\therefore b_n = \frac{-6(-1)^n}{n\pi}$$

Thus,

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$x = \sum \frac{-6(-1)^n}{n\pi}\sin\frac{n\pi x}{3}$$

7. Obtain the Fourier series for the function $f(x) = 2x - 1$ in $0 < x < 3$
[N15/ElexExtcElectBiomInst/5M]
Solution:
We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$

$$= \frac{1}{3}\left\{\int_0^3 (2x-1)dx\right\}$$

$$= \frac{1}{3}\left\{\left[\frac{2x^2}{2} - x\right]_0^3\right\}$$

$$\therefore a_0 = 2$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{3}\left\{\int_0^3 (2x-1)\cos\frac{2n\pi x}{3}\ dx\right\}$$

$$= \frac{2}{3}\left\{\left[(2x-1)\left(\frac{\sin\frac{2n\pi x}{3}}{\frac{2n\pi}{3}}\right) - (2)\left(-\frac{\cos\frac{2n\pi x}{3}}{\frac{4n^2\pi^2}{9}}\right)\right]_0^3\right\}$$

$$= \frac{2}{3}\left[\frac{2\cos 2n\pi}{\frac{4n^2\pi^2}{9}} - \frac{2\cos 0}{\frac{4n^2\pi^2}{9}}\right]$$

$$\therefore a_n = 0$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{3}\left\{\int_0^3 (2x-1)\sin\frac{2n\pi x}{3}\ dx\right\}$$

$$= \frac{2}{3}\left\{\left[(2x-1)\left(-\frac{\cos\frac{2n\pi x}{3}}{\frac{2n\pi}{3}}\right) - (2)\left(-\frac{\sin\frac{2n\pi x}{3}}{\frac{4n^2\pi^2}{9}}\right)\right]_0^3\right\}$$

$$= \frac{2}{3}\left[-\frac{5\cos 2n\pi}{\frac{2n\pi}{3}} - \frac{\cos 0}{\frac{2n\pi}{3}}\right]$$

$$\therefore b_n = -\frac{6}{n\pi}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$2x - 1 = 2 - \frac{6}{\pi}\sum\frac{\sin\left(\frac{2n\pi}{3}\right)}{n}$$

8. Find half range sine series for $f(x) = x + 1$ in $(0, \pi)$
[M17/CompIT/5M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{\pi}\int_0^\pi (x+1)\sin nx\ dx$$
$$= \frac{2}{\pi}\left[(x+1)\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^\pi$$
$$= \frac{-2}{\pi}\left[(\pi+1).\frac{\cos n\pi}{n} - \frac{\cos 0}{n}\right]$$
$$\therefore b_n = -\frac{2[(\pi+1)(-1)^n - 1]}{\pi n}$$

Thus,

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$x + 1 = \sum -\frac{2[(\pi+1)(-1)^n - 1]}{\pi n}\ sinnx$$

9. Find half range sine series for $f(x) = 2x$ in $(0, \pi)$
[N19/Elex/5M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{\pi}\int_0^\pi (2x)\sin nx\ dx$$
$$= \frac{2}{\pi}\left[(2x)\left(-\frac{\cos nx}{n}\right) - (2)\left(-\frac{\sin nx}{n^2}\right)\right]_0^\pi$$
$$= \frac{-2}{\pi}\left[(2\pi).\frac{\cos n\pi}{n}\right]$$
$$\therefore b_n = \frac{-4(-1)^n}{n}$$

Thus,

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$2x = \sum \frac{-4(-1)^n}{n}\ sinnx$$

10. Prove that $\frac{l}{2} - x = \frac{1}{\pi}\sum\frac{1}{n}\sin\left(\frac{2n\pi x}{l}\right), 0 < x < l$
[M18/Chem/6M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{l}\int_0^l \left(\frac{l}{2} - x\right)\sin\frac{n\pi x}{l}\ dx$$

$$= \frac{2}{l}\left[\left(\frac{l}{2}-x\right)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right)-(-1)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_0^l$$

$$= \frac{2}{l}\left[\left(\frac{l}{2}\right).\frac{\cos n\pi}{\frac{n\pi}{l}}+\left(\frac{l}{2}\right).\frac{\cos 0}{\frac{n\pi}{l}}\right]$$

$$\therefore b_n = \frac{1+(-1)^n}{n\pi}$$

Thus,

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$\frac{l}{2}-x = \sum \frac{1+(-1)^n}{n\pi}\sin\frac{n\pi x}{l}$$

$$\frac{l}{2}-x = \frac{1}{\pi}\left[0+\frac{2}{2}.\sin\frac{2\pi x}{l}+0+\frac{2}{4}.\sin\frac{4\pi x}{l}+\cdots\ldots..\right]$$

$$\frac{l}{2}-x = \frac{1}{\pi}\left[\frac{1}{1}.\sin\frac{2\pi x}{l}+\frac{1}{2}.\sin\frac{4\pi x}{l}+\frac{1}{3}.\sin\frac{6\pi x}{l}+\cdots..\right]$$

$$\frac{l}{2}-x = \frac{1}{\pi}\sum\frac{1}{n}.\sin\frac{2n\pi x}{l}$$

11. Find half range sine series for $f(x) = x, 0 < x < 2$

[N17/Extc/5M][M18/Extc/5M]

Solution:

We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{2}\int_0^2 x\sin\frac{n\pi x}{2}\ dx$$

$$= \left[x\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right)-(1)\left(-\frac{\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^2$$

$$= -\frac{4}{n\pi}cosn\pi$$

$$\therefore b_n = \frac{-4[(-1)^n]}{n\pi}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n sinnx$$

$$x = \sum -\frac{4(-1)^n}{n\pi}\sin\left(\frac{n\pi x}{2}\right)$$

12. Find half range cosine series for $f(x) = x, 0 < x < 1$

Using Parseval's identity, deduce that $\frac{\pi^4}{96} = \frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots$

[N19/Elect/8M]

Solution:

We have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{1}\int_0^1 x dx = \left[\frac{x^2}{2}\right]_0^1$$
$$= \frac{1}{2}$$
$$\therefore a_0 = \frac{1}{2}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{1}\int_0^1 x\cos n\pi x \ dx$$
$$= 2\left[x\left(\frac{\sin n\pi x}{n\pi}\right) - (1)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_0^1$$
$$= 2\left[\frac{\cos n\pi}{n^2\pi^2} - \frac{\cos 0}{n^2\pi^2}\right]$$
$$= \frac{2}{n^2\pi^2}[(-1)^n - 1]$$
$$\therefore a_n = \frac{2[(-1)^n - 1]}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$x = \frac{1}{2} + \sum_1^\infty \frac{2[(-1)^n - 1]}{n^2\pi^2}\cos n\pi x$$
$$x = \frac{1}{2} + \frac{2}{\pi^2}\sum_1^\infty \frac{[(-1)^n - 1]}{n^2}\cos n\pi x$$

By Parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2 dx = a_0^2 + \frac{1}{2}\sum_1^\infty a_n^2$$
$$\frac{1}{1}\int_0^1 [x]^2 dx = \frac{1}{4} + \frac{1}{2}\sum_1^\infty \frac{4[(-1)^n - 1]^2}{\pi^4 n^4}$$
$$\int_0^1 x^2 dx = \frac{1}{4} + \frac{2}{\pi^4}\left[\frac{4}{1^4} + \frac{4}{3^4} + \frac{4}{5^4} + \cdots.\right]$$
$$\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{4} + \frac{8}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\ldots\right]$$
$$\frac{1}{3} = \frac{1}{4} + \frac{8}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\ldots\right]$$
$$\frac{1}{3} - \frac{1}{4} = \frac{8}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\ldots\right]$$
$$\frac{1}{12} = \frac{8}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\ldots\right]$$
$$\frac{\pi^4}{96} = \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\ldots\right]$$

13. Obtain Fourier expansion of $f(x) = |x|$ in $(-\pi, \pi)$
[N14/ElexExtcElectBiomInst/5M][M18/AutoMechCivil/6M][M18/Elect/5M][N18/Chem/5M][M19/MTRX/6M]
Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\ldots$
[N17/AutoMechCivil/8M][M18/Elex/8M]
Solution:
We see that, $f(x) = |x|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^\pi xdx$$
$$= \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^\pi$$
$$= \frac{1}{\pi}\left[\frac{\pi^2}{2}\right]$$
$$\therefore a_0 = \frac{\pi}{2}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\int_0^\pi xcosnx\, dx$$
$$= \frac{2}{\pi}\left[x\left(\frac{sinnx}{n}\right) - (1)\left(-\frac{cosnx}{n^2}\right)\right]_0^\pi$$
$$= \frac{2}{\pi n^2}[cosnx]_0^\pi$$
$$= \frac{2}{\pi n^2}[(-1)^n - 1]$$
$$\therefore a_n = \frac{2[(-1)^n-1]}{\pi n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$|x| = \frac{\pi}{2} + \sum_1^\infty \frac{2[(-1)^n-1]}{\pi n^2}cosnx$$

Putting $x = 0$, we get

$$0 = \frac{\pi}{2} + \frac{2}{\pi}\left[-\frac{2}{1^2} - \frac{2}{3^2} - \frac{2}{5^2} - \cdots ..\right]$$
$$-\frac{\pi}{2} = -\frac{4}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots\right]$$
$$\frac{\pi^2}{8} = \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots\right]$$

14. Find the Fourier series of $f(x) = |x|, -k < x < k$. Hence deduce that $\sum \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$

[M15/ElexExtcElectBiomInst/6M]

Solution:

We see that, $f(x) = |x|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{k}\int_0^k xdx$$
$$= \frac{1}{k}\left[\frac{x^2}{2}\right]_0^k$$
$$\therefore a_0 = \frac{k}{2}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{k}\int_0^k x\cos\frac{n\pi x}{k}\ dx$$

$$= \frac{2}{k}\left[x\left(\frac{\sin\frac{n\pi x}{k}}{\frac{n\pi}{k}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{k}}{\frac{n^2\pi^2}{k^2}}\right)\right]_0^k$$

$$= \frac{2k}{\pi^2 n^2}\left[\cos\frac{n\pi x}{k}\right]_0^k$$

$$= \frac{2k}{\pi^2 n^2}[(-1)^n - 1]$$

$$\therefore a_n = \frac{2k[(-1)^n - 1]}{\pi^2 n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$

$$|x| = \frac{k}{2} + \frac{2k}{\pi^2}\sum \frac{[(-1)^n - 1]}{n^2}\cos\left(\frac{n\pi x}{k}\right)$$

By parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2 dx = a_0^2 + \frac{1}{2}\sum_1^\infty a_n^2$$

$$\frac{1}{k}\int_0^k x^2 dx = \frac{k^2}{4} + \frac{1}{2}\sum_1^\infty \frac{4k^2[(-1)^n - 1]^2}{\pi^4 n^4}$$

$$\frac{1}{k}\left[\frac{x^3}{3}\right]_0^k = \frac{k^2}{4} + \frac{4k^2}{2\pi^4}\left[\frac{4}{1^4} + \frac{4}{3^4} + \frac{4}{5^4} + \cdots \ldots\right]$$

$$\frac{k^2}{3} = \frac{k^2}{4} + \frac{8k^2}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots ..\right]$$

$$\frac{k^2}{3} - \frac{k^2}{4} = \frac{8k^2}{\pi^4}\sum_1^\infty \frac{1}{(2n-1)^4}$$

$$\frac{k^2}{12} = \frac{8k^2}{\pi^4}\sum_1^\infty \frac{1}{(2n-1)^4}$$

$$\frac{\pi^4}{96} = \sum_1^\infty \frac{1}{(2n-1)^4}$$

15. Find Fourier series of $f(x) = |x|$ in (-3,3)
[M17/AutoMechCivil/6M]
Solution:
We see that, $f(x) = |x|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{3}\int_0^3 x dx$$

$$= \frac{1}{3}\left[\frac{x^2}{2}\right]_0^3$$

$$\therefore a_0 = \frac{3}{2}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{3}\int_0^3 x\cos\left(\frac{n\pi x}{3}\right) dx$$

$$= \frac{2}{3}\left[x\left(\frac{\sin\left(\frac{n\pi x}{3}\right)}{\frac{n\pi}{3}}\right) - (1)\left(-\frac{\cos\left(\frac{n\pi x}{3}\right)}{\frac{n^2\pi^2}{9}}\right)\right]_0^3$$

$$= \frac{6}{\pi^2 n^2}\left[\cos\left(\frac{n\pi x}{3}\right)\right]_0^3$$
$$= \frac{6}{\pi^2 n^2}[(-1)^n - 1]$$
$$\therefore a_n = \frac{6[(-1)^n - 1]}{\pi^2 n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$|x| = \frac{3}{2} + \frac{6}{\pi^2}\sum \frac{[(-1)^n - 1]}{n^2}\cos\left(\frac{n\pi x}{3}\right)$$

16. Find Fourier series of $f(x) = |x|$ in (-2,2)
[M19/Extc/6M]
Solution:
We see that, $f(x) = |x|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{2}\int_0^2 xdx$$
$$= \frac{1}{2}\left[\frac{x^2}{2}\right]_0^2$$
$$\therefore a_0 = 1$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{2}\int_0^2 x\cos\frac{n\pi x}{2}\ dx$$
$$= \left[x\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^2$$
$$= \frac{4}{\pi^2 n^2}\left[\cos\frac{n\pi x}{2}\right]_0^2$$
$$= \frac{4}{\pi^2 n^2}[(-1)^n - 1]$$
$$\therefore a_n = \frac{4[(-1)^n - 1]}{\pi^2 n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$|x| = 1 + \frac{4}{\pi^2}\sum \frac{[(-1)^n - 1]}{n^2}\cos\left(\frac{n\pi x}{2}\right)$$

17. Find a Fourier series to represent $f(x) = x^2\ in\ (0,2\pi)$
[M14/Chem/5M][N15/Chem/5M]
and hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots \ldots..$
[N17/Elex/5M][N18/AutoMechCivil/8M][M19/AutoMechCivil/8M]
[M19/MTRX/8M][N19/AutoMechCivil/8M]
Solution:

$$a_0 = \frac{1}{2l}\int_0^l f(x)dx$$
$$= \frac{1}{2\pi}\int_0^{2\pi} x^2 dx$$
$$= \frac{1}{2\pi}\left[\frac{x^3}{3}\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left[\frac{8\pi^3}{3}\right]$$
$$\therefore a_0 = \frac{4\pi^2}{3}$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi} x^2 cosnx\, dx$$
$$= \frac{1}{\pi}\left[x^2\left(\frac{sinnx}{n}\right) - (2x)\left(-\frac{cosnx}{n^2}\right) + (2)\left(-\frac{sinnx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{2}{\pi n^2}[xcosnx]_0^{2\pi}$$
$$= \frac{2}{\pi n^2}[2\pi]$$
$$\therefore a_n = \frac{4}{n^2}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\left(\frac{n\pi x}{l}\right)\ dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi} x^2 sinnx\, dx$$
$$= \frac{1}{\pi}\left[x^2\left(-\frac{cosnx}{n}\right) - (2x)\left(-\frac{sinnx}{n^2}\right) + (2)\left(\frac{cosnx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{1}{\pi}\left[-x^2\frac{cosnx}{n} + 2\frac{cosnx}{n^3}\right]_0^{2\pi}$$
$$= \frac{1}{\pi}\left[-\frac{4\pi^2}{n} + \frac{2}{n^3} + 0 - \frac{2}{n^3}\right]$$
$$\therefore b_n = -\frac{4\pi}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx + \sum b_n sinnx$$
$$x^2 = \frac{4\pi^2}{3} + \sum_1^\infty \frac{4}{n^2}cosnx + \sum_1^\infty -\frac{4\pi}{n}sinnx$$

Putting $x = \pi$, we get

$$\pi^2 = \frac{4\pi^2}{3} + 4\sum_1^\infty \frac{(-1)^n}{n^2}$$
$$\pi^2 - \frac{4\pi^2}{3} = 4\left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \cdots \ldots\right]$$
$$-\frac{\pi^2}{3} = -4\ \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots \ldots..\right]$$
$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots \ldots.$$

18. Find the Fourier expansion of $f(x) = 4 - x^2$ in the interval $(0,2)$

[N15/Biot/6M][M16/CompIT/6M][M18/N18/Biom/6M][N18/Elex/6M]

And Prove that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \ldots$

[N14/ElexExtcElectBiomInst/8M][N14/Chem/7M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$

$$= \frac{1}{2}\left\{\int_0^2 (4 - x^2)dx\right\}$$

$$= \frac{1}{2}\left\{\left[4x - \frac{x^3}{3}\right]_0^2\right\}$$

$$\therefore a_0 = \frac{8}{3}$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{1}{1}\left\{\int_0^2 (4 - x^2)\cos n\pi x\ dx\right\}$$

$$= \left\{\left[(4 - x^2)\left(\frac{\sin n\pi x}{n\pi}\right) - (-2x)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right) + (-2)\left(-\frac{\sin n\pi x}{n^3\pi^3}\right)\right]_0^2\right\}$$

$$= \frac{-2}{\pi^2 n^2}\{[2.cos2n\pi]\}$$

$$\therefore a_n = \frac{-4}{n^2\pi^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{1}{1}\left\{\int_0^2 (4 - x^2)\sin n\pi x\ dx\right\}$$

$$= \left\{\left[(4 - x^2)\left(-\frac{\cos n\pi x}{n\pi}\right) - (-2x)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right) + (-2)\left(\frac{\cos n\pi x}{n^3\pi^3}\right)\right]_0^2\right\}$$

$$= \left[-\frac{2}{n^3\pi^3} + \frac{4}{n\pi} + \frac{2}{n^3\pi^3}\right]$$

$$\therefore b_n = \frac{4}{n\pi}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$4 - x^2 = \frac{8}{3} + \sum \frac{-4}{n^2\pi^2} cosn\pi x + \sum \frac{4}{n\pi} sinn\pi x$$

Putting $x = 0$, we get

$$4 = \frac{8}{3} - \frac{4}{\pi^2}\sum\frac{1}{n^2}$$

$$4 - \frac{8}{3} = -\frac{4}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots ..\right]$$

$$\frac{4}{3} = -\frac{4}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots ..\right]$$

$$-\frac{\pi^2}{3} = \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots ..\right] \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Putting $x = 2$ in fourier series, we get

$$0 = \frac{8}{3} - \frac{4}{\pi^2}\sum\frac{1}{n^2}$$

$$-\frac{8}{3}=-\frac{4}{\pi^2}\left[\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots..\right]$$

$$\frac{2\pi^2}{3}=\left[\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots..\right]\text{………………………}(2)$$

Adding (1) & (2), we get

$$-\frac{\pi^2}{3}+\frac{2\pi^2}{3}=\frac{2}{1^2}+\frac{2}{2^2}+\frac{2}{3^2}+\cdots\dots$$

$$\frac{\pi^2}{3}=2\left[\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots\dots\right]$$

$$\frac{\pi^2}{6}=\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\cdots\dots$$

19. Find Fourier expansion of $f(x)=\left(\frac{\pi-x}{2}\right)^2$ in the interval $0\leq x\leq 2\pi$ and $f(x+2\pi)=f(x)$
[M16/Chem/5M][N16/Chem/6M]
Also deduce that

(i) $\frac{\pi^2}{6}=\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\cdots\dots\dots$

[M14/AutoMechCivil/6M][M18/Elex/5M]

(ii) $\frac{\pi^2}{12}=\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots\dots$

(iii) $\frac{\pi^4}{90}=\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+\cdots..$

[M16/AutoMechCivil/8M]

Solution:

We have,

$$a_0=\frac{1}{2l}\int_0^{2l}f(x)dx$$

$$=\frac{1}{2\pi}\int_0^{2\pi}\left(\frac{\pi-x}{2}\right)^2dx$$

$$=\frac{1}{8\pi}\left[\frac{(\pi-x)^3}{-3}\right]_0^{2\pi}$$

$$=\frac{1}{-24\pi}[-\pi^3-\pi^3]$$

$$\therefore a_0=\frac{\pi^2}{12}$$

$$a_n=\frac{1}{l}\int_0^{2l}f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n=\frac{1}{\pi}\int_0^{2\pi}\left(\frac{\pi-x}{2}\right)^2\cos nx\ dx$$

$$=\frac{1}{4\pi}\left[(\pi-x)^2\left(\frac{\sin nx}{n}\right)-2(\pi-x)(-1)\left(-\frac{\cos nx}{n^2}\right)+(2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$$

$$=\frac{-2}{4\pi n^2}[(\pi-x)cosnx]_0^{2\pi}$$

$$=\frac{-1}{2\pi n^2}[-\pi-\pi]$$

$$\therefore a_n=\frac{1}{n^2}$$

$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$

$b_n = \frac{1}{\pi}\int_0^{2\pi}\left(\frac{\pi - x}{2}\right)^2 \sin nx\ \ dx$

$= \frac{1}{4\pi}\left[(\pi - x)^2\left(-\frac{\cos\ nx}{n}\right) - 2(\pi - x)(-1)\left(-\frac{\sin\ nx}{n^2}\right) + (2)\left(\frac{cos nx}{n^3}\right)\right]_0^{2\pi}$

$= \frac{1}{4\pi}\left\{-\frac{\pi^2 cos\,2n\pi}{n} + \frac{2cos\,2n\pi}{n^3} + \frac{\pi^2 cos\,0}{n} - \frac{2cos\,0}{n^3}\right\}$

$\therefore b_n = 0$

Thus, the fourier series is given by

$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$

$\left(\frac{\pi - x}{2}\right)^2 = \frac{\pi^2}{12} + \sum_1^\infty \frac{1}{n^2}\cos nx$

(i) Putting $x = 0$, we get

$\left(\frac{\pi}{2}\right)^2 = \frac{\pi^2}{12} + \sum_1^\infty \frac{1}{n^2}$

$\frac{\pi^2}{4} - \frac{\pi^2}{12} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \ldots..$

$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \ldots \ldots$

(ii) Putting $x = \pi$, we get

$0 = \frac{\pi^2}{12} + \sum_1^\infty \frac{1}{n^2}(-1)^n$

$-\frac{\pi^2}{12} = -\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \cdots \ldots.$

$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots \ldots$

(iii) By Parseval's identity,

$\frac{1}{2l}\int_0^{2l}[f(x)]^2 dx = a_0^2 + \frac{1}{2}\sum_1^\infty (a_n^2 + b_n^2)$

$\frac{1}{2\pi}\left\{\int_0^{2\pi}\left(\frac{\pi - x}{2}\right)^4 dx\right\} = \frac{\pi^4}{144} + \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{1}{32\pi}\left[\frac{(\pi - x)^5}{-5}\right]_0^{2\pi} = \frac{\pi^4}{144} + \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$-\frac{1}{160\pi}[-\pi^5 - \pi^5] = \frac{\pi^4}{144} + \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{80} = \frac{\pi^4}{144} + \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{80} - \frac{\pi^4}{144} = \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{64\pi^4}{80\times 144} = \frac{1}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{90} = \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots.\right]$

20. Find the Fourier series of $f(x) = x^2$, $-\pi \le x \le \pi$ and hence prove that
(i) $\frac{\pi^2}{6} = \sum_1^\infty \frac{1}{n^2}$ (ii) $\frac{\pi^2}{12} = \sum_1^\infty \frac{(-1)^{n+1}}{n^2}$ (iii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$ (iv) $\frac{\pi^4}{90} = \sum_1^\infty \frac{1}{n^4}$
[N13/AutoMechCivil/8M][N14/Chem/5M][M18/Inst/5M][M19/Inst/5M]

Solution:

We see that, $f(x) = x^2$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^\pi x^2 dx$$
$$= \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^\pi$$
$$= \frac{1}{\pi}\left[\frac{\pi^3}{3}\right]$$
$$\therefore a_0 = \frac{\pi^2}{3}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\int_0^\pi x^2 cosnx\, dx$$
$$= \frac{2}{\pi}\left[x^2\left(\frac{sinnx}{n}\right) - (2x)\left(-\frac{cosnx}{n^2}\right) + (2)\left(-\frac{sinnx}{n^3}\right)\right]_0^\pi$$
$$= \frac{4}{\pi n^2}[xcosnx]_0^\pi$$
$$= \frac{4}{\pi n^2}[\pi(-1)^n]$$
$$\therefore a_n = \frac{4(-1)^n}{n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$x^2 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}cosnx$$

(i) Putting $x = \pi$

$$\pi^2 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}cosn\pi$$
$$\pi^2 - \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}(-1)^n$$
$$\frac{2\pi^2}{3} = \sum_1^\infty \frac{4}{n^2}$$
$$\frac{\pi^2}{6} = \sum_1^\infty \frac{1}{n^2} \ldots\ldots\ldots\ldots(1)$$

(ii) Putting $x = 0$

$$0 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}cos0$$
$$0 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}$$
$$-\frac{\pi^2}{3} = 4\sum_1^\infty \frac{(-1)^n}{n^2}$$
$$\frac{\pi^2}{12} = \sum_1^\infty \frac{(-1)^{n+1}}{n^2} \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

(iii) Adding (1) & (2)

$$\frac{\pi^2}{6} + \frac{\pi^2}{12} = \sum_1^\infty \frac{1+(-1)^{n+1}}{n^2}$$
$$\frac{\pi^2}{4} = \left[\frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \cdots \ldots.\right]$$

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots..$

(iv) By parseval's identity

$\frac{1}{l}\int_0^l [f(x)]^2\,dx = a_0^2 + \frac{1}{2}\sum_1^\infty a_n^2$

$\frac{1}{\pi}\int_0^\pi [x^2]^2 dx = \frac{\pi^4}{9} + \frac{1}{2}\sum_1^\infty \frac{16(-1)^{2n}}{n^4}$

$\frac{1}{\pi}\left[\frac{x^5}{5}\right]_0^\pi = \frac{\pi^4}{9} + \frac{16}{2}\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{5} = \frac{\pi^4}{9} + 8\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{5} - \frac{\pi^4}{9} = 8\sum_1^\infty \frac{1}{n^4}$

$\frac{4\pi^4}{45} = 8\sum_1^\infty \frac{1}{n^4}$

$\frac{\pi^4}{90} = \sum_1^\infty \frac{1}{n^4}$

21. Find the Fourier series of $f(x) = x|x|\ ;\ -1 < x < 1$
[M15/Chem/6M][N18/MTRX/6M]
Solution:
We see that, $f(x) = x|x|$ is an odd function.

$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$

$b_n = \frac{2}{1}\int_0^1 x^2 \sin n\pi x\ \ dx$

$= 2\left[x^2\left(-\frac{\cos n\pi x}{n\pi}\right) - (2x)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right) + (2)\left(\frac{\cos n\pi x}{n^3\pi^3}\right)\right]_0^1$

$= 2\left[-\frac{\cos n\pi}{n\pi} + \frac{2\cos n\pi}{n^3\pi^3} - \frac{2\cos 0}{n^3\pi^3}\right]$

$\therefore b_n = 2\left[-\frac{(-1)^n}{n\pi} + \frac{2[(-1)^n - 1]}{n^3\pi^3}\right]$

Thus, the fourier series is given by

$f(x) = a_0 + \sum a_n cosnx$

$x|x| = \sum 2\left[-\frac{(-1)^n}{n\pi} + \frac{2[(-1)^n - 1]}{n^3\pi^3}\right]\sin n\pi x$

22. Find Fourier series for $f(x) = \frac{3x^2 - 6x\pi + 2\pi^2}{12}\ in\ (0, 2\pi)$
Hence find that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots \ldots$
[N14/Biot/8M][M16/CompIT/8M][M16/ElexExtcElectBiomInst/6M]
[M17/ElexExctElectBiomInst/8M][M18/Inst/8M]
Solution:
We have,

$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$

$= \frac{1}{2\pi}\int_0^{2\pi} \frac{3x^2 - 6x\pi + 2\pi^2}{12}\,dx$

$$= \frac{1}{24\pi}\left[3\frac{x^3}{3} - 6\pi\frac{x^2}{2} + 2\pi^2 x\right]_0^{2\pi}$$
$$= \frac{1}{24\pi}[8\pi^3 - 12\pi^3 + 4\pi^3]$$
$$\therefore a_0 = 0$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi} \frac{3x^2-6x\pi+2\pi^2}{12}\cos nx\ dx$$
$$= \frac{1}{12\pi}\left[(3x^2 - 6x\pi + 2\pi^2)\left(\frac{\sin nx}{n}\right) - (6x-6\pi)\left(-\frac{\cos nx}{n^2}\right) + (6)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{1}{12\pi n^2}[(6x-6\pi)cosnx]_0^{2\pi}$$
$$= \frac{1}{12\pi n^2}[6\pi + 6\pi]$$
$$\therefore a_n = \frac{1}{n^2}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi} \frac{3x^2-6x\pi+2\pi^2}{12}\sin nx\ dx$$
$$= \frac{1}{12\pi}\left[(3x^2 - 6x\pi + 2\pi^2)\left(-\frac{\cos nx}{n}\right) - (6x-6\pi)\left(-\frac{\sin nx}{n^2}\right) + (6)\left(\frac{cosnx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{1}{12\pi}\left\{-\frac{2\pi^2 cos\,2n\pi}{n} + \frac{6cos\,2n\pi}{n^3} + \frac{2\pi^2 cos\,0}{n} - \frac{6cos\,0}{n^3}\right\}$$
$$\therefore b_n = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$\frac{3x^2-6x\pi+2\pi^2}{12} = \sum_1^\infty \frac{1}{n^2}\cos nx$$

Putting $x = 0$, we get

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots ….$$

23. Find the fourier series for $f(x) = 9 - x^2$ for $-3 < x < 3$
[M15/CompIT/5M][M16/ElexExtcElectBiomInst/5M][N17/Extc/6M][M18/Extc/6M]

Solution:

We see that, $f(x) = 9 - x^2$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{3}\int_0^3 9 - x^2 dx$$
$$= \frac{1}{3}\left[9x - \frac{x^3}{3}\right]_0^3$$
$$\therefore a_0 = 6$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{3}\int_0^3 (9-x^2)\cos\frac{n\pi x}{3}\ dx$$

$$= \frac{2}{3}\left[(9-x^2)\left(\frac{\sin\frac{n\pi x}{3}}{\frac{n\pi}{3}}\right)-(-2x)\left(-\frac{\cos\frac{n\pi x}{3}}{\frac{n^2\pi^2}{9}}\right)+(-2)\left(-\frac{\sin\frac{n\pi x}{3}}{\frac{n^3\pi^3}{27}}\right)\right]_0^3$$

$$= \frac{2}{3}\left[-\frac{6\cos n\pi}{\frac{n^2\pi^2}{9}}\right]$$

$$\therefore a_n = -\frac{36(-1)^n}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$9 - x^2 = 6 - \frac{36}{\pi^2}\sum\frac{(-1)^n}{n^2}\cos\left(\frac{n\pi x}{3}\right)$$

24. Find the Fourier series of $f(x) = 1 - x^2 in\ (-1,1)$
[M14/Chem/5M][N15/CompIT/5M][M18/MTRX/6M][M19/Biom/6M][N19/Extc/6M]

Solution:

We see that, $f(x) = 1 - x^2$ is an even function

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{1}\int_0^1 1 - x^2 dx$$

$$= \left[x - \frac{x^3}{3}\right]_0^1$$

$$\therefore a_0 = \frac{2}{3}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$= \frac{2}{1}\int_0^1 (1-x^2)\cos n\pi x\ dx$$

$$= 2\left[(1-x^2)\left(\frac{\sin n\pi x}{n\pi}\right)-(-2x)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)+(-2)\left(-\frac{\sin n\pi x}{n^3\pi^3}\right)\right]_0^1$$

$$= 2\left[\frac{-2\cos n\pi}{n^2\pi^2}\right]$$

$$\therefore a_n = \frac{-4(-1)^n}{n^2\pi^2}$$

Thus,

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$1 - x^2 = \frac{2}{3} - \frac{4}{\pi^2}\sum\frac{(-1)^n}{n^2} cos n\pi x$$

25. Find the fourier series for $f(x) = (\pi - x)^2$ in $0 \le x \le 2\pi$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots$
[M18/Extc/8M][N18/Extc/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2\pi}\int_0^{2\pi}(\pi - x)^2 dx$$
$$= \frac{1}{2\pi}\left[\frac{(\pi-x)^3}{-3}\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left[-\frac{\pi^3}{-3} - \frac{\pi^3}{-3}\right]$$
$$\therefore a_0 = \frac{\pi^2}{3}$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\int_0^{2\pi}(\pi - x)^2 \cos nx \ \ dx$$
$$= \frac{1}{\pi}\left[(\pi - x)^2\left(\frac{\sin nx}{n}\right) - 2(\pi - x)(-1)\left(-\frac{\cos nx}{n^2}\right) + (-2)(-1)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{-2}{\pi n^2}[(\pi - x)cosnx]_0^{2\pi}$$
$$= \frac{-2}{\pi n^2}[(-\pi) - \pi]$$
$$\therefore a_n = \frac{4}{n^2}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi}(\pi - x)^2 \sin nx \ \ dx$$
$$= \frac{1}{\pi}\left[(\pi - x)^2\left(-\frac{\cos nx}{n}\right) - 2(\pi - x)(-1)\left(-\frac{\sin nx}{n^2}\right) + (-2)(-1)\left(\frac{cosnx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{1}{\pi}\left\{-\frac{(\pi-2\pi^2)cos\,2n\pi}{n} + \frac{2cos\,2n\pi}{n^3} + \frac{\pi^2 cos\,0}{n} - \frac{2cos\,0}{n^3}\right\}$$
$$= \frac{1}{\pi}\left\{-\frac{\pi^2}{n} + \frac{2}{n^3} + \frac{\pi^2}{n} - \frac{2}{n^3}\right\}$$
$$\therefore b_n = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$(\pi - x)^2 = \frac{\pi^2}{3} + \sum\frac{4}{n^2}cosnx$$

Putting $x = 0$, we get

$$\pi^2 = \frac{\pi^2}{3} + \sum\frac{4}{n^2}$$
$$\pi^2 - \frac{\pi^2}{3} = 4\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots.\right]$$
$$\frac{2\pi^2}{3} = 4\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots.\right]$$
$$\frac{\pi^2}{6} = \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots.\right] \text{……………………. (1)}$$

Putting $x = \pi$ in the fourier series, we get

$$0 = \frac{\pi^2}{3} + \sum\frac{4(-1)^n}{n^2}$$
$$-\frac{\pi^2}{3} = 4\left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \cdots \dots.\right]$$

$\frac{\pi^2}{12} = \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots ..\right]$ …………………… (2)

Adding (1) & (2), we get

$\frac{\pi^2}{6} + \frac{\pi^2}{12} = \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots .\right] + \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots ..\right]$

$\frac{\pi^2}{4} = \frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \cdots ....$

$\frac{\pi^2}{4} = 2\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots ... .\right]$

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots ....$

26. Find the fourier series for $f(x) = 5x^2$ in $(-\pi, \pi)$

[N17/Elect/5M]

Solution:

We see that, $f(x) = 5x^2$ is an even function.

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{\pi}\int_0^\pi 5x^2 dx$

$= \frac{5}{\pi}\left[\frac{x^3}{3}\right]_0^\pi$

$= \frac{5}{\pi}\left[\frac{\pi^3}{3}\right]$

$\therefore a_0 = \frac{5\pi^2}{3}$

$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$

$a_n = \frac{2}{\pi}\int_0^\pi 5x^2 cosnx\, dx$

$= \frac{10}{\pi}\left[x^2\left(\frac{sinnx}{n}\right) - (2x)\left(-\frac{cosnx}{n^2}\right) + (2)\left(-\frac{sinnx}{n^3}\right)\right]_0^\pi$

$= \frac{20}{\pi n^2}[xcosnx]_0^\pi$

$= \frac{20}{\pi n^2}[\pi(-1)^n]$

$\therefore a_n = \frac{20(-1)^n}{n^2}$

Thus, the fourier series is given by

$f(x) = a_0 + \sum a_n cosnx$

$5x^2 = \frac{5\pi^2}{3} + \sum \frac{20(-1)^n}{n^2} cosnx$

27. Find the Fourier series of $f(x) = x^2$ in $(-\pi, \pi)$

[N14/CompIT/5M][M18/Comp/5M]

Solution:

We see that, $f(x) = x^2$ is an even function.

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{\pi}\int_0^{\pi} x^2 dx$

$= \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi}$

$= \frac{1}{\pi}\left[\frac{\pi^3}{3}\right]$

$\therefore a_0 = \frac{\pi^2}{3}$

$a_n = \frac{2}{l}\int_0^{l} f(x)\cos\frac{n\pi x}{l} dx$

$a_n = \frac{2}{\pi}\int_0^{\pi} x^2 cosnx\ dx$

$= \frac{2}{\pi}\left[x^2\left(\frac{sinnx}{n}\right) - (2x)\left(-\frac{cosnx}{n^2}\right) + (2)\left(-\frac{sinnx}{n^3}\right)\right]_0^{\pi}$

$= \frac{4}{\pi n^2}[xcosnx]_0^{\pi}$

$= \frac{4}{\pi n^2}[\pi(-1)^n]$

$\therefore a_n = \frac{4(-1)^n}{n^2}$

Thus, the fourier series is given by

$f(x) = a_0 + \sum a_n cosnx$

$x^2 = \frac{\pi^2}{3} + \sum_1^{\infty}\frac{4(-1)^n}{n^2}cosnx$

28. Find the fourier for $f(x) = x - x^2$ in $(0,2\pi)$

[N17/Elect/6M]

Solution:

We have,

$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$

$= \frac{1}{2\pi}\int_0^{2\pi}(x - x^2)dx$

$= \frac{1}{2\pi}\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{2\pi}$

$= \frac{1}{2\pi}\left[\frac{4\pi^2}{2} - \frac{8\pi^3}{3}\right]$

$\therefore a_0 = \pi - \frac{4\pi^2}{3}$

$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l} dx$

$a_n = \frac{1}{\pi}\int_0^{2\pi}(x - x^2)\cos nx\ \ dx$

$= \frac{1}{\pi}\left[(x - x^2)\left(\frac{\sin nx}{n}\right) - (1 - 2x)\left(-\frac{\cos nx}{n^2}\right) + (-2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$

$= \frac{1}{\pi n^2}[(1 - 2x)cosnx]_0^{2\pi}$

$= \frac{1}{\pi n^2}[1 - 4\pi - 1]$

$\therefore a_n = \frac{-4}{n^2}$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{1}{\pi}\int_0^{2\pi}(x-x^2)\sin nx\ \ dx$$
$$= \frac{1}{\pi}\left[(x-x^2)\left(-\frac{\cos\ nx}{n}\right)-(1-2x)\left(-\frac{\sin\ nx}{n^2}\right)+(-2)\left(\frac{cosnx}{n^3}\right)\right]_0^{2\pi}$$
$$= \frac{1}{\pi}\left\{-\frac{(2\pi-4\pi^2)cos\,2n\pi}{n}-\frac{2cos\,2n\pi}{n^3}+\frac{2cos\,0}{n^3}\right\}$$
$$\therefore b_n = \frac{4\pi-2}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$
$$x - x^2 = \pi - \frac{4\pi^2}{3} + \sum -\frac{4}{n^2}cosnx + \sum\frac{4\pi-2}{n}sinnx$$

29. Find fourier expansion of $f(x) = 2x - x^2, 0 \leq x \leq 3$ whose period is 3
[M15/Biot/6M][M16/Biot/6M][M19/Chem/6M]
Solution:
We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$a_0 = \frac{1}{3}\int_0^{3} 2x - x^2\ dx$$
$$= \frac{1}{3}\left[2\frac{x^2}{2}-\frac{x^3}{3}\right]_0^3$$
$$= \frac{1}{3}[9-9]$$
$$\therefore a_0 = 0$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}\,dx$$
$$a_n = \frac{1}{3}\int_0^{3}(2x-x^2)\cos\frac{2n\pi x}{3}\ \ dx$$
$$= \frac{1}{3}\left[(2x-x^2)\left(\frac{\sin\frac{2n\pi x}{3}}{\frac{2n\pi}{3}}\right)-(2-2x)\left(-\frac{\cos\frac{2n\pi x}{3}}{\frac{4n^2\pi^2}{9}}\right)+(-2)\left(-\frac{\sin\frac{2n\pi x}{3}}{\frac{8n^3\pi^3}{27}}\right)\right]_0^3$$
$$= \frac{1}{3}\left[(2-2x)\left(\frac{\cos\frac{2n\pi x}{3}}{\frac{4n^2\pi^2}{9}}\right)\right]_0^3$$
$$= \frac{3}{4n^2\pi^2}[-4-2]$$
$$\therefore a_n = -\frac{9}{2n^2\pi^2}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{1}{3}\int_0^{3}(2x-x^2)\sin\frac{2n\pi x}{3}\ \ dx$$
$$= \frac{1}{3}\left[(2x-x^2)\left(-\frac{\cos\frac{2n\pi x}{3}}{\frac{2n\pi}{3}}\right)-(2-2x)\left(-\frac{\sin\frac{2n\pi x}{3}}{\frac{4n^2\pi^2}{9}}\right)+(-2)\left(\frac{\cos\frac{2n\pi x}{3}}{\frac{8n^3\pi^3}{27}}\right)\right]_0^3$$

$$= \frac{1}{3}\left[(2x - x^2)\left(-\frac{\cos\frac{2n\pi x}{3}}{\frac{2n\pi}{3}}\right) + (-2)\left(\frac{\cos\frac{2n\pi x}{3}}{\frac{8n^3\pi^3}{27}}\right)\right]_0^3$$

$$= \frac{1}{3}\left[3\frac{1}{\frac{2n\pi}{3}} - 2\frac{1}{\frac{8n^3\pi^3}{27}} + 2\frac{1}{\frac{8n^3\pi^3}{27}}\right]$$

$$\therefore b_n = \frac{3}{2n\pi}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$2x - x^2 = \sum \frac{-9}{2n^2\pi^2}\cos\frac{2n\pi}{3} + \sum\frac{3}{2n\pi}\sin\frac{2n\pi}{3}$$

30. Find the Fourier series for $f(x) = x^2 + 2x$ in $(0,2\pi)$

[M19/Inst/6M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi}(x^2 + 2x)dx$$

$$= \frac{1}{2\pi}\left[\frac{x^3}{3} + \frac{2x^2}{2}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[\frac{8\pi^3}{3} + 4\pi^2\right]$$

$$\therefore a_0 = \frac{4\pi^2}{3} + 2\pi$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi}(x^2 + 2x)\cos nx \ dx$$

$$= \frac{1}{\pi}\left[(x^2 + 2x)\left(\frac{\sin nx}{n}\right) - (2x + 2)\left(-\frac{\cos nx}{n^2}\right) + (2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi n^2}[(2x + 2)cosnx]_0^{2\pi}$$

$$= \frac{1}{\pi n^2}[4\pi + 2 - 2]$$

$$\therefore a_n = \frac{4}{n^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi}(x^2 + 2x)\sin nx \ dx$$

$$= \frac{1}{\pi}\left[(x^2 + 2x)\left(-\frac{\cos nx}{n}\right) - (2x + 2)\left(-\frac{\sin nx}{n^2}\right) + (2)\left(\frac{cosnx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left\{-\frac{(4\pi^2+4\pi)cos\,2n\pi}{n} + \frac{2cos\,2n\pi}{n^3} - \frac{2cos\,0}{n^3}\right\}$$

$$\therefore b_n = -\frac{4(\pi+1)}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$x^2 + 2x = \frac{4\pi^2}{3} + 2\pi + \sum \frac{4}{n^2} cosnx + \sum -\frac{4(\pi+1)}{n} sinnx$$

31. Obtain the Fourier expansion of $f(x) = x^2, -l < x < l$

[N18/Biot/5M]

Solution:

We see that $f(x) = x^2$ is an even function

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{l}\int_0^l x^2 dx$$

$$= \frac{1}{l}\left[\frac{x^3}{3}\right]_0^l$$

$$\therefore a_0 = \frac{l^2}{3}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{l}\int_0^l x^2 \cos\frac{n\pi x}{l}\ dx$$

$$= \frac{2}{l}\left[x^2\left(\frac{\sin\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - 2x\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right) + (2)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^3\pi^3}{l^3}}\right)\right]_0^l$$

$$= \frac{2}{l}\left[\frac{2lcosn\pi}{\frac{n^2\pi^2}{l^2}}\right]$$

$$\therefore a_n = \frac{4l^2(-1)^n}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$x^2 = \frac{l^2}{3} + \sum \frac{4l^2(-1)^n}{n^2\pi^2}\cos\frac{n\pi x}{l}$$

32. Find the Fourier series for $f(x) = x^3$ in $(-\pi, \pi)$

[N17/Biom/5M][N18/Inst/5M][M19/Biom/5M][N19/Comp/5M]

Solution:

We see that $f(x) = x^3$ is an odd function

We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{\pi}\int_0^\pi x^3 \sin nx\ \ dx$$

$$= \frac{2}{\pi}\left[x^3\left(-\frac{\cos nx}{n}\right) - (3x^2)\left(-\frac{\sin nx}{n^2}\right) + (6x)\left(\frac{\cos nx}{n^3}\right) - (6)\left(\frac{sinnx}{n^4}\right)\right]_0^\pi$$

$$= \frac{2}{\pi}\left[x^3\left(-\frac{\cos nx}{n}\right) + (6x)\left(\frac{\cos nx}{n^3}\right)\right]_0^\pi$$

$$= \frac{2}{\pi}\left[-\frac{\pi^3(-1)^n}{n} + \frac{6\pi(-1)^n}{n^3}\right]$$
$$= 2(-1)^n\left[\frac{6}{n^3} - \frac{\pi^2}{n}\right]$$
$$\therefore b_n = (-1)^n\left[\frac{12}{n^3} - \frac{2\pi^2}{n}\right]$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$x^3 = \sum(-1)^n\left[\frac{12}{n^3} - \frac{2\pi^2}{n}\right]\sin nx$$

33. Find half range sine series for $f(x) = (x-1)^2 in\ 0 < x < 1$

Hence find $\sum_{n=1}^{\infty}\frac{1}{n^2}$

[M18/Comp/8M]

[Note: The above question seems to be wrong. It must have been asked as

"Find the half range cosine series for $f(x) = (x-1)^2$ in $0 < x < 1$.

Hence find $\sum\frac{1}{n^2}$"]

Solution:

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{1}\int_0^1 (x-1)^2 dx$$
$$= \left[\frac{(x-1)^3}{3}\right]_0^1$$
$$= 0 - \frac{-1}{3}$$
$$\therefore a_0 = \frac{1}{3}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$
$$a_n = \frac{2}{1}\int_0^1 (x-1)^2\cos n\pi x\ \ dx$$
$$= 2\left[(x-1)^2\left(\frac{\sin n\pi x}{n\pi}\right) - 2(x-1)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right) + (2)\left(-\frac{\sin n\pi x}{n^3\pi^3}\right)\right]_0^1$$
$$= 2\left[\frac{2\cos 0}{n^2\pi^2}\right]$$
$$\therefore a_n = \frac{4}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l}$$
$$(x-1)^2 = \frac{1}{3} + \sum\frac{4}{n^2\pi^2}\cos(n\pi x)$$

Putting $x = 0$, we get

$$1 = \frac{1}{3} + \sum\frac{4}{n^2\pi^2}$$
$$1 - \frac{1}{3} = \frac{4}{\pi^2}\sum\frac{1}{n^2}$$

$$\frac{2}{3} = \frac{4}{\pi^2}\sum\frac{1}{n^2}$$
$$\frac{\pi^2}{6} = \sum\frac{1}{n^2}$$

34. Find half range Fourier sine series of $f(x) = x^3$, $-\pi \le x \le \pi$
[M17/ElexExtcElectBiomInst/5M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{2}{\pi}\int_0^\pi x^3 \sin nx\ \ dx$$
$$= \frac{2}{\pi}\left[x^3\left(-\frac{\cos nx}{n}\right) - (3x^2)\left(-\frac{\sin nx}{n^2}\right) + (6x)\left(\frac{\cos nx}{n^3}\right) - (6)\left(\frac{sinnx}{n^4}\right)\right]_0^\pi$$
$$= \frac{2}{\pi}\left[x^3\left(-\frac{\cos nx}{n}\right) + (6x)\left(\frac{\cos nx}{n^3}\right)\right]_0^\pi$$
$$= \frac{2}{\pi}\left[-\frac{\pi^3(-1)^n}{n} + \frac{6\pi(-1)^n}{n^3}\right]$$
$$= 2(-1)^n\left[\frac{6}{n^3} - \frac{\pi^2}{n}\right]$$
$$\therefore b_n = (-1)^n\left[\frac{12}{n^3} - \frac{2\pi^2}{n}\right]$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$x^3 = \sum(-1)^n\left[\frac{12}{n^3} - \frac{2\pi^2}{n}\right]\sin nx$$

35. Find half range sine series for $f(x) = x^2$ in $0 \le x \le 3$
[N13/ElexExtcElectBiomInst/6M][N16/ElexExtcElectBiomInst/5M]
[M19/Extc/5M]
Solution:
We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{2}{3}\int_0^3 x^2\sin\frac{n\pi x}{3}\ dx$$
$$= \frac{2}{3}\left[x^2\left(-\frac{\cos\frac{n\pi x}{3}}{\frac{n\pi}{3}}\right) - (2x)\left(-\frac{\sin\frac{n\pi x}{3}}{\frac{n^2\pi^2}{9}}\right) + (2)\left(\frac{\cos\frac{n\pi x}{3}}{\frac{n^3\pi^3}{27}}\right)\right]_0^3$$
$$= \frac{2}{3}\left[x^2\left(-\frac{\cos\frac{n\pi x}{3}}{\frac{n\pi}{3}}\right) + (2)\left(\frac{\cos\frac{n\pi x}{3}}{\frac{n^3\pi^3}{27}}\right)\right]_0^3$$
$$= \frac{2}{3}\left[-\frac{9\cos n\pi}{\frac{n\pi}{3}} + \frac{2\cos n\pi}{\frac{n^3\pi^3}{27}} - \frac{2cos\,0}{\frac{n^3\pi^3}{27}}\right]$$
$$= \frac{2}{3}\left[-\frac{27(-1)^n}{n\pi} + \frac{54(-1)^n}{n^3\pi^3} - \frac{54}{n^3\pi^3}\right]$$

$\therefore b_n = 18\left[-\frac{(-1)^n}{n\pi} + \frac{2[(-1)^n - 1]}{n^3\pi^3}\right]$

Thus, the fourier series is given by

$f(x) = \sum b_n \sin\frac{n\pi x}{l}$

$x^2 = 18\sum\left[-\frac{(-1)^n}{n\pi} + \frac{2[(-1)^n - 1]}{n^3\pi^3}\right]\sin\left(\frac{n\pi x}{3}\right)$

36. Find the Fourier expansion of $f(x) = x + x^2$ when $-\pi \le x \le \pi$ and $f(x + 2\pi) = f(x)$

[N16/AutoMechCivil/6M]

Hence deduce that

(i) $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots ..$

(ii) $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots …$

[M15/Biot/8M]

Solution:

Let $f_1(x) = x$ [an odd function]

$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$

$= \frac{2}{\pi}\int_0^\pi x\sin nx\ dx$

$= \frac{2}{\pi}\left[x\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^\pi$

$= \frac{2}{\pi}\left[-\pi.\frac{\cos n\pi}{n}\right]$

$\therefore b_n = -\frac{2(-1)^n}{n}$

Thus, $f_1(x) = \sum b_n \sin\frac{n\pi x}{l}$

$\therefore x = \sum -\frac{2(-1)^n}{n}\sin nx$ ……………….(1)

Let $f_2(x) = x^2$ [an even function]

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{\pi}\int_0^\pi x^2 dx$

$= \frac{1}{\pi}\left[\frac{x^3}{3}\right]_0^\pi$

$\therefore a_0 = \frac{\pi^2}{3}$

$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$

$= \frac{2}{\pi}\int_0^\pi x^2\cos nx\ dx$

$= \frac{2}{\pi}\left[x^2\left(\frac{\sin nx}{n}\right) - (2x)\left(-\frac{\cos nx}{n^2}\right) + (2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^\pi$

$= \frac{2}{\pi}\left[\frac{2\pi\cos n\pi}{n^2}\right]$

$\therefore a_n = \frac{4(-1)^n}{n^2}$

Thus, $f_2(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$

$\therefore x^2 = \frac{\pi^2}{3} + \sum \frac{4(-1)^n}{n^2}\cos nx$ …………………….(2)

Adding (1) & (2), we get

$x + x^2 = \frac{\pi^2}{3} + \sum \frac{4(-1)^n}{n^2}\cos nx + \sum \frac{2(-1)^{n+1}}{n}\sin nx$

Putting $x = \pi$ in eqn (2)

$\pi^2 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2} cosn\pi$

$\pi^2 - \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}(-1)^n$

$\frac{2\pi^2}{3} = \sum_1^\infty \frac{4}{n^2}$

$\frac{\pi^2}{6} = \sum_1^\infty \frac{1}{n^2}$ ………………… (3)

Putting $x = 0$ in eqn (2)

$0 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2} cos0$

$0 = \frac{\pi^2}{3} + \sum_1^\infty \frac{4(-1)^n}{n^2}$

$-\frac{\pi^2}{3} = 4\sum_1^\infty \frac{(-1)^n}{n^2}$

$\frac{\pi^2}{12} = \sum_1^\infty \frac{(-1)^{n+1}}{n^2}$ …………….(4)

Adding (3) & (4)

$\frac{\pi^2}{6} + \frac{\pi^2}{12} = \sum_1^\infty \frac{1+(-1)^{n+1}}{n^2}$

$\frac{\pi^2}{4} = \left[\frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \cdots \ldots .\right]$

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots$

37. Find the Fourier expansion for $f(x) = x - x^2, -1 < x < 1$
[M14/AutoMechCivil/6M][N18/Comp/8M]

Solution:

Let $f_1(x) = x$ [an odd function]

$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$

$= \frac{2}{1}\int_0^1 x\sin n\pi x \ dx$

$= 2\left[x\left(-\frac{\cos n\pi x}{n\pi}\right) - (1)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_0^1$

$= 2\left[-\frac{\cos n\pi}{n\pi}\right]$

$\therefore b_n = -\frac{2(-1)^n}{n\pi}$

Thus, $f_1(x) = \sum b_n \sin\frac{n\pi x}{l}$

$\therefore x = \sum -\frac{2(-1)^n}{n\pi} \sin n\pi x$ ……………….(1)

Let $f_2(x) = x^2$ [an even function]

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{1}\int_0^1 x^2 dx$

$= \left[\frac{x^3}{3}\right]_0^1$

$\therefore a_0 = \frac{1}{3}$

$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$

$= \frac{2}{1}\int_0^1 x^2 \cos n\pi x \ dx$

$= 2\left[x^2\left(\frac{\sin n\pi x}{n\pi}\right) - (2x)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right) + (2)\left(-\frac{\sin n\pi x}{n^3\pi^3}\right)\right]_0^1$

$= 2\left[\frac{2\cos n\pi}{n^2\pi^2}\right]$

$\therefore a_n = \frac{4(-1)^n}{n^2\pi^2}$

Thus, $f_2(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$

$\therefore x^2 = \frac{1}{3} + \sum\frac{4(-1)^n}{n^2\pi^2}\cos n\pi x$ ……………………..(2)

Subtracting (1) & (2), we get

$x - x^2 = -\frac{1}{3} - \frac{4}{\pi^2}\sum\frac{(-1)^n}{n^2}cos n\pi x + \frac{2}{\pi}\sum\frac{(-1)^{n+1}}{n}sin n\pi x$

38. Obtain the expansion of $f(x) = x(\pi - x), 0 < x < \pi$ as a half range cosine series.

[M15/Biot/5M][N19/Extc/5M]

Hence, show that (i) $\sum_1^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$ (ii) $\sum_1^\infty \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$ (iii) $\sum_1^\infty \frac{1}{n^4} = \frac{\pi^4}{90}$

[M14/CompIT/8M][M14/Chem/6M][M19/AutoMechCivil/8M][M19/Comp/8M]

Solution:

We see that,

$a_0 = \frac{1}{l}\int_0^l f(x)dx$

$= \frac{1}{\pi}\int_0^\pi (\pi x - x^2)dx$

$= \frac{1}{\pi}\left[\frac{\pi x^2}{2} - \frac{x^3}{3}\right]_0^\pi$

$= \frac{1}{\pi}\left[\frac{\pi^3}{6}\right]$

$\therefore a_0 = \frac{\pi^2}{6}$

$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$

$$a_n = \frac{2}{\pi}\int_0^{\pi}(\pi x - x^2)\, cosnx\, dx$$
$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{sinnx}{n}\right) - (\pi - 2x)\left(-\frac{cosnx}{n^2}\right) + (-2)\left(-\frac{sinnx}{n^3}\right)\right]_0^{\pi}$$
$$= \frac{2}{\pi n^2}\left[(\pi - 2x)\, cosnx\right]_0^{\pi}$$
$$= \frac{2}{\pi n^2}\left[-\pi(-1)^n - \pi\right]$$
$$\therefore a_n = \frac{-2[(-1)^n + 1]}{n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$x(\pi - x) = \frac{\pi^2}{6} - 2\sum \frac{[1 + (-1)^n]}{n^2} cosnx$$
$$x(\pi - x) = \frac{\pi^2}{6} - 2\left[\frac{2cos\,2x}{2^2} + \frac{2cos\,4x}{4^2} + \frac{2cos\,6x}{6^2} + \cdots..\right]$$
$$x(\pi - x) = \frac{\pi^2}{6} - \left[\frac{cos\,2x}{1^2} + \frac{cos\,4x}{2^2} + \frac{cos\,6x}{3^2} + \cdots..\right]$$

Putting $x = 0$, we get

$$0 = \frac{\pi^2}{6} - \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots..\right]$$
$$\frac{\pi^2}{6} = \sum_1^{\infty} \frac{1}{n^2}$$

Putting $x = \frac{\pi}{2}$, we get

$$\frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right) = \frac{\pi^2}{6} - \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \cdots..\right]$$
$$\frac{\pi^2}{4} = \frac{\pi^2}{6} + \left[\frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} - \cdots..\right]$$
$$\frac{\pi^2}{4} - \frac{\pi^2}{6} = \sum_1^{\infty} \frac{(-1)^{n+1}}{n^2}$$
$$\frac{\pi^2}{12} = \sum_1^{\infty} \frac{(-1)^{n+1}}{n^2}$$

By parseval's identity,

$$\frac{1}{l}\int_0^{l}[f(x)]^2\, dx = a_0^2 + \frac{1}{2}\sum_1^{\infty} a_n^2$$
$$\frac{1}{\pi}\int_0^{\pi}[\pi x - x^2]^2 dx = \frac{\pi^4}{36} + \frac{1}{2}\sum_1^{\infty} \frac{4[(-1)^n + 1]^2}{n^4}$$
$$\frac{1}{\pi}\int_0^{\pi}(\pi^2 x^2 - 2\pi x^3 + x^4)dx = \frac{\pi^4}{36} + 2\left[\frac{4}{2^4} + \frac{4}{4^4} + \frac{4}{6^4} + \cdots.\right]$$
$$\frac{1}{\pi}\left[\frac{\pi^2 x^3}{3} - \frac{2\pi x^4}{4} + \frac{x^5}{5}\right]_0^{\pi} = \frac{\pi^4}{36} + \frac{8}{2^4}\left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots\ldots\right]$$
$$\frac{1}{\pi}\left[\frac{\pi^5}{3} - \frac{\pi^5}{2} + \frac{\pi^5}{5}\right] = \frac{\pi^4}{36} + \frac{1}{2}\sum_1^{\infty} \frac{1}{n^4}$$
$$\frac{1}{\pi}\left[\frac{\pi^5}{30}\right] = \frac{\pi^4}{36} + \frac{1}{2}\sum_1^{\infty} \frac{1}{n^4}$$
$$\frac{\pi^4}{30} - \frac{\pi^4}{36} = \frac{1}{2}\sum_1^{\infty} \frac{1}{n^4}$$
$$\frac{\pi^4}{180} = \frac{1}{2}\sum_1^{\infty} \frac{1}{n^4}$$
$$\frac{\pi^4}{90} = \sum_1^{\infty} \frac{1}{n^4}$$

39. Obtain half range sine series of $f(x) = x(\pi - x)$ in $(0, \pi)$. Hence evaluate $\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^3}$

[M17/AutoMechCivil/8M][N17/MTRX/8M]

Solution:

We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\ sinnx\ dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{-cosnx}{n}\right) - (\pi - 2x)\left(\frac{-sinnx}{n^2}\right) + (-2)\left(\frac{cosnx}{n^3}\right)\right]_0^\pi$$

$$= \frac{-4}{\pi n^3}[cosnx]_0^\pi$$

$$= \frac{-4}{\pi n^3}[(-1)^n - 1]$$

$$\therefore b_n = \frac{-4[(-1)^n - 1]}{\pi n^3}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n sinnx$$

$$x(\pi - x) = \sum \frac{-4[(-1)^n - 1]}{\pi n^3} sinnx$$

$$x(\pi - x) = -\frac{4}{\pi}\left[\frac{-2sinx}{1^3} + \frac{-2\sin 3x}{3^3} + \frac{-2\sin 5x}{5^3} + \cdots..\right]$$

$$x(\pi - x) = \frac{8}{\pi}\left[\frac{sinx}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \cdots..\right]$$

Putting $x = \frac{\pi}{2}$, we get

$$\frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right) = \frac{8}{\pi}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$

$$\frac{\pi^2}{4} = \frac{8}{\pi}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$

$$\frac{\pi^3}{32} = \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$

$$\frac{\pi^3}{32} = \sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+1)^3}$$

40. Obtain half range sine series in $(0, \pi)$ for $x(\pi - x)$ and hence, find the value of $\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots \ldots.. = \frac{\pi^6}{960}$

[N17/Biom/8M]

Solution:

We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\ sinnx\ dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{-cosnx}{n}\right) - (\pi - 2x)\left(\frac{-sinnx}{n^2}\right) + (-2)\left(\frac{cosnx}{n^3}\right)\right]_0^\pi$$

$$= \frac{-4}{\pi n^3}[cosnx]_0^\pi$$

$$= \frac{-4}{\pi n^3}[(-1)^n - 1]$$

$$\therefore b_n = \frac{-4[(-1)^n - 1]}{\pi n^3}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n sinnx$$

$$x(\pi - x) = \sum \frac{-4[(-1)^n - 1]}{\pi n^3} sinnx$$

By parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2\, dx = \frac{1}{2}\sum_1^\infty b_n^2$$

$$\frac{1}{\pi}\int_0^\pi [\pi x - x^2]^2 dx = \frac{1}{2}\sum_1^\infty \frac{16[(-1)^n - 1]^2}{n^6\pi^2}$$

$$\frac{1}{\pi}\int_0^\pi (\pi^2x^2 - 2\pi x^3 + x^4)dx = \frac{8}{\pi^2}\left[\frac{4}{1^6} + \frac{4}{3^6} + \frac{4}{5^6} + \cdots.\right]$$

$$\frac{1}{\pi}\left[\frac{\pi^2x^3}{3} - \frac{2\pi x^4}{4} + \frac{x^5}{5}\right]_0^\pi = \frac{32}{\pi^2}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots\right]$$

$$\frac{1}{\pi}\left[\frac{\pi^5}{3} - \frac{\pi^5}{2} + \frac{\pi^5}{5}\right] = \frac{32}{\pi^2}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots\right]$$

$$\frac{1}{\pi}\left[\frac{\pi^5}{30}\right] = \frac{32}{\pi^2}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots\right]$$

$$\frac{\pi^4}{30} = \frac{32}{\pi^2}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots\right]$$

$$\frac{\pi^6}{960} = \left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots \ldots\right]$$

41. Obtain half range sine series in $(0, \pi)$ for $x(\pi - x)$ and hence, find the value of $\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \cdots \ldots \ldots . . = \frac{\pi^6}{945}$

42. Prove that $x(\pi - x) = \frac{\pi^2}{6} - \left[\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \cdots\right]$

[M18/Chem/6M]

Solution:

We see that,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{\pi}\int_0^\pi (\pi x - x^2)dx$$

$$= \frac{1}{\pi}\left[\frac{\pi x^2}{2} - \frac{x^3}{3}\right]_0^\pi$$

$$= \frac{1}{\pi}\left[\frac{\pi^3}{6}\right]$$

$$\therefore a_0 = \frac{\pi^2}{6}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\, cosnx\, dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{sinnx}{n}\right) - (\pi - 2x)\left(-\frac{cosnx}{n^2}\right) + (-2)\left(-\frac{sinnx}{n^3}\right)\right]_0^\pi$$

$$= \frac{2}{\pi n^2}[(\pi - 2x)\, cosnx]_0^{\pi}$$
$$= \frac{2}{\pi n^2}[-\pi(-1)^n - \pi]$$
$$\therefore a_n = \frac{-2[(-1)^n + 1]}{n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$x(\pi - x) = \frac{\pi^2}{6} - 2\sum \frac{[1+(-1)^n]}{n^2} cosnx$$
$$x(\pi - x) = \frac{\pi^2}{6} - 2\left[\frac{2cos\,2x}{2^2} + \frac{2cos\,4x}{4^2} + \frac{2cos\,6x}{6^2} + \cdots..\right]$$
$$x(\pi - x) = \frac{\pi^2}{6} - \left[\frac{cos\,2x}{1^2} + \frac{cos\,4x}{2^2} + \frac{cos\,6x}{3^2} + \cdots..\right]$$

43. Expand $f(x) = lx - x^2, 0 < x < l$ in half range Sine series
[N18/Elect/5M][N18/Chem/6M]
Solution:
We have, $f(x) = lx - x^2$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l} dx$$
$$= \frac{2}{l}\int_0^l (lx - x^2)\sin\frac{n\pi x}{l}\, dx$$
$$= \frac{2}{l}\left[(lx - x^2)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (l - 2x)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right) + (-2)\left(\frac{\cos\frac{n\pi x}{l}}{\frac{n^3\pi^3}{l^3}}\right)\right]_0^l$$
$$= \frac{2}{l}\left[-2\frac{\cos n\pi}{\frac{n^3\pi^3}{l^3}} + \frac{2\cos 0}{\frac{n^3\pi^3}{l^3}}\right]$$
$$\therefore b_n = -\frac{4l^2[(-1)^n - 1]}{n^3\pi^3}$$

Thus, $f(x) = \sum b_n \sin\frac{n\pi x}{l}$

$$lx - x^2 = \sum -\frac{4l^2[(-1)^n - 1]}{n^3\pi^3}\sin\frac{n\pi x}{l}$$

44. Find half range sine series for $f(x) = lx - x^2$in $(0, l)$. Hence prove that $\frac{1}{1^6} + \frac{1}{3^6} + \cdots ..... = \frac{\pi^6}{960}$
[N14/CompIT/8M][N18/Biot/8M]
Solution:
We have, $f(x) = lx - x^2$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l} dx$$
$$= \frac{2}{l}\int_0^l (lx - x^2)\sin\frac{n\pi x}{l}\, dx$$
$$= \frac{2}{l}\left[(lx - x^2)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (l - 2x)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right) + (-2)\left(\frac{\cos\frac{n\pi x}{l}}{\frac{n^3\pi^3}{l^3}}\right)\right]_0^l$$

$$= \frac{2}{l}\left[-2\frac{\cos n\pi}{\frac{n^3\pi^3}{l^3}} + \frac{2\cos 0}{\frac{n^3\pi^3}{l^3}}\right]$$

$$\therefore b_n = -\frac{4l^2[(-1)^n - 1]}{n^3\pi^3}$$

Thus, $f(x) = \sum b_n \sin\frac{n\pi x}{l}$

$$lx - x^2 = \sum -\frac{4l^2[(-1)^n - 1]}{n^3\pi^3}\sin\frac{n\pi x}{l}$$

By parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2\,dx = \frac{1}{2}\sum_1^\infty b_n^2$$

$$\frac{1}{l}\int_0^l [lx - x^2]^2 dx = \frac{1}{2}\sum_1^\infty \frac{16l^4[(-1)^n - 1]^2}{n^6\pi^6}$$

$$\frac{1}{l}\int_0^l (l^2x^2 - 2lx^3 + x^4)dx = \frac{8l^4}{\pi^6}\left[\frac{4}{1^6} + \frac{4}{3^6} + \frac{4}{5^6} + \cdots.\right]$$

$$\frac{1}{l}\left[\frac{l^2x^3}{3} - \frac{2lx^4}{4} + \frac{x^5}{5}\right]_0^l = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\ldots\right]$$

$$\frac{1}{l}\left[\frac{l^5}{3} - \frac{l^5}{2} + \frac{l^5}{5}\right] = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\ldots\right]$$

$$\frac{1}{l}\left[\frac{l^5}{30}\right] = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\ldots\right]$$

$$\frac{l^4}{30} = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\ldots\right]$$

$$\frac{\pi^6}{960} = \left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\ldots\right]$$

45. Find half range sine series in $(0, \pi)$ for $x(\pi - x)$

[N15/AutoMechCivil/5M][M16/Biot/5M][N18/Elex/5M][N18/Extc/5M]

Solution:

We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\ sinnx\ dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{-cosnx}{n}\right) - (\pi - 2x)\left(\frac{-sinnx}{n^2}\right) + (-2)\left(\frac{cosnx}{n^3}\right)\right]_0^\pi$$

$$= \frac{-4}{\pi n^3}[cosnx]_0^\pi$$

$$= \frac{-4}{\pi n^3}[(-1)^n - 1]$$

$$\therefore b_n = \frac{-4[(-1)^n - 1]}{\pi n^3}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n sinnx$$

$$x(\pi - x) = \sum \frac{-4[(-1)^n - 1]}{\pi n^3} sinnx$$

46. Obtain sine series for $f(x) = x(\pi - x), 0 < x < \pi$ and hence find $\sum \frac{(-1)^n}{(2n-1)^3}$
[M14/ElexExtcElectBiomInst/6M][M19/Elex/6M]
Solution:
We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\ sinnx\ dx$$
$$= \frac{2}{\pi}\left[(\pi x - x^2)\left(\frac{-cosnx}{n}\right) - (\pi - 2x)\left(\frac{-sinnx}{n^2}\right) + (-2)\left(\frac{cosnx}{n^3}\right)\right]_0^\pi$$
$$= \frac{-4}{\pi n^3}[cosnx]_0^\pi$$
$$= \frac{-4}{\pi n^3}[(-1)^n - 1]$$
$$\therefore b_n = \frac{-4[(-1)^n - 1]}{\pi n^3}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n sinnx$$
$$x(\pi - x) = \sum \frac{-4[(-1)^n - 1]}{\pi n^3} sinnx$$
$$x(\pi - x) = -\frac{4}{\pi}\left[\frac{-2sinx}{1^3} + \frac{-2\sin 3x}{3^3} + \frac{-2\sin 5x}{5^3} + \cdots..\right]$$
$$x(\pi - x) = \frac{8}{\pi}\left[\frac{sinx}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \cdots..\right]$$

Putting $x = \frac{\pi}{2}$, we get

$$\frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right) = \frac{8}{\pi}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$
$$\frac{\pi^2}{4} = \frac{8}{\pi}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$
$$\frac{\pi^3}{32} = \left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots..\right]$$

47. Obtain the half range cosine series for $f(x) = lx - x^2$ in $(0, l)$
[N15/AutoMechCivil/6M][N17/Inst/6M]
Solution:
We see that,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{l}\int_0^l (lx - x^2)dx$$
$$= \frac{1}{l}\left[\frac{lx^2}{2} - \frac{x^3}{3}\right]_0^l$$
$$= \frac{1}{l}\left[\frac{l^3}{6}\right]$$
$$\therefore a_0 = \frac{l^2}{6}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$
$$a_n = \frac{2}{l}\int_0^l (lx - x^2)\cos\frac{n\pi x}{l}\ dx$$
$$= \frac{2}{l}\left[(lx - x^2)\left(\frac{\sin\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (l - 2x)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right) + (-2)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^3\pi^3}{l^3}}\right)\right]_0^l$$
$$= \frac{2}{l}\left[-\frac{l\cos n\pi}{\frac{n^2\pi^2}{l^2}} - \frac{l\cos 0}{\frac{n^2\pi^2}{l^2}}\right]$$
$$= \frac{-2l^2}{\pi^2 n^2}[(-1)^n + 1]$$
$$\therefore a_n = \frac{-2l^2[(-1)^n + 1]}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$lx - x^2 = \frac{l^2}{6} - \frac{2l^2}{\pi^2}\sum\frac{[1 + (-1)^n]}{n^2}\cos\left(\frac{n\pi x}{l}\right)$$

48. Obtain half range sine series for $f(x) = x(2 - x)$ in $(0,2)$
[M19/Chem/6M]
Solution:
We see that,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$
$$b_n = \frac{2}{2}\int_0^2 (2x - x^2)\sin\frac{n\pi x}{2}\ dx$$
$$= \left[(2x - x^2)\left(\frac{-\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (2 - 2x)\left(\frac{-\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right) + (-2)\left(\frac{\cos\frac{n\pi x}{2}}{\frac{n^3\pi^3}{8}}\right)\right]_0^2$$
$$= -\frac{2\cos n\pi}{\frac{n^3\pi^3}{8}} + \frac{2cos\,0}{\frac{n^3\pi^3}{8}}$$
$$= \frac{8}{\pi^3 n^3}[-2(-1)^n + 2]$$
$$\therefore b_n = \frac{-16[(-1)^n - 1]}{n^3\pi^3}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$2x - x^2 = \sum\frac{-16[(-1)^n - 1]}{n^3\pi^3}\sin\left(\frac{n\pi x}{2}\right)$$

49. Find half range Fourier cosine series of the function $f(x) = 2x - x^2$, $x \in (0, 2)$
[N14/ElexExtcElectBiomInst/6M]
Solution:
We see that,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{2}\int_0^2 (2x - x^2)dx$$
$$= \frac{1}{2}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^2$$
$$\therefore a_0 = \frac{2}{3}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{2}\int_0^2 (2x - x^2)\cos\frac{n\pi x}{2}\ dx$$
$$= \left[(2x - x^2)\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (2 - 2x)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right) + (-2)\left(-\frac{\sin\frac{n\pi x}{2}}{\frac{n^3\pi^3}{8}}\right)\right]_0^2$$
$$= -\frac{2\cos n\pi}{\frac{n^2\pi^2}{4}} - \frac{2cos\, 0}{\frac{n^2\pi^2}{4}}$$
$$= \frac{4}{\pi^2 n^2}[-2(-1)^n - 2]$$
$$\therefore a_n = \frac{-8[(-1)^n + 1]}{n^2\pi^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$2x - x^2 = \frac{2}{3} - \frac{8}{\pi^2}\sum\frac{[1 + (-1)^n]}{n^2}\cos\left(\frac{n\pi x}{2}\right)$$

50. Find Fourier series for $f(x) = \begin{cases} x & 0 \le x \le \pi \\ 2\pi - x & \pi \le x \le 2\pi \end{cases}$

hence deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots$.

[M14/Biot/8M][N16/CompIT/8M][M18/MTRX/8M][M19/Elex/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2\pi}\left\{\int_0^\pi x dx + \int_\pi^{2\pi}(2\pi - x)dx\right\}$$
$$= \frac{1}{2\pi}\left\{\left[\frac{x^2}{2}\right]_0^\pi + \left[2\pi x - \frac{x^2}{2}\right]_\pi^{2\pi}\right\}$$
$$= \frac{1}{2\pi}\left\{\left[\frac{\pi^2}{2}\right] + \left[4\pi^2 - \frac{4\pi^2}{2} - 2\pi^2 + \frac{\pi^2}{2}\right]\right\}$$
$$= \frac{1}{2\pi}\left\{\frac{\pi^2}{2} + \frac{\pi^2}{2}\right\}$$
$$\therefore a_0 = \frac{\pi}{2}$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\left\{\int_0^\pi x\cos nx\ dx + \int_\pi^{2\pi}(2\pi - x)\cos nx\, dx\right\}$$

$= \frac{1}{\pi}\left\{\left[x\left(\frac{\sin nx}{n}\right)-(1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi}+\left[(2\pi-x)\left(\frac{\sin nx}{n}\right)-(-1)\left(-\frac{\cos nx}{n^2}\right)\right]_{\pi}^{2\pi}\right\}$

$= \frac{1}{\pi}\left\{\left[\frac{cosnx}{n^2}\right]_0^{\pi}-\left[\frac{cosnx}{n^2}\right]_{\pi}^{2\pi}\right\}$

$= \frac{1}{\pi n^2}\{[(-1)^n-1]-[1-(-1)^n]\}$

$\therefore a_n = \frac{2[(-1)^n-1]}{n^2\pi}$

$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$

$b_n = \frac{1}{\pi}\left\{\int_0^{\pi} x\sin nx\ dx+\int_{\pi}^{2\pi}(2\pi-x)\sin nx\, dx\right\}$

$= \frac{1}{\pi}\left\{\left[x\left(-\frac{\cos nx}{n}\right)-(1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{\pi}+\left[(2\pi-x)\left(-\frac{\cos nx}{n}\right)-(-1)\left(-\frac{\sin nx}{n^2}\right)\right]_{\pi}^{2\pi}\right\}$

$= \frac{1}{\pi}\left\{\left[-\frac{xcosnx}{n}\right]_0^{\pi}-\left[\frac{(2\pi-x)cosnx}{n}\right]_{\pi}^{2\pi}\right\}$

$= \frac{1}{\pi n}\{[-\pi(-1)^n]-[-\pi(-1)^n]\}$

$\therefore b_n = 0$

Thus, the fourier series is given by

$f(x) = a_0+\sum a_n\cos\frac{n\pi x}{l}+\sum b_n\sin\frac{n\pi x}{l}$

$f(x) = \frac{\pi}{2}+\sum_1^{\infty}\frac{2[(-1)^n-1]}{n^2\pi}\cos nx$

By Parseval's identity,

$\frac{1}{2l}\int_0^{2l}[f(x)]^2dx = a_0^2+\frac{1}{2}\sum_1^{\infty}(a_n^2+b_n^2)$

$\frac{1}{2\pi}\left\{\int_0^{\pi}x^2dx+\int_{\pi}^{2\pi}(2\pi-x)^2dx\right\} = \frac{\pi^2}{4}+\frac{1}{2}\sum_1^{\infty}\frac{4[(-1)^n-1]^2}{n^4\pi^2}$

$\frac{1}{2\pi}\left\{\left[\frac{x^3}{3}\right]_0^{\pi}+\left[\frac{(2\pi-x)^3}{-3}\right]_{\pi}^{2\pi}\right\} = \frac{\pi^2}{4}+\frac{4}{2\pi^2}\left[\frac{4}{1^4}+\frac{4}{3^4}+\frac{4}{5^4}+\cdots\ldots\right]$

$\frac{1}{6\pi}\{\pi^3+\pi^3\} = \frac{\pi^2}{4}+\frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots.\right]$

$\frac{\pi^2}{3} = \frac{\pi^2}{4}+\frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots.\right]$

$\frac{\pi^2}{3}-\frac{\pi^2}{4} = \frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots.\right]$

$\frac{\pi^2}{12} = \frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots.\right]$

$\frac{\pi^4}{96} = \left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots.\right]$

51. Obtain Fourier series for $f(x) = \begin{cases} x+\frac{\pi}{2} & -\pi < x < 0 \\ \frac{\pi}{2}-x & 0 < x < \pi \end{cases}$

Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\ldots$

[M16/Biot/6M][N16/ElexExtcElectBiomInst/8M][N17/MTRX/6M]

Also deduce that $\frac{\pi^4}{96} = \frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots.$

[M14/Biot/8M][N14/Biot/8M][N15/CompIT/8M][N18/Extc/8M]
[N18/Biot/8M][M19/Elect/8M][N19/Extc/8M]

Solution:

We have,

$$f(x)=\begin{cases} x+\frac{\pi}{2} & -\pi<x<0 \\ \frac{\pi}{2}-x & 0<x<\pi \end{cases}$$

Putting $x=-x$, we get

$$f(-x)=\begin{cases} \frac{\pi}{2}-(-x) & -\pi<x<0 \\ -x+\frac{\pi}{2} & 0<x<\pi \end{cases}$$

$$f(-x)=\begin{cases} x+\frac{\pi}{2} & -\pi<x<0 \\ \frac{\pi}{2}-x & 0<x<\pi \end{cases}$$

$\therefore f(-x)=f(x)$ i.e. $f(x)$ is an even function.

Now,

$$a_0=\frac{1}{l}\int_0^l f(x)dx$$
$$=\frac{1}{\pi}\int_0^{\pi}\left(\frac{\pi}{2}-x\right)dx$$
$$=\frac{1}{\pi}\left[\frac{\pi}{2}x-\frac{x^2}{2}\right]_0^{\pi}$$

$\therefore a_0=0$

$$a_n=\frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n=\frac{2}{\pi}\int_0^{\pi}\left(\frac{\pi}{2}-x\right)\cos nx\ dx$$
$$=\frac{2}{\pi}\left[\left(\frac{\pi}{2}-x\right)\left(\frac{\sin nx}{n}\right)-(-1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi}$$
$$=\frac{-2}{\pi n^2}[cosnx]_0^{\pi}$$
$$=\frac{-2}{\pi n^2}[(-1)^n-1]$$

$$\therefore a_n=\frac{-2[(-1)^n-1]}{\pi n^2}$$

Thus, the fourier series is given by

$$f(x)=a_0+\sum a_n\cos\frac{n\pi x}{l}$$
$$\frac{\pi}{2}-x=\sum_1^{\infty}\frac{-2[(-1)^n-1]}{n^2\pi}\cos nx$$

Putting $x=0$,

$$\frac{\pi}{2}=-\frac{2}{\pi}\left[-\frac{2}{1^2}-\frac{2}{3^2}-\frac{2}{5^2}-\cdots.\right]$$
$$\frac{\pi}{2}=\frac{4}{\pi}\left[\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\dots.\right]$$
$$\frac{\pi^2}{8}=\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots\dots\dots$$

By parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2\,dx = a_0^2 + \frac{1}{2}\sum_1^\infty a_n^2$$
$$\frac{1}{\pi}\int_0^\pi \left[\frac{\pi}{2}-x\right]^2 dx = \frac{1}{2}\sum_1^\infty \frac{4[(-1)^n-1]^2}{n^4\pi^2}$$
$$\frac{1}{\pi}\left[\frac{\left(\frac{\pi}{2}-x\right)^3}{-3}\right]_0^\pi = \frac{4}{2\pi^2}\left[\frac{4}{1^4}+\frac{4}{3^4}+\frac{4}{5^4}+\cdots\ldots\right]$$
$$-\frac{1}{3\pi}\left[\left(-\frac{\pi}{2}\right)^3-\left(\frac{\pi}{2}\right)^3\right] = \frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$-\frac{1}{3\pi}\left[-\frac{\pi^3}{4}\right] = \frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{\pi^2}{12} = \frac{8}{\pi^2}\left[\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots\right]$$
$$\frac{\pi^4}{96} = \frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\ldots$$

52. Find the Fourier expansion of $f(x) = \begin{cases} 0 & -2<x<-1 \\ 1+x & -1<x<0 \\ 1-x & 0<x<1 \\ 0 & 1<x<2 \end{cases}$

[N16/Chem/6M]

Solution:

$$f(x) = \begin{cases} 0 & -2<x<-1 \\ 1+x & -1<x<0 \\ 1-x & 0<x<1 \\ 0 & 1<x<2 \end{cases}$$

Putting $x=-x$, we get

$$f(-x) = \begin{cases} 0 & -2<x<-1 \\ 1-(-x) & -1<x<0 \\ 1+(-x) & 0<x<1 \\ 0 & 1<x<2 \end{cases}$$
$$f(-x) = \begin{cases} 0 & -2<x<-1 \\ 1+x & -1<x<0 \\ 1-x & 0<x<1 \\ 0 & 1<x<2 \end{cases}$$

$\therefore f(-x) = f(x)$ i.e. $f(x)$ is an even function.

Now,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{2}\int_0^1 (1-x)dx$$
$$= \frac{1}{2}\left[x-\frac{x^2}{2}\right]_0^1$$
$$\therefore a_0 = \frac{1}{4}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{2}\int_0^1 (1-x)\cos\frac{n\pi x}{2}\,dx$$

$$= \left[(1-x)\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right)-(-1)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^1$$

$$= \frac{-4}{\pi^2 n^2}\left[\cos\frac{n\pi x}{2}\right]_0^1$$

$$= \frac{-4}{\pi^2 n^2}\left[\cos\frac{n\pi}{2}-1\right]$$

$$\therefore a_n = \frac{-4\left[\cos\frac{n\pi}{2}-1\right]}{\pi^2 n^2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$f(x) = \frac{1}{4} + \frac{4}{\pi^2}\sum\frac{\left[1-cosn\frac{\pi}{2}\right]}{n^2}\cos\left(\frac{n\pi x}{2}\right)$$

53. Find the Fourier series for periodic function $f(x) = \begin{cases} -\pi & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$

[M16/Chem/7M][N16/Chem/8M]

state the value of $f(x) at\ x = 0$ and hence deduce that $\sum_1^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

[N17/Elex/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)dx$$

$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0} -\pi dx + \int_0^{\pi} x dx\right\}$$

$$= \frac{1}{2\pi}\left\{[-\pi x]_{-\pi}^{0} + \left[\frac{x^2}{2}\right]_0^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{0-\pi^2+\frac{\pi^2}{2}-0\right\}$$

$$\therefore a_0 = -\frac{\pi}{4}$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -\pi\cos nx\ dx + \int_0^{\pi} x\cos nx\, dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[-\pi\frac{sinnx}{n}\right]_{-\pi}^{0} + \left[x\left(\frac{\sin nx}{n}\right)-(1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi n^2}\{[\cos nx]_0^{\pi}\}$$

$$\therefore a_n = \frac{(-1)^n-1}{\pi n^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} -\pi\sin nx\ dx + \int_0^{\pi} x\sin nx\, dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[-\pi\left(-\frac{cosnx}{n}\right)\right]_{-\pi}^{0} + \left[x\left(-\frac{\cos nx}{n}\right)-(1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi}\left\{\frac{\pi \cos 0}{n} - \frac{\pi cos n\pi}{n} - \frac{\pi \cos n\pi}{n}\right\}$$

$$\therefore b_n = \frac{1-2(-1)^n}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = -\frac{\pi}{4} + \frac{1}{\pi}\sum\frac{[(-1)^n-1]}{n^2} cosnx + \sum\frac{[1-2(-1)^n]}{n} sinnx$$

We see that $x = 0$ is a point of discontinuity, thus value of $f(x)$ at $x = 0$ is

$$f(x) = \frac{[f_1(x)+f_2(x)]}{2} = \frac{[-\pi+x]}{2}$$

$$f(0) = -\frac{\pi}{2}$$

Now, putting $x = 0$ in the fourier series, we get

$$f(0) = -\frac{\pi}{4} + \frac{1}{\pi}\sum\frac{[(-1)^n-1]}{n^2}$$

$$-\frac{\pi}{2} = -\frac{\pi}{4} + \left[-\frac{2}{1^2} - \frac{2}{3^2} - \frac{2}{5^2} - \cdots \dots.\right]$$

$$-\frac{\pi}{2} + \frac{\pi}{4} = -2\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \dots\right]$$

$$-\frac{\pi}{4} = -2\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \dots\right]$$

$$\frac{\pi}{8} = \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \dots\right]$$

54. Find half range sine series for $f(x)$ where $f(x) = \begin{cases} x & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$

[M14/AutoMechCivil/6M][N18/MTRX/8M]

Solution:

We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} x\sin nx\ dx + \int_{\frac{\pi}{2}}^{\pi}(\pi - x)\sin nx\ dx\right\}$$

$$= \frac{2}{\pi}\left\{\left[x\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{\frac{\pi}{2}} + \left[(\pi - x)\left(-\frac{\cos nx}{n}\right) - (-1)\left(-\frac{\sin nx}{n^2}\right)\right]_{\frac{\pi}{2}}^{\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\left[\frac{\pi}{2}.\left(-\frac{\cos\frac{n\pi}{2}}{n}\right) + \frac{\sin\frac{n\pi}{2}}{n^2}\right] + \left[\frac{\pi}{2}.\frac{\cos\frac{n\pi}{2}}{n} + \frac{\sin\frac{n\pi}{2}}{n^2}\right]\right\}$$

$$= \frac{2}{\pi}\left[2\frac{\sin\frac{n\pi}{2}}{n^2}\right]$$

$$\therefore b_n = \frac{4}{n^2\pi}\sin\frac{n\pi}{2}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \sum\frac{4}{n^2\pi}\sin\frac{n\pi}{2}.\sin nx$$

$f(x)=\frac{4}{\pi}\sum\frac{\sin\left(n\frac{\pi}{2}\right)}{n^2}sinnx$

55. Find Fourier series $f(x)=\begin{cases}a(x-l) & -l<x<0\\ a(x+l) & 0<x<l\end{cases}$

[N17/Biom/6M]

Solution:

We have,

$f(x)=\begin{cases}a(x-l) & -l<x<0\\ a(x+l) & 0<x<l\end{cases}$

Putting $x=-x$, we get

$f(-x)=\begin{cases}a(-x-l) & -l<-x<0\\ a(-x+l) & 0<-x<l\end{cases}$

$f(-x)=\begin{cases}-a(x+l) & l>x>0\\ -a(x-l) & 0>x>-l\end{cases}$

$f(-x)=\begin{cases}-a(x-l) & -l<x<0\\ -a(x+l) & 0<x<l\end{cases}$

$\therefore f(-x)=-f(x)$ i.e. $f(x)$ is an odd function.

Now,

$b_n=\frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$

$b_n=\frac{2}{l}\int_0^l a(x+l)\sin\frac{n\pi x}{l}\,dx$

$=\frac{2a}{l}\left[(x+l)\left(\frac{-\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right)-(1)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_0^l$

$=\frac{2a}{l}\left[2l\left(-\frac{\cos n\pi}{\frac{n\pi}{l}}\right)-l\left(-\frac{\cos 0}{\frac{n\pi}{l}}\right)\right]$

$=\frac{2a}{\frac{n\pi}{l}}[-(-1)^n+1]$

$\therefore b_n=\frac{-2al[(-1)^n-1]}{n\pi}$

Thus, the fourier series is given by

$f(x)=\sum b_n\sin\frac{n\pi x}{l}$

$f(x)=\sum_1^\infty\frac{-2al[(-1)^n-1]}{n\pi}\sin\frac{n\pi x}{l}$

56. Find Fourier expansion of $f(x)=\begin{cases}x & -1<x<0\\ x+2 & 0<x<1\end{cases}$

[N19/Comp/8M]

Solution:

We have,

$a_0=\frac{1}{2l}\int_{-l}^l f(x)dx$

$$= \frac{1}{2}\left\{\int_{-1}^{0} x dx + \int_{0}^{1}(x+2)dx\right\}$$
$$= \frac{1}{2}\left\{\left[\frac{x^2}{2}\right]_{-1}^{0} + \left[\frac{x^2}{2} + 2x\right]_{0}^{1}\right\}$$
$$= \frac{1}{2}\left\{0 - \frac{1}{2} + \frac{1}{2} + 2 - 0\right\}$$
$$\therefore a_0 = 1$$
$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{1}\left\{\int_{-1}^{0} x\cos n\pi x \ dx + \int_{0}^{1}(x+2)\cos n\pi x \, dx\right\}$$
$$= \left\{\left[x\left(\frac{\sin n\pi x}{n\pi}\right) - (1)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_{-1}^{0} + \left[(x+2)\left(\frac{\sin n\pi x}{n\pi}\right) - (1)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_{0}^{1}\right\}$$
$$= \left\{\left[\frac{\cos n\pi x}{n^2\pi^2}\right]_{-1}^{0} + \left[\frac{\cos n\pi x}{n^2\pi^2}\right]_{0}^{1}\right\}$$
$$= \frac{1}{n^2\pi^2} - \frac{(-1)^n}{n^2\pi^2} + \frac{(-1)^n}{n^2\pi^2} - \frac{1}{n^2\pi^2}$$
$$\therefore a_n = 0$$
$$b_n = \frac{1}{l}\int_{0}^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{1}\left\{\int_{-1}^{0} x\sin n\pi x \ dx + \int_{0}^{1}(x+2)\sin n\pi x \, dx\right\}$$
$$= \left\{\left[x\left(-\frac{\cos n\pi x}{n\pi}\right) - (1)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_{-1}^{0} + \left[(x+2)\left(-\frac{\cos n\pi x}{n\pi}\right) - (1)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_{0}^{1}\right\}$$
$$= \left\{-\frac{cos n\pi}{n\pi} - \frac{3\cos n\pi}{n\pi} + \frac{2cos\, 0}{n\pi}\right\}$$
$$= \frac{-4(-1)^n + 2}{n\pi}$$
$$\therefore b_n = \frac{2[1-2(-1)^n]}{n\pi}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = 1 + \sum \frac{2[1-2(-1)^n]}{n\pi}\sin\frac{n\pi x}{l}$$

57. Find the Fourier series of $f(x) = \begin{cases} 0 & -\pi \le x \le 0 \\ x & 0 \le x \le \pi \end{cases}$. Hence find the value of $\frac{\pi^2}{8}$

[N18/Inst/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)dx$$
$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0} 0dx + \int_{0}^{\pi} xdx\right\}$$
$$= \frac{1}{2\pi}\left\{\left[\frac{x^2}{2}\right]_{0}^{\pi}\right\}$$
$$\therefore a_0 = \frac{\pi}{4}$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx$$

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cos nx\ \ dx + \int_0^{\pi} x\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[x\left(\frac{\sin nx}{n}\right) - (1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi n^2}\{[\cos nx]_0^{\pi}\}$$

$$\therefore a_n = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\ \ dx + \int_0^{\pi} x\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[x\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi}\left\{-\frac{\pi\cos n\pi}{n}\right\}$$

$$\therefore b_n = -\frac{(-1)^n}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$

$$f(x) = \frac{\pi}{4} + \sum\frac{(-1)^n - 1}{\pi n^2}cosnx + \sum -\frac{(-1)^n}{n}sinnx$$

58. Find fourier series of $f(x) = \begin{cases}\pi + x & 0 < x < \pi\\ \pi - x & -\pi < x < 0\end{cases}$

[N19/Elex/6M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)dx$$

$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0}(\pi + x)dx + \int_0^{\pi}(\pi - x)dx\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\pi x + \frac{x^2}{2}\right]_{-\pi}^{0} + \left[\pi x - \frac{x^2}{2}\right]_0^{\pi}\right\}$$

$$= \frac{1}{2\pi}\left\{0 + \pi^2 - \frac{\pi^2}{2} + \pi^2 - \frac{\pi^2}{2} - 0\right\}$$

$$\therefore a_0 = \frac{\pi}{2}$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx$$

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi + x)\cos nx\ \ dx + \int_0^{\pi}(\pi - x)\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[(\pi + x)\left(\frac{sinnx}{n}\right) - (1)\left(-\frac{cosnx}{n^2}\right)\right]_{-\pi}^{0} + \left[(\pi - x)\left(\frac{\sin nx}{n}\right) - (-1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi}\left\{\left[\frac{cos\,0}{n^2} - \frac{cosn\pi}{n^2}\right] + \left[-\frac{cosn\pi}{n^2} + \frac{cos\,0}{n^2}\right]\right\}$$

$$\therefore a_n = \frac{2[1 - (-1)^n]}{\pi n^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi+x)\sin nx\ dx + \int_{0}^{\pi}(\pi-x)\sin nx\, dx\right\}$$
$$= \frac{1}{\pi}\left\{\left[(\pi+x)\left(\frac{-cosnx}{n}\right)-(1)\left(\frac{-sinnx}{n^2}\right)\right]_{-\pi}^{0} + \left[(\pi-x)\left(\frac{-\cos nx}{n}\right)-(-1)\left(\frac{-\sin nx}{n^2}\right)\right]_{0}^{\pi}\right\}$$
$$= \frac{1}{\pi}\left\{\frac{-\pi cos\, 0}{n} + \frac{\pi cos\, 0}{n}\right\}$$
$$\therefore b_n = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = \frac{\pi}{2} + \frac{1}{\pi}\sum\frac{2[1-(-1)^n]}{n^2}cosnx$$

59. Obtain the Fourier series of $f(x) = \begin{cases} 0 & -\pi \le x \le 0 \\ x^2 & 0 \le x \le \pi \end{cases}$

[M14/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)dx$$
$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0} 0dx + \int_{0}^{\pi} x^2 dx\right\}$$
$$= \frac{1}{2\pi}\left\{\left[\frac{x^3}{3}\right]_0^{\pi}\right\}$$
$$\therefore a_0 = \frac{\pi^2}{6}$$
$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cos nx\ dx + \int_{0}^{\pi} x^2\cos nx\, dx\right\}$$
$$= \frac{1}{\pi}\left\{\left[x^2\left(\frac{\sin nx}{n}\right)-(2x)\left(-\frac{\cos nx}{n^2}\right)+(2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^{\pi}\right\}$$
$$= \frac{2}{\pi n^2}\{[x\cos nx]_0^{\pi}\}$$
$$\therefore a_n = \frac{2(-1)^n}{n^2}$$
$$b_n = \frac{1}{l}\int_{0}^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\ dx + \int_{0}^{\pi} x^2\sin nx\, dx\right\}$$
$$= \frac{1}{\pi}\left\{\left[x^2\left(-\frac{\cos nx}{n}\right)-(2x)\left(-\frac{\sin nx}{n^2}\right)+(2)\left(\frac{\cos nx}{n^3}\right)\right]_0^{\pi}\right\}$$
$$= \frac{1}{\pi}\left\{-\frac{\pi^2\cos n\pi}{n} + \frac{2\cos n\pi}{n^3} - 2\frac{cos\,0}{n^3}\right\}$$
$$\therefore b_n = \frac{1}{\pi}\left\{-\frac{\pi^2(-1)^n}{n} + \frac{2[(-1)^n-1]}{n^3}\right\}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = \frac{\pi^2}{6} + 2\sum\frac{(-1)^n}{n^2}cosnx + \frac{1}{\pi}\sum\left[-\frac{\pi(-1)^n}{n} + \frac{2[(-1)^n-1]}{n^3}\right]sinnx$$

60. Find Fourier series for the following function

$$f(x)=\begin{cases}(x-\pi)^2 & 0\le x\le \pi\\ \pi^2 & \pi\le x\le 2\pi\end{cases}$$

[N18/Inst/6M]

Solution:

We have,

$$a_0=\frac{1}{2l}\int_0^{2l}f(x)dx$$
$$=\frac{1}{2\pi}\left\{\int_0^{\pi}(x-\pi)^2dx+\int_{\pi}^{2\pi}\pi^2dx\right\}$$
$$=\frac{1}{2\pi}\left\{\left[\frac{(x-\pi)^3}{3}\right]_0^{\pi}+[\pi^2x]_{\pi}^{2\pi}\right\}$$
$$=\frac{1}{2\pi}\left\{\left[0-\frac{-\pi^3}{3}\right]+[2\pi^3-\pi^3]\right\}$$
$$=\frac{1}{2\pi}\left\{\frac{\pi^3}{3}+\pi^3\right\}$$
$$\therefore a_0=\frac{2\pi^2}{3}$$
$$a_n=\frac{1}{l}\int_0^{2l}f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n=\frac{1}{\pi}\left\{\int_0^{\pi}(x-\pi)^2\cos nx\ \ dx+\int_{\pi}^{2\pi}(\pi^2)\cos nx\,dx\right\}$$
$$=\frac{1}{\pi}\left\{\left[(x-\pi)^2\left(\frac{\sin nx}{n}\right)-2(x-\pi)\left(-\frac{\cos nx}{n^2}\right)+2\left(-\frac{\sin nx}{n^3}\right)\right]_0^{\pi}+\left[\pi^2\left(\frac{\sin nx}{n}\right)\right]_{\pi}^{2\pi}\right\}$$
$$=\frac{1}{\pi}\left\{\left[2(x-\pi)\frac{cosnx}{n^2}\right]_0^{\pi}\right\}$$
$$=\frac{2}{\pi n^2}\{0+\pi\cos 0\}$$
$$\therefore a_n=\frac{2}{n^2}$$
$$b_n=\frac{1}{l}\int_0^{2l}f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n=\frac{1}{\pi}\left\{\int_0^{\pi}(x-\pi)^2\sin nx\ \ dx+\int_{\pi}^{2\pi}(\pi^2)\sin nx\,dx\right\}$$
$$=\frac{1}{\pi}\left\{\left[(x-\pi)^2\left(\frac{-\cos nx}{n}\right)-2(x-\pi)\left(\frac{-\sin nx}{n^2}\right)+2\left(\frac{cosnx}{n^3}\right)\right]_0^{\pi}+\left[(\pi^2)\left(-\frac{\cos nx}{n}\right)\right]_{\pi}^{2\pi}\right\}$$
$$=\frac{1}{\pi}\left\{\left[-(x-\pi)^2\left(\frac{\cos nx}{n}\right)+2\left(\frac{cosnx}{n^3}\right)\right]_0^{\pi}-\left[\frac{\pi^2cosnx}{n}\right]_{\pi}^{2\pi}\right\}$$
$$=\frac{1}{\pi}\left\{\left[0+\frac{2cosn\pi}{n^3}+\frac{\pi^2}{n}-\frac{2}{n^3}\right]-\left[\frac{\pi^2}{n}-\frac{\pi^2cosn\pi}{n}\right]\right\}$$
$$\therefore b_n=\frac{1}{\pi}\left[\frac{2((-1)^n-1)}{n^3}+\frac{\pi^2(-1)^n}{n}\right]$$

Thus, the fourier series is given by

$$f(x)=a_0+\sum a_n\cos\frac{n\pi x}{l}+\sum b_n\sin\frac{n\pi x}{l}$$
$$f(x)=\frac{2\pi^2}{3}+\sum_1^{\infty}\frac{2}{n^2}\cos nx+\sum_1^{\infty}\frac{1}{\pi}\left[\frac{2((-1)^n-1)}{n^3}+\frac{\pi^2(-1)^n}{n}\right]\sin nx$$

61. Find Fourier series for the following function

$$f(x)=\begin{cases}(x-\pi)^2 & 0\le x\le \pi\\ 0 & \pi\le x\le 2\pi\end{cases}$$

[N19/Elect/6M]

Solution:

We have,

$$a_0=\frac{1}{2l}\int_0^{2l}f(x)dx$$

$$=\frac{1}{2\pi}\left\{\int_0^{\pi}(x-\pi)^2dx+\int_{\pi}^{2\pi}0dx\right\}$$

$$=\frac{1}{2\pi}\left\{\left[\frac{(x-\pi)^3}{3}\right]_0^{\pi}\right\}$$

$$=\frac{1}{2\pi}\left\{\left[0-\frac{-\pi^3}{3}\right]\right\}$$

$$=\frac{1}{2\pi}\left\{\frac{\pi^3}{3}\right\}$$

$$\therefore a_0=\frac{\pi^2}{6}$$

$$a_n=\frac{1}{l}\int_0^{2l}f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n=\frac{1}{\pi}\left\{\int_0^{\pi}(x-\pi)^2\cos nx\ dx+\int_{\pi}^{2\pi}(0)\cos nx\,dx\right\}$$

$$=\frac{1}{\pi}\left\{\left[(x-\pi)^2\left(\frac{\sin nx}{n}\right)-2(x-\pi)\left(-\frac{\cos nx}{n^2}\right)+2\left(-\frac{\sin nx}{n^3}\right)\right]_0^{\pi}\right\}$$

$$=\frac{1}{\pi}\left\{\left[2(x-\pi)\frac{cosnx}{n^2}\right]_0^{\pi}\right\}$$

$$=\frac{2}{\pi n^2}\{0+\pi\cos 0\}$$

$$\therefore a_n=\frac{2}{n^2}$$

$$b_n=\frac{1}{l}\int_0^{2l}f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n=\frac{1}{\pi}\left\{\int_0^{\pi}(x-\pi)^2\sin nx\ dx+\int_{\pi}^{2\pi}0\sin nx\,dx\right\}$$

$$=\frac{1}{\pi}\left\{\left[(x-\pi)^2\left(\frac{-\cos nx}{n}\right)-2(x-\pi)\left(\frac{-\sin nx}{n^2}\right)+2\left(\frac{cosnx}{n^3}\right)\right]_0^{\pi}\right\}$$

$$=\frac{1}{\pi}\left\{\left[-(x-\pi)^2\left(\frac{\cos nx}{n}\right)+2\left(\frac{cosnx}{n^3}\right)\right]_0^{\pi}\right\}$$

$$=\frac{1}{\pi}\left\{\left[0+\frac{2cosn\pi}{n^3}+\frac{\pi^2}{n}-\frac{2}{n^3}\right]\right\}$$

$$\therefore b_n=\frac{1}{\pi}\left[\frac{2((-1)^n-1)}{n^3}+\frac{\pi^2}{n}\right]$$

Thus, the fourier series is given by

$$f(x)=a_0+\sum a_n\cos\frac{n\pi x}{l}+\sum b_n\sin\frac{n\pi x}{l}$$

$$f(x)=\frac{\pi^2}{6}+\sum_1^{\infty}\frac{2}{n^2}\cos nx+\sum_1^{\infty}\frac{1}{\pi}\left[\frac{2((-1)^n-1)}{n^3}+\frac{\pi^2}{n}\right]\sin nx$$

62. If $f(x)=\begin{cases}\pi x & 0\le x\le 1\\ \pi(2-x) & 1\le x\le 2\end{cases}$ with period 2.

[N14/Biot/6M][M17/CompIT/8M]

show that, $f(x) = \frac{\pi}{2} - \frac{4}{\pi}\sum_0^\infty \frac{1}{(2n+1)^2}\cos(2n+1)\,\pi x$

Deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots$

[N14/CompIT/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2}\left\{\int_0^1 \pi x dx + \int_1^2 \pi(2-x)dx\right\}$$
$$= \frac{1}{2}\left\{\left[\pi\frac{x^2}{2}\right]_0^1 + \left[\pi\left(2x - \frac{x^2}{2}\right)\right]_1^2\right\}$$
$$= \frac{\pi}{2}\left\{\frac{1}{2} + 4 - \frac{4}{2} - 2 + \frac{1}{2}\right\}$$
$$\therefore a_0 = \frac{\pi}{2}$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{1}\left\{\int_0^1 \pi x\cos n\pi x\ \ dx + \int_1^2 \pi(2-x)\cos n\pi x\, dx\right\}$$
$$= \left\{\left[\pi x\left(\frac{\sin n\pi x}{n\pi}\right) - (\pi)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_0^1 + \left[\pi(2-x)\left(\frac{\sin n\pi x}{n\pi}\right) - (-\pi)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_1^2\right\}$$
$$= \frac{1}{\pi n^2}\{[(-1)^n - 1] - [1 - (-1)^n]\}$$
$$\therefore a_n = \frac{2[(-1)^n - 1]}{n^2\pi}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{1}\left\{\int_0^1 \pi x\sin n\pi x\ \ dx + \int_1^2 \pi(2-x)\sin n\pi x\, dx\right\}$$
$$= \left\{\left[\pi x\left(-\frac{\cos n\pi x}{n\pi}\right) - (\pi)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_0^1 + \left[\pi(2-x)\left(-\frac{\cos n\pi x}{n\pi}\right) - (-\pi)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_1^2\right\}$$
$$= -\frac{1}{n}[(-1)^n - (-1)^n]$$
$$\therefore b_n = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$
$$f(x) = \frac{\pi}{2} + \sum\frac{2[(-1)^n - 1]}{n^2\pi} cosn\pi x$$
$$f(x) = \frac{\pi}{2} + \frac{2}{\pi}\left[-\frac{2\cos\pi x}{1^2} - \frac{2\cos 3\pi x}{3^2} - \frac{2\cos 5\pi x}{5^2} - \cdots\ldots\right]$$

Putting $x = 1$,

$$f(1) = \frac{\pi}{2} + \frac{2}{\pi}\left[\frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \cdots.\right]$$

We see that, at $x = 1$ the function is discontinuos

$\therefore f(x) = \frac{f_1(x) + f_2(x)}{2}$ at $x = 1$

$$\therefore f(x) = \frac{\pi x + \pi(2-x)}{2} = \pi$$

Thus, $f(1) = \pi$

$$\pi = \frac{\pi}{2} + \frac{2}{\pi}\left[\frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \cdots.\right]$$
$$\frac{\pi}{2} = \frac{4}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$$
$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \ldots$$

63. Expand $f(x) = \begin{cases} \pi x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$ period 2 into a Fourier series.

[M14/CompIT/6M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2}\left\{\int_0^1 \pi x dx + \int_1^2 0 dx\right\}$$
$$= \frac{1}{2}\left\{\left[\pi\frac{x^2}{2}\right]_0^1\right\}$$
$$\therefore a_0 = \frac{\pi}{4}$$
$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{1}{1}\left\{\int_0^1 \pi x\cos n\pi x\ \ dx + \int_1^2 0\cos n\pi x\, dx\right\}$$
$$= \left\{\left[\pi x\left(\frac{\sin n\pi x}{n\pi}\right) - (\pi)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_0^1\right\}$$
$$= \frac{1}{\pi n^2}\{[(-1)^n - 1]\}$$
$$\therefore a_n = \frac{[(-1)^n - 1]}{n^2\pi^2}$$
$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n = \frac{1}{1}\left\{\int_0^1 \pi x\sin n\pi x\ \ dx + \int_1^2 0\sin n\pi x\, dx\right\}$$
$$= \left\{\left[\pi x\left(-\frac{\cos n\pi x}{n\pi}\right) - (\pi)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_0^1\right\}$$
$$= -\frac{1}{n}[(-1)^n]$$
$$\therefore b_n = \frac{-(-1)^n}{n}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$
$$f(x) = \frac{\pi}{4} + \sum\frac{[(-1)^n - 1]}{n^2\pi}cosn\pi x - \sum\frac{(-1)^n}{n}sinn\pi x$$

64. Find the fourier series for $f(x)=\begin{cases}\pi x & 0\le x\le 1\\ 0 & x=1\\ \pi(x-2) & 1\le x\le 2\end{cases}$

show that, $f(x)=\sum_1^\infty \frac{2(-1)^{n+1}}{n} sinn\pi x$.

Also deduce that $\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots$

[M15/Chem/8M]

Solution:

$$a_0=\frac{1}{2l}\int_0^{2l} f(x)dx$$
$$=\frac{1}{2}\left\{\int_0^1 \pi x dx+\int_1^2 \pi(x-2)dx\right\}$$
$$=\frac{1}{2}\left\{\left[\pi\frac{x^2}{2}\right]_0^1+\left[\pi\left(\frac{x^2}{2}-2x\right)\right]_1^2\right\}$$
$$=\frac{\pi}{2}\left\{\frac{1}{2}+\frac{4}{2}-4-\frac{1}{2}+2\right\}=0$$
$$a_n=\frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n=\frac{1}{1}\left\{\int_0^1 \pi x\cos n\pi x\ \ dx+\int_1^2 \pi(x-2)\cos n\pi x\, dx\right\}$$
$$=\left\{\left[\pi x\left(\frac{\sin n\pi x}{n\pi}\right)-(\pi)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_0^1+\left[\pi(x-2)\left(\frac{\sin n\pi x}{n\pi}\right)-(\pi)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_1^2\right\}$$
$$\therefore a_n=\frac{1}{\pi n^2}\{[(-1)^n-1]+[1-(-1)^n]\}=0$$
$$b_n=\frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$b_n=\frac{1}{1}\left\{\int_0^1 \pi x\sin n\pi x\ \ dx+\int_1^2 \pi(x-2)\sin n\pi x\, dx\right\}$$
$$=\left\{\left[\pi x\left(-\frac{\cos n\pi x}{n\pi}\right)-(\pi)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_0^1+\left[\pi(x-2)\left(-\frac{\cos n\pi x}{n\pi}\right)-(\pi)\left(-\frac{\sin n\pi x}{n^2\pi^2}\right)\right]_1^2\right\}$$
$$=-\frac{1}{n}[(-1)^n+(-1)^n]=-\frac{2(-1)^n}{n}$$

Thus, the fourier series is given by

$$f(x)=a_0+\sum a_n\cos\frac{n\pi x}{l}+\sum b_n\sin\frac{n\pi x}{l}$$
$$f(x)=\sum-\frac{2(-1)^n}{n}sinn\pi x$$
$$f(x)=-2\left[-\frac{\sin\pi x}{1}+\frac{\sin 2\pi x}{2}-\frac{\sin 3\pi x}{3}+\cdots\ldots\right]$$

Putting $x=\frac{1}{2}$,

$$f\left(\frac{1}{2}\right)=-2\left[\frac{-1}{1}+\frac{1}{3}-\frac{1}{5}+\cdots.\right]$$
$$\pi\left(\frac{1}{2}\right)=2\left[\frac{1}{1}-\frac{1}{3}+\frac{1}{5}-\cdots.\right]$$
$$\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\cdots.$$

65. Obtain Fourier series for the function $f(x) = \begin{cases} 1 + \frac{2x}{\pi} & -\pi \le x \le 0 \\ 1 - \frac{2x}{\pi} & 0 \le x \le \pi \end{cases}$

[N15/ElextExtcElectBiomInst/6M]

deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$..

[N16/CompIT/8M][M17/ElexExtcElectBiomInst/8M]
[M18/N18/Biom/8M][N18/Elect/8M]

Solution:

We have,

$$f(x) = \begin{cases} 1 + \frac{2x}{\pi} & -\pi < x < 0 \\ 1 - \frac{2x}{\pi} & 0 < x < \pi \end{cases}$$

Putting $x = -x$, we get

$$f(-x) = \begin{cases} 1 - \frac{2(-x)}{\pi} & -\pi < x < 0 \\ 1 + \frac{2(-x)}{\pi} & 0 < x < \pi \end{cases}$$

$$f(-x) = \begin{cases} 1 + \frac{2x}{\pi} & -\pi < x < 0 \\ 1 - \frac{2x}{\pi} & 0 < x < \pi \end{cases}$$

$\therefore f(-x) = f(x)$ i.e. $f(x)$ is an even function.

Now,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{\pi}\int_0^\pi \left(1 - \frac{2x}{\pi}\right)dx$$

$$= \frac{1}{\pi}\left[x - \frac{2x^2}{2\pi}\right]_0^\pi$$

$\therefore a_0 = 0$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{\pi}\int_0^\pi \left(1 - \frac{2x}{\pi}\right)\cos nx\ dx$$

$$= \frac{2}{\pi}\left[\left(1 - \frac{2x}{\pi}\right)\left(\frac{\sin nx}{n}\right) - \left(-\frac{2}{\pi}\right)\left(-\frac{\cos nx}{n^2}\right)\right]_0^\pi$$

$$= \frac{-4}{\pi^2 n^2}[cosnx]_0^\pi$$

$$= \frac{-4}{\pi^2 n^2}[(-1)^n - 1]$$

$\therefore a_n = \frac{-4[(-1)^n - 1]}{\pi^2 n^2}$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$1 - \frac{2x}{\pi} = \sum_1^\infty \frac{-4[(-1)^n - 1]}{\pi^2 n^2}\cos nx$$

Putting $x = 0$,

$$1 = -\frac{4}{\pi^2}\sum_1^\infty \frac{[(-1)^n - 1]}{n^2}$$

$$-\frac{\pi^2}{4} = \left[-\frac{2}{1^2} - \frac{2}{3^2} - \frac{2}{5^2} - \cdots.\right]$$

$$-\frac{\pi^2}{4} = -2\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots..$$

66. Find the Fourier series of $f(x) = \begin{cases} 2 & -2 < x < 0 \\ x & 0 < x < 2 \end{cases}$

[M14/Biot/6M][N16/Chem/8M][N18/Elex/8M][M19/Chem/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_{-l}^{l} f(x)dx$$

$$= \frac{1}{4}\left\{\int_{-2}^{0} 2dx + \int_0^2 xdx\right\}$$

$$= \frac{1}{4}\left\{[2x]_{-2}^{0} + \left[\frac{x^2}{2}\right]_0^2\right\}$$

$$= \frac{1}{4}\left\{[0 + 4] + \left[\frac{4}{2}\right]\right\}$$

$$\therefore a_0 = \frac{3}{2}$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{1}{2}\left\{\int_{-2}^{0} 2\cos\frac{n\pi x}{2}\,dx + \int_0^2 x\cos\frac{n\pi x}{2}dx\right\}$$

$$= \frac{1}{2}\left\{\left[\frac{2\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_{-2}^{0} + \left[x\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^2\right\}$$

$$= \frac{1}{2}\left\{\frac{4}{n^2\pi^2}[(-1)^n - 1]\right\}$$

$$\therefore a_n = \frac{2[(-1)^n - 1]}{n^2\pi^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{1}{2}\left\{\int_{-2}^{0} 2\sin\frac{n\pi x}{2}\,dx + \int_0^2 x\sin\frac{n\pi x}{2}dx\right\}$$

$$= \frac{1}{2}\left\{\left[\frac{-2\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right]_{-2}^{0} + \left[x\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (1)\left(-\frac{\sin\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_0^2\right\}$$

$$= \frac{-1}{n\pi}\left\{\left[2\cos\frac{n\pi x}{2}\right]_{-2}^{0} + \left[x\cos\frac{n\pi x}{2}\right]_0^2\right\}$$

$$= -\frac{1}{n\pi}\{[2 - 2(-1)^n] + [2(-1)^n]\}$$

$$\therefore b_n = \frac{-2}{n\pi}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \frac{3}{2} + \frac{2}{\pi^2}\sum\frac{[(-1)^n - 1]}{n^2} cosn\frac{\pi}{2}x - \frac{2}{\pi}\sum\frac{sinn\frac{\pi}{2}x}{n}$$

67. Find the half range cosine series for $f(x) = \begin{cases} 1 & 0 \le x \le 1 \\ x & 1 \le x \le 2 \end{cases}$

[M18/Inst/6M]

Solution:

Solution:

We have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{2}\left\{\int_0^1 1dx + \int_1^2 xdx\right\}$$

$$= \frac{1}{2}\left\{[x]_0^1 + \left[\frac{x^2}{2}\right]_1^2\right\}$$

$$= \frac{1}{2}\left\{1 - 0 + \frac{4}{2} - \frac{1}{2}\right\}$$

$$= \frac{1}{2}\left\{\frac{5}{2}\right\}$$

$$\therefore a_0 = \frac{5}{4}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{2}\left\{\int_0^1 1\cos\frac{n\pi x}{2}\ dx + \int_1^2 x\cos\frac{n\pi x}{2}\ dx\right\}$$

$$= \left\{\left[\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right)\right]_0^1 + \left[(x)\left(\frac{\sin\frac{n\pi x}{2}}{\frac{n\pi}{2}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{2}}{\frac{n^2\pi^2}{4}}\right)\right]_1^2\right\}$$

$$= \left\{\left[\left(\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}}\right)\right] + \left[\frac{\cos\ n\pi}{\frac{n^2\pi^2}{4}} - \frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{2}} - \frac{\cos\frac{n\pi}{2}}{\frac{n^2\pi^2}{4}}\right]\right\}$$

$$= \frac{(-1)^n - \cos\frac{n\pi}{2}}{\frac{n^2\pi^2}{4}}$$

$$\therefore a_n = \frac{4}{n^2\pi^2}\left[(-1)^n - \cos\frac{n\pi}{2}\right]$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$f(x) = \frac{5}{4} + \sum\frac{4}{n^2\pi^2}\left[(-1)^n - \cos\frac{n\pi}{2}\right].\cos\frac{n\pi x}{2}$$

68. Obtain half range cosine series for $f(x) = \begin{cases} kx & 0 < x < \frac{l}{2} \\ k(l-x) & \frac{l}{2} < x < l \end{cases}$

Deduce the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$.

[M15/CompIT/8M]

Solution:

We have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$= \frac{1}{l}\left\{\int_0^{\frac{l}{2}} kx dx + \int_{\frac{l}{2}}^{l} k(l-x)dx\right\}$$

$$= \frac{k}{l}\left\{\left[\frac{x^2}{2}\right]_0^{\frac{l}{2}} + \left[lx - \frac{x^2}{2}\right]_{\frac{l}{2}}^{l}\right\}$$

$$= \frac{k}{l}\left\{\frac{l^2}{8} + l^2 - \frac{l^2}{2} - \frac{l^2}{2} + \frac{l^2}{8}\right\}$$

$$= \frac{k}{l}\left\{\frac{l^2}{4}\right\}$$

$$\therefore a_0 = \frac{kl}{4}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{l}\left\{\int_0^{\frac{l}{2}} kx\cos\frac{n\pi x}{l}\ dx + \int_{\frac{l}{2}}^{l} k(l-x)\cos\frac{n\pi x}{l}\ dx\right\}$$

$$= \frac{2k}{l}\left\{\left[x\left(\frac{\sin\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (1)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_0^{\frac{l}{2}} + \left[(l-x)\left(\frac{\sin\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (-1)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_{\frac{l}{2}}^{l}\right\}$$

$$= \frac{2k}{l}\left\{\left[\frac{l}{2}.\left(\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{l}}\right) + \frac{\cos\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}} - \frac{\cos 0}{\frac{n^2\pi^2}{l^2}}\right] + \left[-\frac{\cos n\pi}{\frac{n^2\pi^2}{l^2}} - \frac{l}{2}.\frac{\sin\frac{n\pi}{2}}{\frac{n\pi}{l}} + \frac{\cos\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}}\right]\right\}$$

$$= \frac{2k}{l}\left[\frac{2\cos\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}} - \frac{1}{\frac{n^2\pi^2}{l^2}} - \frac{(-1)^n}{\frac{n^2\pi^2}{l^2}}\right]$$

$$\therefore a_n = \frac{2kl}{n^2\pi^2}\left[2\cos\frac{n\pi}{2} - 1 - (-1)^n\right]$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l}$$

$$f(x) = \frac{kl}{4} + \sum\frac{2kl}{n^2\pi^2}\left[2\cos\frac{n\pi}{2} - 1 - (-1)^n\right].\cos\frac{n\pi x}{l}$$

$$f(x) = \frac{kl}{4} + \frac{2kl}{\pi^2}\left[-\frac{4\cos\frac{2\pi x}{l}}{2^2} - \frac{4\cos\frac{6\pi x}{l}}{6^2} - \frac{4\cos\frac{10\pi x}{l}}{10^2} - \cdots\right]$$

$$f(x) = \frac{kl}{4} - \frac{2kl}{\pi^2}\left[\frac{\cos\frac{2\pi x}{l}}{1^2} + \frac{\cos\frac{6\pi x}{l}}{3^2} + \frac{\cos\frac{10\pi x}{l}}{5^2} + \cdots\right]$$

Putting, $x = 0$

$$f(0) = \frac{kl}{4} - \frac{2kl}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$$

$$0 = \frac{kl}{4} - \frac{2kl}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$$

$$-\frac{kl}{4} = -\frac{2kl}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\ldots$$

69. Find half range cosine series for $f(x)$ where $f(x) = \begin{cases} x & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$

[N14/AutoMechCivil/6M][M16/ElexExtcElectBiomInst/6M]
[N17/Elex/6M][M18/Elex/6M]

Solution:

We have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\left\{\int_0^{\frac{\pi}{2}} x dx + \int_{\frac{\pi}{2}}^{\pi} (\pi - x)dx\right\}$$
$$= \frac{1}{\pi}\left\{\left[\frac{x^2}{2}\right]_0^{\frac{\pi}{2}} + \left[\pi x - \frac{x^2}{2}\right]_{\frac{\pi}{2}}^{\pi}\right\}$$
$$= \frac{1}{\pi}\left\{\frac{\pi^2}{8} + \pi^2 - \frac{\pi^2}{2} - \frac{\pi^2}{2} + \frac{\pi^2}{8}\right\}$$
$$= \frac{1}{\pi}\left\{\frac{\pi^2}{4}\right\}$$
$$\therefore a_0 = \frac{\pi}{4}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} x\cos nx \ dx + \int_{\frac{\pi}{2}}^{\pi} (\pi - x)\cos nx \ dx\right\}$$
$$= \frac{2}{\pi}\left\{\left[x\left(\frac{\sin nx}{n}\right) - (1)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\frac{\pi}{2}} + \left[(\pi - x)\left(\frac{\sin nx}{n}\right) - (-1)\left(-\frac{\cos nx}{n^2}\right)\right]_{\frac{\pi}{2}}^{\pi}\right\}$$
$$= \frac{2}{\pi}\left\{\left[\frac{\pi}{2}.\left(\frac{\sin\frac{n\pi}{2}}{n}\right) + \frac{\cos\frac{n\pi}{2}}{n^2} - \frac{\cos 0}{n^2}\right] + \left[-\frac{\cos n\pi}{n^2} - \frac{\pi}{2}.\frac{\sin\frac{n\pi}{2}}{n} + \frac{\cos\frac{n\pi}{2}}{n^2}\right]\right\}$$
$$= \frac{2}{\pi}\left[\frac{2\cos\frac{n\pi}{2}}{n^2} - \frac{1}{n^2} - \frac{(-1)^n}{n^2}\right]$$
$$\therefore a_n = \frac{2}{\pi n^2}\left[2\cos\frac{n\pi}{2} - 1 - (-1)^n\right]$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$f(x) = \frac{\pi}{4} + \sum\frac{2}{\pi n^2}\left[2\cos\frac{n\pi}{2} - 1 - (-1)^n\right].\cos nx$$

70. Find half range sine series for $f(x)$ where $f(x) = \begin{cases} x & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$

Hence find the sum of $\sum_{2n-1}^{\infty}\frac{1}{n^4}$

[N14/Chem/6M][N15/Chem/6M][M16/Chem/7M][N16/Chem/6M]
[N17/Comp/8M]

Solution:

We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} x\sin nx\ dx + \int_{\frac{\pi}{2}}^{\pi}(\pi - x)\sin nx\ dx\right\}$$

$$= \frac{2}{\pi}\left\{\left[x\left(-\frac{\cos nx}{n}\right) - (1)\left(-\frac{\sin nx}{n^2}\right)\right]_0^{\frac{\pi}{2}} + \left[(\pi - x)\left(-\frac{\cos nx}{n}\right) - (-1)\left(-\frac{\sin nx}{n^2}\right)\right]_{\frac{\pi}{2}}^{\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\left[-\frac{\pi}{2}.\left(\frac{\cos\frac{n\pi}{2}}{n}\right) + \frac{\sin\frac{n\pi}{2}}{n^2}\right] + \left[\frac{\pi}{2}.\frac{\cos\frac{n\pi}{2}}{n} + \frac{\sin\frac{n\pi}{2}}{n^2}\right]\right\}$$

$$= \frac{2}{\pi}\left[\frac{2\sin\frac{n\pi}{2}}{n^2}\right]$$

$$\therefore b_n = \frac{4}{\pi n^2}\left[\sin\frac{n\pi}{2}\right]$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \sum \frac{4}{\pi n^2}\left[\sin\frac{n\pi}{2}\right].\sin nx$$

By Parseval's identity,

$$\frac{1}{l}\int_0^l [f(x)]^2 dx = \frac{1}{2}\sum_1^\infty b_n^2$$

$$\frac{1}{\pi}\{\int_0^{\frac{\pi}{2}} x^2 dx + \int_{\frac{\pi}{2}}^{\pi}(\pi - x)^2 dx = \frac{1}{2}\sum_1^\infty \frac{16\sin^2 n\frac{\pi}{2}}{\pi^2 n^4}$$

$$\frac{1}{\pi}\left\{\left[\frac{x^3}{3}\right]_0^{\frac{\pi}{2}} + \left[\frac{(\pi - x)^2}{-2}\right]_{\frac{\pi}{2}}^{\pi}\right\} = \frac{8}{\pi^2}\left[\frac{\sin^2\frac{\pi}{2}}{1^4} + \frac{\sin^2\frac{3\pi}{2}}{3^4} + \frac{\sin^2\frac{5\pi}{2}}{5^4} + \cdots..\right]$$

$$\frac{1}{\pi}\left\{\frac{\pi^3}{24} + \frac{\pi^3}{24}\right\} = \frac{8}{\pi^2}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots..\right]$$

$$\frac{1}{\pi}.\frac{\pi^3}{12} = \frac{8}{\pi^2}\sum_{2n-1}^\infty \frac{1}{n^4}$$

$$\frac{\pi^2}{12} = \frac{8}{\pi^2}\sum_{2n-1}^\infty \frac{1}{n^4}$$

$$\frac{\pi^4}{96} = \sum_{2n-1}^\infty \frac{1}{n^4}$$

71. Find half range sine series for $f(x)$ where $f(x) = \begin{cases} x & 0 < x < 2 \\ 4 - x & 2 < x < 4 \end{cases}$

[M15/Chem/6M][N15/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$$

$$b_n = \frac{2}{4}\left\{\int_0^2 x\sin\frac{n\pi x}{4}\ dx + \int_2^4 (4 - x)\sin\frac{n\pi x}{4}\ dx\right\}$$

$$= \frac{1}{2}\left\{\left[x\left(-\frac{\cos\frac{n\pi x}{4}}{\frac{n\pi}{4}}\right) - (1)\left(-\frac{\sin\frac{n\pi x}{4}}{\frac{n^2\pi^2}{16}}\right)\right]_0^2 + \left[(4 - x)\left(-\frac{\cos\frac{n\pi x}{4}}{\frac{n\pi}{4}}\right) - (-1)\left(-\frac{\sin\frac{n\pi x}{4}}{\frac{n^2\pi^2}{16}}\right)\right]_2^4\right\}$$

$$= \frac{1}{2}\left\{\left[2\left(-\frac{\cos\frac{n\pi}{2}}{\frac{n\pi}{4}}\right) + \frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{16}}\right] + \left[2\ \frac{\cos\frac{n\pi}{2}}{\frac{n\pi}{4}} + \frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{16}}\right]\right\}$$

$$= \frac{1}{2}\left[2\frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{16}}\right]$$

$$\therefore b_n = \frac{16}{n^2\pi^2}\sin\frac{n\pi}{2}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \sum \frac{16}{n^2\pi^2}\sin\frac{n\pi}{2}.\sin\frac{n\pi x}{4}$$

72. Find the half range sine series for the function

$$f(x) = \begin{cases} \frac{2kx}{l} & 0 \le x \le \frac{l}{2} \\ \frac{2k}{l}(l-x) & \frac{l}{2} \le x \le l \end{cases}$$

[N13/CompIT/6M]

Solution:

We have,

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{2}{l}\left\{\int_0^{\frac{l}{2}} \frac{2kx}{l}\sin\frac{n\pi x}{l}\,dx + \int_{\frac{l}{2}}^{l}\frac{2k(l-x)}{l}\sin\frac{n\pi x}{l}\,dx\right\}$$

$$= \frac{4k}{l^2}\left\{\left[x\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (1)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_0^{\frac{l}{2}} + \left[(l-x)\left(-\frac{\cos\frac{n\pi x}{l}}{\frac{n\pi}{l}}\right) - (-1)\left(-\frac{\sin\frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}}\right)\right]_{\frac{l}{2}}^{l}\right\}$$

$$= \frac{4k}{l^2}\left\{\left[\frac{l}{2}.\left(-\frac{\cos\frac{n\pi}{2}}{\frac{n\pi}{l}}\right) + \frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}}\right] + \left[\frac{l}{2}.\frac{\cos\frac{n\pi}{2}}{\frac{n\pi}{l}} + \frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}}\right]\right\}$$

$$= \frac{4k}{l^2}\left[2\frac{\sin\frac{n\pi}{2}}{\frac{n^2\pi^2}{l^2}}\right]$$

$$\therefore b_n = \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$

$$f(x) = \sum \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}.\sin\frac{n\pi x}{l}$$

$$f(x) = \frac{8k}{\pi^2}\left[\frac{\sin\left(\frac{\pi x}{l}\right)}{1^2} - \frac{\sin\left(\frac{3\pi x}{l}\right)}{3^2} + \cdots..\right]$$

Type IV: Trigonometric functions $(sinx, cosx, etc)$

1. Obtain Fourier expansion of $f(x) = |sinx|$ in $(-\pi, \pi)$
   [N13/ElexExtcElectBiomInst/5M][N14/Biot/6M]
   Solution:
   We see that, $f(x) = |sinx|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^\pi sinxdx$$
$$= \frac{1}{\pi}[-cosx]_0^\pi$$
$$= \frac{1}{\pi}[-cos\pi + cos0] = \frac{2}{\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\int_0^\pi sinxcosnx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi 2sinx\ cosnx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi \sin(1+n)x + \sin(1-n)x\ dx$$
$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n}\right]_0^\pi$$
$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n} + \frac{cos\,0}{1+n} + \frac{cos\,0}{1-n}\right]$$
$$= \frac{1}{\pi}\left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n} + \frac{1}{1+n} + \frac{1}{(1-n)}\right]$$
$$= \frac{[(-1)^n+1]}{\pi}\left[\frac{1}{1+n} + \frac{1}{1-n}\right]$$
$$= \frac{[(-1)^n+1]}{\pi}\left[\frac{2}{1-n^2}\right]$$
$$\therefore a_n = \frac{2[(-1)^n+1]}{\pi(1-n^2)}, n \neq 1$$
$$a_1 = \frac{2}{\pi}\int_0^\pi sinx\ cosx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi sin2x\ dx$$
$$= \frac{1}{\pi}\left[-\frac{cos\,2x}{2}\right]_0^\pi$$
$$= -\frac{1}{2\pi}[\cos 2\pi - \cos 0] = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$f(x) = a_0 + a_1 cosx + \sum_2^\infty \frac{2[(-1)^n+1]}{\pi(1-n^2)}\cos nx$$
$$|sinx| = \frac{2}{\pi} - \frac{2}{\pi}\sum_2^\infty \frac{[(-1)^n+1]}{n^2-1}cosnx$$

2. Find half range cosine series of $f(x) = sinx$ in $0 \le x \le \pi$.
[N13/CompIT/5M][M14/Chem/7M][M15/ElexExtcElectBiomInst/5M]
Hence deduce that $\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots \ldots.. = \frac{1}{2}$
[M14/ElexExtcElectBiomInst/6M]
Hence deduce that $\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots \ldots = \frac{\pi^2-8}{16}$
[M19/MTRX/8M][N19/Comp/8M]
Solution:
We see have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^\pi sinxdx$$
$$= \frac{1}{\pi}[-cosx]_0^\pi$$
$$= \frac{1}{\pi}[-cos\pi + cos0]$$
$$\therefore a_0 = \frac{2}{\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\int_0^\pi sinxcosnx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi 2sinx\ cosnx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi \sin(1+n)x + \sin(1-n)x\ dx$$
$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n}\right]_0^\pi$$
$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n} + \frac{cos\,0}{1+n} + \frac{cos\,0}{1-n}\right]$$
$$= \frac{1}{\pi}\left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n} + \frac{1}{1+n} + \frac{1}{(1-n)}\right]$$
$$= \frac{[(-1)^n+1]}{\pi}\left[\frac{1}{1+n} + \frac{1}{1-n}\right]$$
$$= \frac{[(-1)^n+1]}{\pi}\left[\frac{2}{1-n^2}\right]$$
$$\therefore a_n = \frac{2[(-1)^n+1]}{\pi(1-n^2)}, n \ne 1$$
$$a_1 = \frac{2}{\pi}\int_0^\pi sinx\ cosx\ dx$$
$$= \frac{1}{\pi}\int_0^\pi sin2x\ dx$$
$$= \frac{1}{\pi}\left[-\frac{cos\,2x}{2}\right]_0^\pi$$
$$= -\frac{1}{2\pi}[\cos2\pi - \cos0]$$
$$\therefore a_1 = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$

$$f(x) = a_0 + a_1 cosx + \sum_2^\infty \frac{2[(-1)^n+1]}{\pi(1-n^2)} \cos nx$$
$$sinx = \frac{2}{\pi} - \frac{2}{\pi}\sum_2^\infty \frac{[(-1)^n+1]}{n^2-1} cosnx$$
$$sinx = \frac{2}{\pi} - \frac{2}{\pi}\left[\frac{2cos\,2x}{3} + \frac{2cos\,4x}{15} + \frac{2cos\,6x}{35} + \cdots \ldots\right]$$
$$sinx = \frac{2}{\pi} - \frac{4}{\pi}\left[\frac{cos\,2x}{1\times3} + \frac{cos\,4x}{3\times5} + \frac{cos\,6x}{5\times7} + \cdots \ldots\right]$$
Putting, $x = 0$
$$sin0 = \frac{2}{\pi} - \frac{4}{\pi}\left[\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots \ldots\right]$$
$$0 = \frac{2}{\pi} - \frac{4}{\pi}\left[\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots \ldots\right]$$
$$-\frac{2}{\pi} = -\frac{4}{\pi}\left[\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots \ldots\right]$$
$$\frac{1}{2} = \left[\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots \ldots\right]$$
By parseval's identity,
$$\frac{1}{l}\int_0^l [f(x)]^2\,dx = a_0^2 + \frac{a_1^2}{2} + \frac{1}{2}\sum_2^\infty a_n^2$$
$$\frac{1}{\pi}\int_0^\pi [sinx]^2 dx = \frac{4}{\pi^2} + \frac{1}{2}\sum_2^\infty \frac{4[(-1)^n+1]^2}{\pi^2(1-n^2)^2}$$
$$\frac{1}{\pi}\int_0^\pi \left[\frac{1-cos\,2x}{2}\right] dx = \frac{4}{\pi^2} + \frac{4}{2\pi^2}\left[\frac{4}{3^2} + \frac{4}{15^2} + \frac{4}{35^2} + \cdots\right]$$
$$\frac{1}{\pi}\left[\frac{x}{2} - \frac{sin\,2x}{4}\right]_0^\pi = \frac{4}{\pi^2} + \frac{16}{2\pi^2}\left[\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots\right]$$
$$\frac{1}{\pi}\left[\frac{\pi}{2}\right] = \frac{4}{\pi^2} + \frac{8}{\pi^2}\left[\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots\right]$$
$$\frac{1}{2} - \frac{4}{\pi^2} = \frac{8}{\pi^2}\left[\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots\right]$$
$$\frac{\pi^2-8}{2\pi^2} = \frac{8}{\pi^2}\left[\frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots\right]$$
$$\frac{\pi^2-8}{16} = \frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots \ldots$$

3. Find the Fourier expansion for $f(x) = \sqrt{1-cosx}\ \ in\ (0,2\pi)$ hence, deduce that $\frac{1}{2} = \sum_1^\infty \frac{1}{4n^2-1}$
[N13/ElexExtcElectBiomInst/8M][N13/CompIT/8M][N15/CompIT/8M][N15/Chem/6M]
Solution:

We have, $f(x) = \sqrt{1-cosx} = \sqrt{2\sin^2\frac{x}{2}} = \sqrt{2}\sin\frac{x}{2}$
$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$
$$= \frac{1}{2\pi}\left\{\int_0^{2\pi} \sqrt{2}\sin\frac{x}{2}\,dx\right\}$$
$$= \frac{\sqrt{2}}{2\pi}\left\{\left[-\frac{\cos\frac{x}{2}}{\frac{1}{2}}\right]_0^{2\pi}\right\}$$

$$= -\frac{\sqrt{2}}{\pi}[\cos\pi - \cos 0]$$

$$\therefore a_0 = \frac{2\sqrt{2}}{\pi}$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}\,dx$$

$$= \frac{1}{\pi}\left\{\int_0^{2\pi} \sqrt{2}\sin\frac{x}{2}\cos nx\ \ dx\right\}$$

$$= \frac{\sqrt{2}}{2\pi}\int_0^{2\pi} 2\sin\frac{x}{2}\cos nx\,dx$$

$$= \frac{\sqrt{2}}{2\pi}\int_0^{2\pi} \sin\left(\frac{1}{2}+n\right)x + \sin\left(\frac{1}{2}-n\right)x\,dx$$

$$= \frac{\sqrt{2}}{2\pi}\left[-\frac{\cos\left(\frac{1}{2}+n\right)x}{\frac{1}{2}+n} - \frac{\cos\left(\frac{1}{2}-n\right)x}{\frac{1}{2}-n}\right]_0^{2\pi}$$

$$= \frac{\sqrt{2}}{2\pi}\left[-\frac{\cos(1+2n)\pi}{\frac{1}{2}+n} - \frac{\cos(1-2n)\pi}{\frac{1}{2}-n} + \frac{\cos 0}{\frac{1}{2}+n} + \frac{\cos 0}{\frac{1}{2}-n}\right]$$

$$= \frac{\sqrt{2}}{2\pi}\left[\frac{1}{\frac{1}{2}+n} + \frac{1}{\frac{1}{2}-n} + \frac{1}{\frac{1}{2}+n} + \frac{1}{\frac{1}{2}-n}\right]$$

$$= \frac{2\sqrt{2}}{2\pi}\left[\frac{1}{\frac{1}{2}+n} + \frac{1}{\frac{1}{2}-n}\right]$$

$$= \frac{\sqrt{2}}{\pi}\left[\frac{1}{\frac{1}{4}-n^2}\right]$$

$$\therefore a_n = \frac{4\sqrt{2}}{\pi(1-4n^2)}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$

$$= \frac{1}{\pi}\left\{\int_0^{2\pi} \sqrt{2}\sin\frac{x}{2}\sin nx\ \ dx\right\}$$

$$= \frac{\sqrt{2}}{2\pi}\int_0^{2\pi} 2\sin\frac{x}{2}\sin nx\,dx$$

$$= \frac{\sqrt{2}}{2\pi}\int_0^{2\pi} \cos\left(\frac{1}{2}-n\right)x - \cos\left(\frac{1}{2}+n\right)x\,dx$$

$$= \frac{\sqrt{2}}{2\pi}\left[\frac{\sin\left(\frac{1}{2}-n\right)x}{\frac{1}{2}-n} - \frac{\sin\left(\frac{1}{2}+n\right)x}{\frac{1}{2}+n}\right]_0^{2\pi}$$

$$= \frac{\sqrt{2}}{2\pi}\left[\frac{\sin(1-2n)\pi}{\frac{1}{2}-n} - \frac{\sin(1+2n)\pi}{\frac{1}{2}+n} - \frac{\sin 0}{\frac{1}{2}+n} + \frac{\sin 0}{\frac{1}{2}-n}\right]$$

$$\therefore b_n = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$

$$\sqrt{1-cosx} = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\sum\frac{cosnx}{4n^2-1}$$

Putting, $x = 0$

$$\sqrt{1-cos0} = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\sum\frac{1}{4n^2-1}$$

$$0 = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\sum\frac{1}{4n^2-1}$$
$$-\frac{2\sqrt{2}}{\pi} = -\frac{4\sqrt{2}}{\pi}\sum\frac{1}{4n^2-1}$$
$$\frac{1}{2} = \sum\frac{1}{4n^2-1}$$

4. Find the Fourier expansion for $f(x) = \sqrt{1-cosx}\ in\ (-\pi,\pi)$
[M19/Elex/5M]
Solution:

We have, $f(x) = \sqrt{1-cosx} = \sqrt{2\sin^2\frac{x}{2}} = \sqrt{2}\sin\frac{x}{2}$

We see that $f(x)$ is an even function

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\left\{\int_0^\pi \sqrt{2}\sin\frac{x}{2}dx\right\}$$
$$= \frac{\sqrt{2}}{\pi}\left\{\left[-\frac{\cos\frac{x}{2}}{\frac{1}{2}}\right]_0^\pi\right\}$$
$$= -\frac{2\sqrt{2}}{\pi}\left[\cos\frac{\pi}{2} - \cos 0\right]$$
$$\therefore a_0 = \frac{2\sqrt{2}}{\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$= \frac{2}{\pi}\left\{\int_0^\pi \sqrt{2}\sin\frac{x}{2}\cos nx\ \ dx\right\}$$
$$= \frac{\sqrt{2}}{\pi}\int_0^\pi 2\sin\frac{x}{2}\cos nx\,dx$$
$$= \frac{\sqrt{2}}{\pi}\int_0^\pi \sin\left(\frac{1}{2}+n\right)x + \sin\left(\frac{1}{2}-n\right)x\,dx$$
$$= \frac{\sqrt{2}}{\pi}\left[-\frac{\cos\left(\frac{1}{2}+n\right)x}{\frac{1}{2}+n} - \frac{\cos\left(\frac{1}{2}-n\right)x}{\frac{1}{2}-n}\right]_0^\pi$$
$$= \frac{\sqrt{2}}{\pi}\left[-\frac{\cos\left(\frac{1}{2}+n\right)\pi}{\frac{1}{2}+n} - \frac{\cos\left(\frac{1}{2}-n\right)\pi}{\frac{1}{2}-n} + \frac{\cos\ 0}{\frac{1}{2}+n} + \frac{\cos\ 0}{\frac{1}{2}-n}\right]$$

We see that

$$\cos\left(\frac{1}{2}\pm n\right)\pi = \cos\left(\frac{\pi}{2}\pm n\pi\right) = \cos\frac{\pi}{2}\ cosn\pi \mp \sin\frac{\pi}{2}\ sinn\pi = 0$$
$$\therefore a_n = \frac{\sqrt{2}}{\pi}\left[\frac{1}{\frac{1}{2}+n} + \frac{1}{\frac{1}{2}-n}\right]$$
$$= \frac{\sqrt{2}}{\pi}\left[\frac{1}{\frac{1}{4}-n^2}\right]$$
$$\therefore a_n = \frac{4\sqrt{2}}{\pi(1-4n^2)}$$

Thus, the Fourier series is given by

$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$

$\sqrt{1 - cosx} = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\sum\frac{cosnx}{4n^2-1}$

5. Expand $f(x) = xsinx$ in the interval $0 \le x \le 2\pi$.
[M18/Elect/8M][N18/Biot/6M]
Hence deduce that $\sum_2^{\infty}\frac{1}{n^2-1} = \frac{3}{4}$
[M14/Biot/8M][M15/Biot/8M][M16/Biot/8M]
[N16/ElexExtcElectBiomInst/8M]
Solution:
We have,

$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$

$= \frac{1}{2\pi}\left\{\int_0^{2\pi} x\ sinxdx\right\}$

$= \frac{1}{2\pi}\{[x\ (-cosx) - (1)(-sinx)]_0^{2\pi}\}$

$= -\frac{1}{2\pi}[2\pi]$

$\therefore a_0 = -1$

$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$

$= \frac{1}{\pi}\left\{\int_0^{2\pi} x\ sinx\cos nx\ \ dx\right\}$

$= \frac{1}{2\pi}\int_0^{2\pi} x.2sinx\cos nx\ dx$

$= \frac{1}{2\pi}\int_0^{2\pi} x[\sin(1+n)x + \sin(1-n)\ x]\ dx$

$= \frac{1}{2\pi}\left[x\left(-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n}\right) - (1)\left(-\frac{\sin(1+n)x}{(1+n)^2} - \frac{\sin(1-n)x}{(1-n)^2}\right)\right]_0^{2\pi}$

$= \frac{1}{2\pi}\left[2\pi\left(-\frac{\cos(1+n)2\pi}{1+n} - \frac{\cos(1-n)2\pi}{1-n}\right)\right]$

$= \left[\frac{-1}{1+n} - \frac{1}{1-n}\right]$

$= -\frac{2}{1-n^2}$

$\therefore a_n = \frac{2}{(n^2-1)}, n \ne 1$

$a_1 = \frac{1}{\pi}\int_0^{2\pi} x\ sinx\ cosx\ dx$

$= \frac{1}{2\pi}\int_0^{2\pi} x\ sin2x\ dx$

$= \frac{1}{2\pi}\left[x\ \left(-\frac{cos\,2x}{2}\right) - (1)\left(-\frac{sin\,2x}{4}\right)\right]_0^{2\pi}$

$= \frac{1}{2\pi}\left[-2\pi\frac{cos\,2\pi}{2}\right]$

$\therefore a_1 = -\frac{1}{2}$

$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$

$$= \frac{1}{\pi}\left\{\int_0^{2\pi} x\, sinx \sin nx\ \ dx\right\}$$
$$= \frac{1}{2\pi}\int_0^{2\pi} x(2sinx \sin nx)dx$$
$$= \frac{1}{2\pi}\int_0^{2\pi} x(\cos(1-n)x - \cos(1+n)\,x)\,dx$$
$$= \frac{1}{2\pi}\left[x\left(\frac{\sin(1-n)x}{1-n} - \frac{\sin(1+n)x}{1+n}\right) - (1)\left(-\frac{\cos(1-n)x}{(1-n)^2} + \frac{\cos(1+n)x}{(1+n)^2}\right)\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left[\frac{\cos(1-n)2\pi}{(1-n)^2} - \frac{\cos(1+n)2\pi}{(1+n)^2} - \frac{cos\,0}{(1-n)^2} + \frac{cos\,0}{(1+n)^2}\right]$$

$\therefore b_n = 0, n \neq 1$

$$b_1 = \frac{1}{\pi}\int_0^{2\pi} x\, sinx\, sinx\, dx$$
$$= \frac{1}{\pi}\int_0^{2\pi} x\sin^2 x\ \ dx$$
$$= \frac{1}{2\pi}\int_0^{2\pi} x(1 - cos2x)dx$$
$$= \frac{1}{2\pi}\left[x\ \left(x - \frac{sin\,2x}{2}\right) - (1)\left(\frac{x^2}{2} + \frac{cos\,2x}{4}\right)\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left[2\pi(2\pi) - \left(\frac{4\pi^2}{2} + \frac{cos\,4\pi}{4}\right) + \frac{cos\,0}{4}\right]$$

$\therefore b_1 = \pi$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = a_0 + a_1 cosx + \sum_2^\infty a_n cosnx + b_1 sinx + \sum_2^\infty b_n sinnx$$
$$xsinx = -1 - \frac{cosx}{2} + 2\sum_2^\infty \frac{cosnx}{n^2-1} + \pi sinx$$

Putting, $x = 0$

$$0 = -1 - \frac{1}{2} + 2\sum_2^\infty \frac{1}{n^2-1}$$
$$0 = -\frac{3}{2} + 2\sum_2^\infty \frac{1}{n^2-1}$$
$$\frac{3}{2} = 2\sum_2^\infty \frac{1}{n^2-1}$$
$$\frac{3}{4} = \sum_2^\infty \frac{1}{n^2-1}$$

6. Find half range sine series for $f(x) = xsinx$ in $(0, \pi)$
[N14/Biot/5M][N16/AutoMechCivil/6M]
hence deduce that $\frac{\pi^2}{8\sqrt{2}} = \frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \cdots$
[M16/AutoMechCivil/6M]

Solution:

We have,

$$b_n = \frac{2}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$
$$= \frac{2}{\pi}\left\{\int_0^{\pi} x\, sinx \sin nx\ \ dx\right\}$$
$$= \frac{2}{2\pi}\int_0^{\pi} x(2sinx \sin nx)dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x(\cos(1-n)x - \cos(1+n)\,x)\,dx$$
$$= \frac{1}{\pi}\left[x\left(\frac{\sin(1-n)x}{1-n} - \frac{\sin(1+n)x}{1+n}\right) - (1)\left(-\frac{\cos(1-n)x}{(1-n)^2} + \frac{\cos(1+n)x}{(1+n)^2}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\frac{\cos(1-n)\pi}{(1-n)^2} - \frac{\cos(1+n)\pi}{(1+n)^2} - \frac{\cos 0}{(1-n)^2} + \frac{\cos 0}{(1+n)^2}\right]$$
$$= \frac{1}{\pi}\left[-\frac{(-1)^n}{(1-n)^2} + \frac{(-1)^n}{(1+n)^2} - \frac{1}{(1-n)^2} + \frac{1}{(1+n)^2}\right]$$
$$= -\frac{[(-1)^n+1]}{\pi}\left[\frac{1}{(1-n)^2} - \frac{1}{(1+n)^2}\right]$$
$$\therefore b_n = -\frac{[(-1)^n+1]}{\pi}\left[\frac{1}{(1-n)^2} - \frac{1}{(1+n)^2}\right], n \neq 1$$
$$b_1 = \frac{2}{\pi}\int_0^{\pi} x\,sinx\,sinx\,dx$$
$$= \frac{2}{\pi}\int_0^{\pi} x\sin^2 x\ dx$$
$$= \frac{2}{2\pi}\int_0^{\pi} x(1-cos2x)dx$$
$$= \frac{1}{\pi}\left[x\left(x - \frac{sin\,2x}{2}\right) - (1)\left(\frac{x^2}{2} + \frac{cos\,2x}{4}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\pi(\pi) - \left(\frac{\pi^2}{2} + \frac{cos\,2\pi}{4}\right) + \frac{cos0}{4}\right]$$
$$\therefore b_1 = \frac{\pi}{2}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = b_1 sinx + \sum_2^{\infty} b_n sinnx$$
$$xsinx = \frac{\pi}{2}sinx + \sum_2^{\infty} -\frac{[(-1)^n+1]}{\pi}\left[\frac{1}{(1-n)^2} - \frac{1}{(1+n)^2}\right]sinnx$$
$$xsinx = \frac{\pi}{2}sinx - \frac{1}{\pi}\left[2\left(\frac{1}{1^2} - \frac{1}{3^2}\right)sin2x + 2\left(\frac{1}{3^2} - \frac{1}{5^2}\right)sin4x + \cdots\ldots\right]$$

Putting, $x = \frac{\pi}{4}$

$$\frac{\pi}{4}\sin\frac{\pi}{4} = \frac{\pi}{2}\sin\frac{\pi}{4} - \frac{2}{\pi}\left[\left(\frac{1}{1^2} - \frac{1}{3^2}\right) + \left(\frac{1}{5^2} - \frac{1}{7^2}\right)(-1) + \left(\frac{1}{9^2} - \frac{1}{11^2}\right) + \cdots.\right]$$
$$\frac{\pi}{4\sqrt{2}} = \frac{\pi}{2\sqrt{2}} - \frac{2}{\pi}\left[\frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \frac{1}{11^2} - \cdots..\right]$$
$$\frac{\pi}{4\sqrt{2}} - \frac{\pi}{2\sqrt{2}} = -\frac{2}{\pi}\left[\frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \frac{1}{11^2} - \cdots..\right]$$
$$-\frac{\pi}{4\sqrt{2}} = -\frac{2}{\pi}\left[\frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \frac{1}{11^2} - \cdots..\right]$$
$$\frac{\pi^2}{8\sqrt{2}} = \frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \cdots\ldots\ldots$$

7. Obtain Fourier expansion of $f(x) = |cosx|$ in $(-\pi, \pi)$
[M15/AutoMechCivil/8M]
Solution:
We see that, $f(x) = |cosx|$ is an even function.

$$a_0 = \frac{1}{l}\int_0^{l} f(x)dx$$

$$=\frac{1}{\pi}\left\{\int_0^{\frac{\pi}{2}} cosxdx + \int_{\frac{\pi}{2}}^{\pi} -cosxdx\right\}$$

$$=\frac{1}{\pi}\{[sinx]_0^{\frac{\pi}{2}} - [sinx]_{\frac{\pi}{2}}^{\pi}\} \ = \frac{1}{\pi}[(1-0)-(0-1)]$$

$$\therefore a_0 = \frac{2}{\pi}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$a_n = \frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} cosxcosnx\ dx + \int_{\frac{\pi}{2}}^{\pi} -cosx\ cosnxdx\right\}$$

$$=\frac{1}{\pi}\{\int_0^{\frac{\pi}{2}} 2cosx\ cosnx\ dx - \int_{\frac{\pi}{2}}^{\pi} 2cosx\ cosnx$$

$$=\frac{1}{\pi}\left\{\int_0^{\frac{\pi}{2}} \cos(1+n)\,x + \cos(1-n)\,x\,dx - \int_{\frac{\pi}{2}}^{\pi} \cos(1+n)\,x + \cos(1-n)\,x\,dx\right\}$$

$$=\frac{1}{\pi}\left\{\left[\frac{\sin(1+n)x}{1+n} + \frac{\sin(1-n)x}{1-n}\right]_0^{\frac{\pi}{2}} - \left[\frac{\sin(1+n)x}{1+n} + \frac{\sin(1-n)x}{1-n}\right]_{\frac{\pi}{2}}^{\pi}\right\}$$

$$=\frac{1}{\pi}\left[\frac{\cos\frac{n\pi}{2}}{1+n} + \frac{\cos\frac{n\pi}{2}}{1-n} + \frac{\cos\frac{n\pi}{2}}{1+n} + \frac{\cos\frac{n\pi}{2}}{1-n}\right] \qquad \left[\because \sin(1\pm n)\frac{\pi}{2} = \sin\left(\frac{\pi}{2}\pm\frac{n\pi}{2}\right) = \cos\frac{n\pi}{2}\right]$$

$$=\frac{2\cos\frac{n\pi}{2}}{\pi}\left[\frac{1}{1+n} + \frac{1}{1-n}\right] = \frac{2\cos\frac{n\pi}{2}}{\pi}\left[\frac{2}{1-n^2}\right]$$

$$\therefore a_n = \frac{4\cos\frac{n\pi}{2}}{\pi(1-n^2)}, n \neq 1$$

$$a_1 = \frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} \cos^2 x\ \ dx + \int_{\frac{\pi}{2}}^{\pi} -\cos^2 x\,dx\right\}$$

$$=\frac{2}{\pi}\left\{\int_0^{\frac{\pi}{2}} \frac{1+cos\,2x}{2}\ dx - \int_{\frac{\pi}{2}}^{\pi} \frac{1+cos\,2x}{2}\,dx\right\}$$

$$=\frac{1}{\pi}\left\{\left[x + \frac{sin\,2x}{2}\right]_0^{\frac{\pi}{2}} - \left[x + \frac{sin\,2x}{2}\right]_{\frac{\pi}{2}}^{\pi}\right\} = \frac{1}{\pi}\left[\frac{\pi}{2} - \pi + \frac{\pi}{2}\right]$$

$$\therefore a_1 = 0$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$

$$f(x) = a_0 + a_1 cosx + \sum_2^{\infty} a_n \cos nx$$

$$|cosx| = \frac{2}{\pi} - \frac{4}{\pi}\sum_2^{\infty}\frac{\cos\left(\frac{n\pi}{2}\right)}{n^2-1}cosnx$$

8. Obtain half range cosine series for $cosax$ in $(0,\pi)$ where a is not an integer and hence show that $\sum_1^{\infty}\frac{1}{a^2-n^2} = \frac{a\pi cota\pi}{1+2a^2}$

[N17/Elect/8M][M19/Inst/8M]

Solution:

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$

$$=\frac{1}{\pi}\int_0^{\pi} cosaxdx$$

$$= \frac{1}{\pi}\left[\frac{sinax}{a}\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\frac{sina\pi}{a}\right]$$
$$\therefore a_0 = \frac{sina\pi}{a\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$a_n = \frac{2}{\pi}\int_0^{\pi} cosaxcosnx\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} 2cosax\,cosnx\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} \cos(a+n)x + \cos(a-n)x\,dx$$
$$= \frac{1}{\pi}\left[\frac{\sin(a+n)x}{a+n} + \frac{\sin(a-n)x}{a-n}\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\frac{\sin(a+n)\pi}{a+n} + \frac{\sin(a-n)\pi}{a-n}\right] \quad [\because \sin(a\pm n)\pi = sina\pi cosn\pi \pm cosa\pi sinn\pi = (-1)^n sina\pi]$$
$$= \frac{1}{\pi}\left[\frac{(-1)^n sina\pi}{a+n} + \frac{(-1)^n sina\pi}{a-n}\right]$$
$$= \frac{(-1)^n sina\pi}{\pi}\left[\frac{1}{a+n} + \frac{1}{a-n}\right]$$
$$= \frac{(-1)^n sina\pi}{\pi}\left[\frac{2a}{a^2-n^2}\right]$$
$$\therefore a_n = \frac{2a(-1)^n sina\pi}{\pi(a^2-n^2)}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n cosnx$$
$$cosax = \frac{sina\pi}{a\pi} + \sum\frac{2a(-1)^n sina\pi}{\pi(a^2-n^2)}cosnx$$

Putting $x = \pi$, we get

$$cosa\pi = \frac{sina\pi}{a\pi} + \frac{2asina\pi}{\pi}\sum\frac{(-1)^n}{a^2-n^2}(-1)^n$$
$$cosa\pi = sina\pi\left[\frac{1}{a\pi} + \frac{2a}{\pi}\sum\frac{1}{a^2-n^2}\right]$$
$$cota\pi = \frac{1}{a\pi} + \frac{2a}{\pi}\sum\frac{1}{a^2-n^2}$$
$$a\pi cota\pi = 1 + 2a^2\sum\frac{1}{a^2-n^2}$$

9. Find Fourier series for $f(x) = cosmx$ in $(-\pi,\pi)$ where m is not an integer. Deduce that $cotm\pi = \frac{2m}{\pi}\left[\frac{1}{2m^2} + \frac{1}{m^2-1^2} + \frac{1}{m^2-2^2} + \cdots\frac{1}{m^2-n^2}\right]$ hence show that $\sum_1^{\infty}\frac{1}{9n^2-1} = \frac{1}{2} - \frac{\pi\sqrt{3}}{18}$

[N14/AutoMechCivil/8M]

Solution:

We see that, $f(x) = \cos mx$ is an even function.

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^{\pi} cosmxdx$$

$$= \frac{1}{\pi}\left[\frac{\sin mx}{m}\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\frac{\sin m\pi}{m}\right]$$
$$\therefore a_0 = \frac{\sin m\pi}{m\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$
$$a_n = \frac{2}{\pi}\int_0^{\pi} \cos mx \cos nx\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} 2\cos mx\cos nx\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} \cos(m+n)x + \cos(m-n)x\,dx$$
$$= \frac{1}{\pi}\left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n}\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\frac{\sin(m+n)\pi}{m+n} + \frac{\sin(m-n)\pi}{m-n}\right] \quad [\because \sin(m\pm n)\pi = \sin m\pi\cos n\pi \pm \cos m\pi \sin n\pi = (-1)^n \sin m\pi]$$
$$= \frac{1}{\pi}\left[\frac{(-1)^n\sin m\pi}{m+n} + \frac{(-1)^n\sin m\pi}{m-n}\right]$$
$$= \frac{(-1)^n\sin m\pi}{\pi}\left[\frac{1}{m+n} + \frac{1}{m-n}\right]$$
$$= \frac{(-1)^n\sin m\pi}{\pi}\left[\frac{2m}{m^2-n^2}\right]$$
$$\therefore a_n = \frac{2m(-1)^n\sin m\pi}{\pi(m^2-n^2)}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos nx$$
$$\cos mx = \frac{\sin m\pi}{m\pi} + \sum_1^{\infty}\frac{2m(-1)^n\sin m\pi}{\pi(m^2-n^2)}\cos nx$$
$$\cos mx = \frac{\sin m\pi}{m\pi} + \frac{2m\sin m\pi}{\pi}\left[-\frac{\cos x}{m^2-1} + \frac{\cos 2x}{m^2-4} - \frac{\cos 3x}{m^2-9} + \cdots..\right]$$
$$\cos mx = \frac{2m\sin m\pi}{\pi}\left[\frac{1}{2m^2} - \frac{\cos x}{m^2-1^2} + \frac{\cos 2x}{m^2-2^2} - \frac{\cos 3x}{m^2-3^2} + \cdots.\right]$$

Putting $x = \pi$,

$$\cos m\pi = \frac{2m\sin m\pi}{\pi}\left[\frac{1}{2m^2} + \frac{1}{m^2-1^2} + \frac{1}{m^2-2^2} + \frac{1}{m^2-3^2} + \cdots..\right]$$
$$\frac{\cos m\pi}{\sin m\pi} = \frac{2m}{\pi}\left[\frac{1}{2m^2} + \frac{1}{m^2-1^2} + \frac{1}{m^2-2^2} + \frac{1}{m^2-3^2} + \cdots..\right]$$
$$\cot m\pi = \frac{2m}{\pi}\left[\frac{1}{2m^2} + \frac{1}{m^2-1^2} + \frac{1}{m^2-2^2} + \cdots\frac{1}{m^2-n^2}\right]$$

Now,

$$\cos mx = \frac{\sin m\pi}{m\pi} + \sum_1^{\infty}\frac{2m(-1)^n\sin m\pi}{\pi(m^2-n^2)}\cos nx$$

Putting $m = \frac{1}{3}$ and $x = \pi$,

$$\cos\frac{\pi}{3} = \frac{\frac{\sin\pi}{3}}{\frac{\pi}{3}} + \frac{2}{3\pi}\sum_1^{\infty}\frac{(-1)^n\sin\frac{\pi}{3}}{\frac{1}{9}-n^2}(-1)^n$$
$$\frac{1}{2} = \frac{\frac{\sqrt{3}}{2}}{\frac{\pi}{3}} + \frac{2}{3\pi}.\frac{\sqrt{3}}{2}\sum_1^{\infty}\frac{1}{\frac{1}{9}-n^2}$$

$\frac{1}{2}=\frac{3\sqrt{3}}{2\pi}+\frac{\sqrt{3}}{3\pi}\sum_1^\infty \frac{9}{1-9n^2}$
$\frac{1}{2}-\frac{3\sqrt{3}}{2\pi}=\frac{3\sqrt{3}}{\pi}\sum_1^\infty \frac{1}{1-9n^2}$
$\frac{\pi}{6\sqrt{3}}-\frac{1}{2}=\sum_1^\infty \frac{1}{1-9n^2}$
$\frac{1}{2}-\frac{\pi\sqrt{3}}{18}=\sum_1^\infty \frac{1}{9n^2-1}$

10. Find the Fourier series for $f(x)=\begin{cases} cosx & -\pi<x<0 \\ sinx & 0<x<\pi \end{cases}$

[M16/AutoMechCivil/8M]

Solution:

We have,

$a_0=\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx$
$=\frac{1}{2\pi}\left\{\int_{-\pi}^{0} cosx\,dx+\int_0^\pi sinxdx\right\}$
$=\frac{1}{2\pi}\{[sinx]_{-\pi}^0+[-cosx]_0^\pi\}$
$=\frac{1}{2\pi}\{-cos\pi+cos0\}$
$\therefore a_0=\frac{1}{\pi}$

$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos\frac{n\pi x}{l}dx$
$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} cosx\,cosnxdx+\int_0^\pi sinx\,cosnxdx\right\}$
$=\frac{1}{2\pi}\left\{\int_{-\pi}^{0} 2\,cosx\,cosnxdx+\int_0^\pi 2\,sinx\,cosnxdx\right\}$
$=\frac{1}{2\pi}\left\{\int_{-\pi}^{0} \cos(1+n)x+\cos(1-n)xdx+\int_0^\pi \sin(1+n)x+\sin(1-n)x\,dx\right\}$
$=\frac{1}{2\pi}\left\{\left[\frac{\sin(1+n)x}{1+n}+\frac{\sin(1-n)x}{1-n}\right]_{-\pi}^0+\left[-\frac{\cos(1+n)x}{1+n}-\frac{\cos(1-n)x}{1-n}\right]_0^\pi\right\}$
$=\frac{1}{2\pi}\left[\frac{(-1)^n}{1+n}+\frac{(-1)^n}{1-n}+\frac{1}{1+n}+\frac{1}{1-n}\right]$
$=\frac{[(-1)^n+1]}{2\pi}\left[\frac{1}{1+n}+\frac{1}{1-n}\right]$
$=\frac{[(-1)^n+1]}{2\pi}\left[\frac{2}{1-n^2}\right]$
$\therefore a_n=\frac{[(-1)^n+1]}{\pi(1-n^2)}, n\neq 1$

$a_1=\frac{1}{\pi}\left\{\int_{-\pi}^{0}\cos^2 x\,dx+\int_0^\pi sinx\,cosxdx\right\}$
$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}\frac{1+cos\,2x}{2}dx+\int_0^\pi \frac{sin\,2x}{2}dx\right\}$
$=\frac{1}{2\pi}\left\{\left[x+\frac{sin\,2x}{2}\right]_{-\pi}^0+\left[-\frac{cos\,2x}{2}\right]_0^\pi\right\}$
$=\frac{1}{2\pi}\left[\pi-\frac{1}{2}+\frac{1}{2}\right]$
$\therefore a_1=\frac{1}{2}$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin\frac{n\pi x}{l}dx$$
$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} cosx\ sinnxdx + \int_{0}^{\pi} sinx\ sinnxdx\right\}$$
$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0} 2\ cosx\ sinnxdx + \int_{0}^{\pi} 2\ sinx\ sinnxdx\right\}$$
$$= \frac{1}{2\pi}\left\{\int_{-\pi}^{0} \sin(1+n)x - \sin(1-n)x dx + \int_{0}^{\pi} \cos(1-n)x - \cos(1+n)x\ dx\right\}$$
$$= \frac{1}{2\pi}\left\{\left[-\frac{\cos(1+n)x}{1+n} + \frac{\cos(1-n)x}{1-n}\right]_{-\pi}^{0} + \left[\frac{\sin(1-n)x}{1-n} - \frac{\sin(1+n)x}{1+n}\right]_{0}^{\pi}\right\}$$
$$= \frac{1}{2\pi}\left[-\frac{1}{1+n} + \frac{1}{1-n} - \frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n}\right]$$
$$= \frac{[(-1)^n+1]}{2\pi}\left[-\frac{1}{1+n} + \frac{1}{1-n}\right]$$
$$= \frac{[(-1)^n+1]}{2\pi}\left[\frac{2n}{1-n^2}\right]$$
$$\therefore b_n = \frac{n[(-1)^n+1]}{\pi(1-n^2)}, n \neq 1$$
$$b_1 = \frac{1}{\pi}\left\{\int_{-\pi}^{0} cosx\ sinx\ dx + \int_{0}^{\pi} \sin^2 x\ dx\right\}$$
$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} \frac{sin\,2x}{2}dx + \int_{0}^{\pi} \frac{1-cos\,2x}{2}dx\right\}$$
$$= \frac{1}{2\pi}\left\{\left[-\frac{cos\,2x}{2}\right]_{-\pi}^{0} + \left[x - \frac{sin\,2x}{2}\right]_{0}^{\pi}\right\}$$
$$= \frac{1}{2\pi}\left[-\frac{1}{2} + \frac{1}{2} + \pi\right]$$
$$\therefore b_1 = \frac{1}{2}$$

Thus, the fourier series is given by,

$$f(x) = a_0 + \sum a_n\cos\frac{n\pi x}{l} + \sum b_n\sin\frac{n\pi x}{l}$$
$$f(x) = a_0 + a_1 cosx + \sum_2^\infty a_n cosnx + b_1 sinx + \sum_2^\infty b_n sinnx$$
$$f(x) = \frac{1}{\pi} + \frac{cosx}{2} + \sum_2^\infty \frac{[(-1)^n+1]}{\pi(1-n^2)} cosnx + \frac{sinx}{2} + \sum_2^\infty \frac{n[(-1)^n+1]}{\pi(1-n^2)} sinnx$$

11. Find half range cosine series for $f(x) = xsinx$ in $(0,\pi)$ and hence show that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \cdots \ldots. = \frac{\pi-2}{4}$

[N18/Comp/8M]

Solution:

We have,

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\left\{\int_0^\pi x\ sinxdx\right\}$$
$$= \frac{1}{\pi}\{[x\ (-cosx) - (1)(-sinx)]_0^\pi\}$$
$$= -\frac{1}{\pi}[-\pi]$$
$$\therefore a_0 = 1$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$

$$= \frac{2}{\pi}\left\{\int_0^{\pi} x\ sinx \cos nx\ \ dx\right\}$$
$$= \frac{2}{2\pi}\int_0^{\pi} x.2sinx \cos nx\ dx$$
$$= \frac{1}{\pi}\int_0^{\pi} x[\sin(1+n)x + \sin(1-n)\ x]\ dx$$
$$= \frac{1}{\pi}\left[x\left(-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n}\right) - (1)\left(-\frac{\sin(1+n)x}{(1+n)^2} - \frac{\sin(1-n)x}{(1-n)^2}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\pi\left(-\frac{\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n}\right)\right]$$
$$= \left[\frac{(-1)^n}{1+n} + \frac{(-1)^n}{1-n}\right]$$
$$= \frac{2(-1)^n}{1-n^2}$$
$$\therefore a_n = \frac{-2(-1)^n}{(n^2-1)}, n \neq 1$$
$$a_1 = \frac{2}{\pi}\int_0^{\pi} x\ sinx\ cosx\ dx$$
$$= \frac{2}{2\pi}\int_0^{\pi} x\ sin2x\ dx$$
$$= \frac{1}{\pi}\left[x\ \left(-\frac{cos\,2x}{2}\right) - (1)\left(-\frac{sin\,2x}{4}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[-\pi\frac{cos\,2\pi}{2}\right]$$
$$\therefore a_1 = -\frac{1}{2}$$

Thus, the fourier series is given by

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$f(x) = a_0 + a_1 cosx + \sum_2^{\infty} a_n cosnx$$
$$xsinx = 1 - \frac{cosx}{2} - 2\sum_2^{\infty}\frac{(-1)^n\,cosnx}{n^2-1}$$
$$xsinx = 1 - \frac{cosx}{2} - 2\left[\frac{cos\,2x}{3} - \frac{cos\,3x}{8} + \frac{cos\,4x}{15} - \frac{cos\,5x}{24} + \cdots.\right]$$

Putting, $x = \frac{\pi}{2}$

$$\frac{\pi}{2} = 1 - 0 - 2\left[\frac{cos\pi}{3} - 0 + \frac{cos\,2\pi}{15} - 0 + \cdots.\right]$$
$$\frac{\pi}{2} - 1 = -2\left[-\frac{1}{1\cdot3} + \frac{1}{3\cdot5} - \frac{1}{5\cdot7} + \cdots..\right]$$
$$\frac{\pi-2}{2} = 2\left[\frac{1}{1\cdot3} - \frac{1}{3\cdot5} + \frac{1}{5\cdot7} - \cdots\ldots\right]$$
$$\frac{\pi-2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \cdots \ldots.$$

12. Obtain Fourier series of $xcosx\ in\ (-\pi,\pi)$
[M15/Biot/6M][N15/AutoMechCivil/8M][N19/Elex/8M]
Solution:
We have, $f(x) = xcosx$ is an odd function

$$b_n = \frac{2}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx$$
$$= \frac{2}{\pi}\left\{\int_0^{\pi} x\ cosx \sin nx\ \ dx\right\}$$

$$= \frac{1}{\pi}\int_0^{\pi} x(2cosx\,\sin nx)dx$$
$$= \frac{1}{\pi}\int_0^{\pi} x(\sin(1+n)x - \sin(1-n)\,x)\,dx$$
$$= \frac{1}{\pi}\left[x\left(-\frac{\cos(1+n)x}{1+n} + \frac{\cos(1-n)x}{1-n}\right) - (1)\left(-\frac{\sin(1+n)x}{(1+n)^2} + \frac{\sin(1-n)x}{(1-n)^2}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[\pi\left(-\frac{\cos(1+n)\pi}{1+n} + \frac{\cos(1-n)\pi}{1-n}\right)\right]$$
$$= \left[\frac{(-1)^n}{1+n} - \frac{(-1)^n}{1-n}\right]$$
$$= (-1)^n\left[\frac{1}{1+n} - \frac{1}{1-n}\right] = -\frac{2n(-1)^n}{1-n^2}, n \neq 1$$
$$b_1 = \frac{2}{\pi}\int_0^{\pi} x\,cosx\,sinx\,dx$$
$$= \frac{1}{\pi}\int_0^{\pi} xsin2x\,dx$$
$$= \frac{1}{\pi}\left[x\left(-\frac{cos\,2x}{2}\right) - (1)\left(-\frac{sin\,2x}{4}\right)\right]_0^{\pi}$$
$$= \frac{1}{\pi}\left[-\frac{\pi}{2}\right]$$
$$\therefore b_1 = -\frac{1}{2}$$

Thus, the fourier series is given by

$$f(x) = \sum b_n \sin\frac{n\pi x}{l}$$
$$f(x) = b_1 sinx + \sum_2^{\infty} b_n sinnx$$
$$xcosx = -\frac{1}{2}sinx + 2\sum_2^{\infty} n.\frac{(-1)^n}{n^2-1}\,sinnx$$

Type V: Exponential functions (e^{ax}, e^{-ax}, etc)

1. State Dirichlet's condition for a function $f(x)$ to have a Fourier series over (a,b). Obtain the Fourier series for $f(x) = e^{-x}$ over $0 < x < 2\pi$ and $f(x+2\pi) = f(x)$

Ans. $e^{-x} = \frac{1-e^{-2\pi}}{2\pi} + \frac{1-e^{-2\pi}}{\pi}\sum\frac{cosnx}{1+n^2} + \frac{1-e^{-2\pi}}{\pi}\sum\frac{n.sinnx}{1+n^2}$

2. Find half range cosine series for $f(x) = e^x$ in $0 < x < 1$
[N13/AutoMechCivil/6M][M15/AutoMechCivil/6M][N16/CompIT/5M]
Solution:

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{1}\int_0^1 e^x dx$$
$$= [e^x]_0^1$$
$$\therefore a_0 = (e-1)$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$= \frac{2}{1}\int_0^1 e^x \cos n\pi x \ dx$$
$$= 2\left[\frac{e^x}{1^2+n^2\pi^2}(1.\cos n\pi x + n\pi.\sin n\pi x)\right]_0^1$$
$$= 2\left[\frac{e}{1+n^2\pi^2}(-1)^n - \frac{1}{1+n^2\pi^2}(1)\right]$$
$$\therefore a_n = \frac{2[e(-1)^n-1]}{n^2\pi^2}$$

Thus,

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$
$$e^x = (e-1) + 2\sum\frac{[e(-1)^n-1]}{1+n^2\pi^2}\cos(n\pi x)$$

3. Find the Fourier series for $f(x) = e^{-|x|}$ in the interval $(-\pi, \pi)$
[M19/Comp/5M]
Solution:
We see that, $f(x) = e^{-|x|}$ is an even function

$$a_0 = \frac{1}{l}\int_0^l f(x)dx$$
$$= \frac{1}{\pi}\int_0^\pi e^{-x}dx$$
$$= \frac{1}{\pi}\left[\frac{e^{-x}}{-1}\right]_0^\pi$$
$$\therefore a_0 = \frac{1-e^{-\pi}}{\pi}$$
$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$$
$$= \frac{2}{\pi}\int_0^\pi e^{-x}\cos nx \ dx$$
$$= \frac{2}{\pi}\left[\frac{e^{-x}}{(-1)^2+n^2}(-1.\cos nx + n.\sin nx)\right]_0^\pi$$

$$= \frac{2}{\pi}\left[\frac{e^{-\pi}}{1+n^2}(-1)(-1)^n - \frac{e^0}{1+n^2}(-1)\right]$$

$$\therefore a_n = \frac{2[1-(-1)^n e^{-\pi}]}{\pi(1+n^2)}$$

Thus,

$$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l}$$

$$e^{-|x|} = \frac{1-e^{-\pi}}{\pi} + \sum \frac{2[1-(-1)^n e^{-\pi}]}{\pi(1+n^2)}\cos(nx)$$

4. Prove that in the interval $0 < x < \pi$,
$\frac{e^{ax}-e^{-ax}}{e^{a\pi}-e^{-a\pi}} = \frac{2}{\pi}\left[\frac{sinx}{a^2+1} - \frac{2sin\,2x}{a^2+4} + \frac{3sin\,3x}{a^2+9} - ..\right]$
5. Show that $e^x = 2\pi\left[\frac{1+e}{1+\pi^2}.sin\pi x + \frac{2(1+e)}{1+4\pi^2}.sin2\pi x + \frac{3(1+e)}{1+9\pi^2}.sin3\pi x + \cdots\right]$
6. Find the Fourier series of the function $f(x) = e^{-x}, 0 < x < 2\pi$ and $f(x+2\pi) = f(x)$. Hence deduce that the value of $\sum_{n=2}^{\infty}\frac{(-1)^n}{n^2+1}$

[M15/ElexExtcElectBiomInst/8M]

Solution:

We have,

$$a_0 = \frac{1}{2l}\int_0^{2l} f(x)dx$$

$$= \frac{1}{2\pi}\left\{\int_0^{2\pi} e^{-x}dx\right\}$$

$$= \frac{1}{2\pi}\left\{\left[\frac{e^{-x}}{-1}\right]_0^{2\pi}\right\}$$

$$= \frac{1}{2\pi}\{-e^{-2\pi} + e^0\}$$

$$\therefore a_0 = \frac{1-e^{-2\pi}}{2\pi}$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}dx$$

$$= \frac{1}{\pi}\left\{\int_0^{2\pi} e^{-x}\,cosnxdx\right\}$$

$$= \frac{1}{\pi}\left[\frac{e^{-x}}{(-1)^2+n^2}(-1.cosnx + n.sinnx)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{e^{-2\pi}}{1+n^2}(-1) - \frac{e^0}{1+n^2}(-1)\right]$$

$$\therefore a_n = \frac{1-e^{-2\pi}}{\pi(1+n^2)}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}dx$$

$$= \frac{1}{\pi}\left\{\int_0^{2\pi} e^{-x}\,sinnxdx\right\}$$

$$= \frac{1}{\pi}\left[\frac{e^{-x}}{(-1)^2+n^2}(-1.sinnx - n.cosnx)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{e^{-2\pi}}{1+n^2}(-n) - \frac{e^0}{1+n^2}(-n)\right]$$

$\therefore b_n = \frac{n(1-e^{-2\pi})}{\pi(1+n^2)}$

Thus, the fourier series is given by,

$f(x) = a_0 + \sum a_n \cos\frac{n\pi x}{l} + \sum b_n \sin\frac{n\pi x}{l}$

$e^{-x} = \frac{1-e^{-2\pi}}{2\pi} + \sum_1^\infty \frac{1-e^{-2\pi}}{\pi(1+n^2)}\ cosnx + \sum_1^\infty \frac{n(1-e^{-2\pi})}{\pi(1+n^2)}\ sinnx$

$\frac{e^{-x}}{1-e^{-2\pi}} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{cosnx}{1+n^2} + \frac{1}{\pi}\sum_1^\infty \frac{nsinnx}{1+n^2}$

Putting $x = \pi$

$\frac{e^{-\pi}}{1-e^{-2\pi}} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{(-1)^n}{1+n^2}$

$\frac{1}{e^{\pi}(1-e^{-2\pi})} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{(-1)^n}{1+n^2}$

$\frac{1}{(e^{\pi}-e^{-\pi})} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{(-1)^n}{1+n^2}$

$\frac{1}{2sin\,h\pi} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{(-1)^n}{1+n^2}$

$\frac{cosec\ h\pi}{2} = \frac{1}{2\pi} + \frac{1}{\pi}\sum_1^\infty \frac{(-1)^n}{1+n^2}$

$\frac{cosec\ h\pi}{2} = \frac{1}{2\pi} - \frac{1}{2\pi} + \frac{1}{\pi}\sum_2^\infty \frac{(-1)^n}{1+n^2}$

$\frac{cosec\ h\pi}{2} = \frac{1}{\pi}\sum_2^\infty \frac{(-1)^n}{1+n^2}$

$\frac{\pi}{2}\ cosech\pi = \sum_2^\infty \frac{(-1)^n}{1+n^2}$

7. Find the Fourier series of the function $f(x) = e^x\ in\ (0,2\pi)$

Ans.$e^x = \frac{e^{2\pi}-1}{2\pi} + \frac{e^{2\pi}-1}{\pi}\sum\frac{cosnx}{1+n^2} - \frac{e^{2\pi}-1}{\pi}\sum\frac{n.sinnx}{1+n^2}$

8. Obtain the fourier series for $f(x) = e^x$ in $(-\pi, \pi)$. Hence derive the series for $\frac{\pi}{sin\,h\pi}$

Ans.$e^x = \frac{sin\,h\pi}{\pi} + \frac{2sinh\pi}{\pi}\sum\frac{(-1)^n}{1+n^2}cosnx - \frac{2sin\,h\pi}{\pi}\sum\frac{n(-1)^n}{1+n^2}sinnx$

Type VI: Fourier Integral theorem (Not for AutoMechCivilComp)

1. Express the function $f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$ as a fourier integral and hence evaluate $\int_0^\infty \frac{sin\omega cos\omega x}{\omega} d\omega$

[M17/CompIT/8M][N17/Elex/8M][M18/Elex/8M][N18/Elex/6M] [N19/Elex/8M]

Solution:

$$f(x) = \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty f(s)\cos w(s-x)\ dw\ ds$$

$$= \frac{1}{\pi}\int_0^\infty \int_{-1}^1 1\ \cos w(s-x)\ dw\ ds$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{sinw\,(s-x)}{w}\right]_{-1}^1 dw$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin(w-wx)}{w} - \frac{\sin(-w-wx)}{w}\right] dw$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin(w-wx)+\sin(w+wx)}{w}\right] dw$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{2\sin w\cdot coswx}{w}\right] dw$$

$$= \frac{2}{\pi}\int_0^\infty \left[\frac{sinwcoswx}{w}\right] dw$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{sinwcoswx}{w} dw$$

$$\therefore \int_0^\infty \frac{sinwcoswx}{w} dw = \frac{\pi}{2} f(x)$$

$$\therefore \int_0^\infty \frac{sinwcoswx}{w} dw = \begin{cases} \frac{\pi}{2} & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

2. Find Fourier cosine integral for $f(x) = \begin{cases} 1 - x^2 & 0 \le x \le 1 \\ 0 & x > 1 \end{cases}$.

Hence evaluate $\int_0^\infty \left(\frac{xcosx - sinx}{x^3}\right).\cos\frac{x}{2} dx$

[M14/Chem/6M][N14/CompIT/8M][N15/Biot/8M][M18/Inst/8M]

Solution:

By Fourier Cosine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s)\cos ws\, dw\ ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \int_0^1 (1-s^2)\cos ws\ \ dw\ ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx\ \left[(1-s^2)\left(\frac{sinws}{w}\right) - (-2s)\left(-\frac{cosws}{w^2}\right) + (-2)\left(-\frac{sinws}{w^3}\right)\right]_0^1 dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[-\frac{2cosw}{w^2} + \frac{2sinw}{w^3}\right] dw$$

$$= \frac{4}{\pi}\int_0^\infty \cos wx \left[\frac{sinw - wcosw}{w^3}\right] dw$$

$$\therefore f(x) = \frac{4}{\pi}\int_0^\infty \cos wx \left[\frac{sinw - wcosw}{w^3}\right] dw$$

Putting $x = \frac{1}{2}$,

$$f\left(\frac{1}{2}\right) = \frac{4}{\pi}\int_0^\infty \cos\frac{w}{2}\left[\frac{sinw - wcosw}{w^3}\right]dw$$

We see that, at $x = \frac{1}{2}, f(x) = 1 - x^2$

$$\therefore f\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

Thus,

$$\frac{3}{4} = \frac{4}{\pi}\int_0^\infty \left[\frac{sinw - wcosw}{w^3}\right]\cos\frac{w}{2}dw$$

$$-\frac{\pi}{3} = \int_0^\infty \left[\frac{wcosw - sinw}{w^3}\right]\cos\frac{w}{2}\ dw$$

Also,

$$\int_0^\infty \left(\frac{xcosx - sinx}{x^3}\right).\cos\frac{x}{2}dx = -\frac{\pi}{3}$$

3. Express the function $f(x) = \begin{cases} \frac{\pi}{2} & 0 \le x \le \pi \\ 0 & x > \pi \end{cases}$ as fourier integral and show that $\int_0^\infty \left(\frac{1-cos\pi\omega}{\omega}\right).\sin\omega x\, d\omega = \frac{\pi}{2}; 0 < x < \pi$

[N15/Biot/6M]

Solution:

By Fourier Sine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin wx \int_0^\infty f(s)\sin ws\, dw\, ds$$

$$= \frac{2}{\pi}\int_0^\infty \sin wx \int_0^\pi \frac{\pi}{2}\ sinws\ ds\, dw$$

$$= \int_0^\infty \sin wx \int_0^\pi\ sinws\ ds\, dw$$

$$= \int_0^\infty \sin wx\left[-\frac{cosws}{w}\right]_0^\pi dw$$

$$= \int_0^\infty \sin wx\left[\frac{-cos\pi w + cos\,0}{w}\right]dw$$

$$= \int_0^\infty \sin wx\left[\frac{1 - cos\pi w}{w}\right]dw$$

$$f(x) = \int_0^\infty \sin wx\left[\frac{1 - cos\pi w}{w}\right]dw$$

Thus,

$$\int_0^\infty \sin wx\left[\frac{1 - cos\pi w}{w}\right]dw = f(x) = \frac{\pi}{2}; 0 < x < \pi$$

4. Express $f(x) = 1$ when $-1 < x < 1$ and $f(x) = 0$ when $|x| > 1$ as a fourier integral.

[N16/Chem/5M]

Solution:

$$f(x) = \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty f(s)\cos w(s-x)\ dw\, ds$$

$$= \frac{1}{\pi}\int_0^\infty \int_{-1}^1 1\ \cos w(s-x)\ dw\, ds$$

$$= \frac{1}{\pi}\int_0^\infty \left[\frac{sinw\,(s-x)}{w}\right]_{-1}^{1} dw$$
$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin(w-wx)}{w} - \frac{\sin(-w-wx)}{w}\right] dw$$
$$= \frac{1}{\pi}\int_0^\infty \left[\frac{\sin(w-wx)+\sin(w+wx)}{w}\right] dw$$
$$= \frac{1}{\pi}\int_0^\infty \left[\frac{2\sin w \cdot coswx}{w}\right] dw$$
$$= \frac{2}{\pi}\int_0^\infty \left[\frac{sinwcoswx}{w}\right] dw$$
$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{sinwcoswx}{w} dw$$

5. Find Fourier integral representation of $f(x) = 1 - x^2, -1 \le x \le 1$ and is zero otherwise.
[M16/Chem/7M][N16/Chem/6M][N18/Extc/6M][N18/Elect/6M] [N19/Extc/8M]
Solution:
We see that, $f(x) = 1 - x^2$ is an even function
By Fourier Cosine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s)\cos ws\, dw\, ds$$
$$= \frac{2}{\pi}\int_0^\infty \cos wx \int_0^1 (1-s^2)\cos ws\ dw\, ds$$
$$= \frac{2}{\pi}\int_0^\infty \cos wx\ \left[(1-s^2)\left(\frac{sinws}{w}\right) - (-2s)\left(-\frac{cosws}{w^2}\right) + (-2)\left(-\frac{sinws}{w^3}\right)\right]_0^1 dw$$
$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[-\frac{2cosw}{w^2} + \frac{2sinw}{w^3}\right] dw$$
$$= \frac{4}{\pi}\int_0^\infty \cos wx \left[\frac{sinw - wcosw}{w^3}\right] dw$$
$$\therefore f(x) = \frac{4}{\pi}\int_0^\infty \cos wx \left[\frac{sinw - wcosw}{w^3}\right] dw$$

6. Find Fourier sine integral of $f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & x > 2 \end{cases}$
[M15/Chem/6M][N16/CompIT/8M]
Solution:
By Fourier Sine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin wx \int_0^\infty f(s)\sin ws\, dw\, ds$$
$$= \frac{2}{\pi}\int_0^\infty \sin wx \left\{\int_0^1 s\sin ws\ ds + \int_1^2 (2-s)sinws\, ds\right\} dw$$
$$= \frac{2}{\pi}\int_0^\infty \sin wx \left\{\left[s\left(\frac{-cosws}{w}\right) - (1)\left(\frac{-sinws}{w^2}\right)\right]_0^1 + \left[(2-s)\left(\frac{-cosws}{w}\right) - (-1)\left(\frac{-sinws}{w^2}\right)\right]_1^2\right\} dw$$
$$= \frac{2}{\pi}\int_0^\infty \sin wx \left[-\frac{cosw}{w} + \frac{sinw}{w^2} - \frac{\sin 2w}{w^2} + \frac{cosw}{w} + \frac{sinw}{w^2}\right] dw$$
$$= \frac{2}{\pi}\int_0^\infty \sin wx \left[\frac{2sinw - \sin 2w}{w^2}\right] dw$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{2sinw - \sin 2w}{w^2} sinwxdw$$

7. Find Fourier cosine integral of $f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 < x < 2 \\ 0 & x > 2 \end{cases}$

[N14/Chem/6M][N18/Biom/8M]

Solution:

By Fourier Coine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s)\cos ws\, dw\, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left\{\int_0^1 s\cos ws\ \ ds + \int_1^2 (2-s)cosws\ ds\right\} dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left\{\left[s\left(\frac{sinws}{w}\right) - (1)\left(\frac{-cosws}{w^2}\right)\right]_0^1 + \left[(2-s)\left(\frac{sinws}{w}\right) - (-1)\left(\frac{-cosws}{w^2}\right)\right]_1^2\right\} dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[\frac{sinw}{w} + \frac{cosw}{w^2} - \frac{1}{w^2} - \frac{\cos 2w}{w^2} - \frac{sinw}{w} + \frac{cosw}{w^2}\right] dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[\frac{2cosw - 1 - \cos 2w}{w^2}\right] dw$$

$$f(x) = \frac{1}{\pi}\int_0^\infty \frac{2cosw - (1+\cos 2w)}{w^2} coswxdw$$

$$f(x) = \frac{1}{\pi}\int_0^\infty \frac{2cosw - 2\cos^2 w}{w^2} coswxdw$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{cosw - \cos^2 w}{w^2} coswxdw$$

8. Find Fourier cosine integral representation for $f(x) = e^{-ax}, x > 0$. Hence show that $\int_0^\infty \frac{cos\omega x}{1+\omega^2} d\omega = \frac{\pi}{2} e^{-x}, x \geq 0$

[M17/ElexExctElectBiomInst/8M][N18/Biot/6M][N18/Inst/6M]

Solution:

Let $f(x) = \frac{\pi}{2} e^{-ax}$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s)\cos ws\, dw\, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty e^{-as} \cos ws\ \ dw\, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx\ \left[\frac{e^{-as}}{(-a)^2 + w^2}(-a.cosws + w.sinws)\right]_0^\infty dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[-\frac{e^0}{a^2+w^2}(-a)\right] dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[\frac{a}{a^2+w^2}\right] dw$$

$$\therefore f(x) = \frac{2}{\pi}\int_0^\infty \frac{acoswx}{w^2+a^2} dw$$

$$e^{-ax} = \frac{2}{\pi}\int_0^\infty \frac{acoswx}{w^2+a^2} dw$$

Put $a = 1$,

$$e^{-x} = \frac{2}{\pi}\int_0^\infty \frac{coswx}{w^2+1} dw$$

$$\int_0^\infty \frac{coswx}{w^2+1} dw = \frac{\pi}{2} e^{-x}$$

9. Show that $\int_0^\infty \frac{cos\lambda x}{\lambda^2+1} d\lambda = \frac{\pi}{2} e^{-x}$, $x \geq 0$ by definition of Fourier cosine integral.
[M14/CompIT/6M]
Solution:
Let $f(x) = \frac{\pi}{2} e^{-x}$

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s) \cos ws \, dw \, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty \frac{\pi}{2} e^{-s} \cos ws \; dw \, ds$$

$$= \int_0^\infty \cos wx \left[\frac{e^{-s}}{(-1)^2+w^2}(-1.cosws + w.sinws)\right]_0^\infty dw$$

$$= \int_0^\infty \cos wx \left[-\frac{e^0}{1+w^2}(-1)\right] dw$$

$$= \int_0^\infty \cos wx \left[\frac{1}{1+w^2}\right] dw$$

$$\therefore f(x) = \int_0^\infty \frac{coswx}{w^2+1} dw$$

Since, $f(x) = \frac{\pi}{2} e^{-x}$, we get

$$\frac{\pi}{2} e^{-x} = \int_0^\infty \frac{coswx}{w^2+1} dw$$

10. Find Fourier Integral representation of $f(x) = e^{-|x|}, -\infty < x < \infty$
[M14/Biot/6M]
Solution:
We see that, $f(x) = e^{-|x|}$ is an even function
By Fourier Cosine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty f(s) \cos ws \, dw \, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \int_0^\infty e^{-s} \cos ws \; dw \, ds$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[\frac{e^{-s}}{(-1)^2+w^2}(-1.cosws + w.sinws)\right]_0^\infty dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[-\frac{e^0}{1+w^2}(-1)\right] dw$$

$$= \frac{2}{\pi}\int_0^\infty \cos wx \left[\frac{1}{1+w^2}\right] dw$$

$$\therefore f(x) = \frac{2}{\pi}\int_0^\infty \frac{coswx}{w^2+1} dw$$

11. Find Fourier integral representation of $f(x) = \begin{cases} e^{ax} & x \leq 0, a > 0 \\ e^{-ax} & x \geq 0, a > 0 \end{cases}$
Hence show that $\int_0^\infty \frac{cos\omega x}{\omega^2+a^2} d\omega = \frac{\pi}{2a} e^{-ax}$
[N14/Biot/6M][N17/Inst/8M][M18/Elect/8M][M19/Chem/6M]
Solution:
We have,

$$f(x)=\begin{cases} e^{ax} & x\le 0, a>0 \\ e^{-ax} & x\ge 0, a>0\end{cases}$$

Putting $x=-x$, we get

$$f(-x)\ =\begin{cases} e^{-a(-x)} & x\le 0, a>0 \\ e^{a(-x)} & x\ge 0, a>0\end{cases}$$

$$f(-x)\ =\begin{cases} e^{ax} & x\le 0, a>0 \\ e^{-ax} & x\ge 0, a>0\end{cases}$$

$$\therefore f(-x)=f(x)$$

Thus, $f(x)$ is an even function

By Fourier Cosine Integral,

$$f(x)=\frac{2}{\pi}\int_0^\infty \cos wx\int_0^\infty f(s)\cos ws\,dw\,ds$$

$$=\frac{2}{\pi}\int_0^\infty \cos wx\int_0^\infty e^{-as}\cos ws\ dw\,ds$$

$$=\frac{2}{\pi}\int_0^\infty \cos wx\ \left[\frac{e^{-as}}{(-a)^2+w^2}(-a.cosws+w.sinws)\right]_0^\infty dw$$

$$=\frac{2}{\pi}\int_0^\infty \cos wx\left[-\frac{e^0}{a^2+w^2}(-a)\right]dw$$

$$=\frac{2}{\pi}\int_0^\infty \cos wx\left[\frac{a}{a^2+w^2}\right]dw$$

$$\therefore f(x)=\frac{2a}{\pi}\int_0^\infty \frac{coswx}{w^2+a^2}dw$$

12. Find Fourier sine integral representation for $f(x)=\frac{e^{-ax}}{x}$

[N14/Biot/6M][N15/CompIT/8M][N16/ElexExtcElectBiomInst/8M]

Solution:

By Fourier Sine Integral,

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty f(s)\sin ws\,dw\,ds$$

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty \frac{e^{-as}}{s}\sin ws\,dw\,ds$$

Consider,

$I(a)=\int_0^\infty \frac{e^{-as}}{s}sinws\ ds$ ……………(1)

Diff w.r.t a by the rule of DUIS.

$$I'(a)=\int_0^\infty \frac{-e^{-as}\times s}{s}sinws\ ds$$

$$I'(a)=-\int_0^\infty e^{-as}sinws\ ds$$

$$I'(a)=-\left[\frac{e^{-as}}{(-a)^2+w^2}(-a\ sinws-w\ cosws)\right]_0^\infty$$

$$I'(a)=-\left[-\frac{e^0}{a^2+w^2}(-w)\right]$$

$$I'(a)=-\frac{w}{a^2+w^2}$$

Integrating w.r.t a,

$I(a)=-\tan^{-1}\left(\frac{a}{w}\right)+C$ ………….(2)

Put $a = \infty$ in eqn (1) & (2),

$I(\infty) = 0$ & $I(\infty) = -\tan^{-1}(\infty) + C = -\frac{\pi}{2} + C$

$\therefore 0 = -\frac{\pi}{2} + C,\ C = \frac{\pi}{2}$

Thus,

$I(a) = \frac{\pi}{2} - \tan^{-1}\left(\frac{a}{w}\right) = \tan^{-1}\left(\frac{w}{a}\right)$

Now,

$f(x) = \frac{2}{\pi}\int_0^\infty \sin wx \tan^{-1}\frac{w}{a}\, dw$

13. Prove that $e^{-x}cosx = \frac{2}{\pi}\int_0^\infty \frac{(u^2+2)cosux}{u^4+4}du$

[M15/Biot/6M]

Solution:

Let $f(x) = e^{-x}cosx$

Consider, Fourier cosine integral

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos ux \int_0^\infty f(s)\cos us\, du\, ds$$
$$= \frac{2}{\pi}\int_0^\infty \cos ux \int_0^\infty e^{-s}\cos s \cos us\ du\, ds$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \int_0^\infty e^{-s}(2\cos s\cos us)\, du\, ds$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \int_0^\infty e^{-s}(\cos(1+u)s + \cos(1-u)\,s)\, du\, ds$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \int_0^\infty e^{-s}\cos(1+\mathrm{u})s + e^{-s}\cos(1-u)s\ du\, ds$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \left[\frac{e^{-s}(-\cos(1+u)s + (1+u)\sin(1+u)s)}{(-1)^2+(1+u)^2} + \frac{e^{-s}(-\cos(1-u)s+(1-u)\sin(1-u)s)}{(-1)^2+(1-u)^2}\right]_0^\infty du$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \left[\frac{1}{1+(1+u)^2} + \frac{1}{1+(1-u)^2}\right]du$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \left[\frac{1}{2+2u+u^2} + \frac{1}{2-2u+u^2}\right]du$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \left[\frac{2-2u+u^2+2+2u+u^2}{(2+u^2)^2-(2u)^2}\right]du$$
$$= \frac{1}{\pi}\int_0^\infty \cos ux \left[\frac{2u^2+4}{4+4u^2+u^4-4u^2}\right]du$$
$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{(u^2+2)cosux}{u^4+4}du$$

Since, $f(x) = e^{-x}cosx$, we get

$$e^{-x}cosx = \frac{2}{\pi}\int_0^\infty \frac{(u^2+2)cosux}{u^4+4}du$$

14. Express the function $f(x) = \begin{cases} -e^{kx} & for\ x < 0 \\ e^{-kx} & for\ x > 0 \end{cases}$ as a Fourier integral and hence prove that $\int_0^\infty \frac{wsinwx}{w^2+k^2}dw = \frac{\pi}{2}e^{-kx}$ if $x > 0, k > 0$

[M19/Elex/8M][M19/Elect/6M][M19/Biom/8M]

Solution:

We have,

$$f(x)=\begin{cases}-e^{kx} & x<0\\ e^{-kx} & x>0\end{cases}$$

Putting $x=-x$, we get

$$f(-x)=\begin{cases}e^{kx} & x<0\\ -e^{-kx} & x>0\end{cases}$$

$$\therefore f(-x)=-f(x)$$

Thus, $f(x)$ is an odd function

By Fourier Sine Integral,

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty f(s)\sin ws\,dw\,ds$$

$$=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty e^{-ks}\sin ws\,dw\,ds$$

$$=\frac{2}{\pi}\int_0^\infty \sin wx\left[\frac{e^{-ks}}{(-k)^2+w^2}(-k.sinws-w.cosws)\right]_0^\infty dw$$

$$=\frac{2}{\pi}\int_0^\infty \sin wx\left[0-\frac{e^0}{k^2+w^2}(-w)\right]dw$$

$$=\frac{2}{\pi}\int_0^\infty \sin wx\left[\frac{w}{k^2+w^2}\right]dw$$

$$\therefore f(x)=\frac{2}{\pi}\int_0^\infty \frac{wsinwx}{k^2+w^2}dw$$

15. Express $f(x)=\frac{\pi}{2}e^{-x}cosx, x>0$ as a Fourier sine integral and show that $\int_0^\infty \frac{w^3 sinwx}{w^4+4}dw=\frac{\pi}{2}e^{-x}cosx, x>0$

[N13/CompIT/6M]

Solution:

We have,

$$f(x)=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty f(s)\sin ws\,dw\,ds$$

$$=\frac{2}{\pi}\int_0^\infty \sin wx\int_0^\infty \frac{\pi}{2}e^{-s}\cos s\sin ws\ \,dw\,ds$$

$$=\frac{1}{2}\int_0^\infty \sin wx\int_0^\infty e^{-s}(2\cos s\sin ws)\,dw\,ds$$

$$=\frac{1}{2}\int_0^\infty \sin wx\int_0^\infty e^{-s}(\sin(1+w)s-\sin(1-w)\,s)\,dw\,ds$$

$$=\frac{1}{2}\int_0^\infty \sin wx\int_0^\infty e^{-s}\sin(1+w)\,s-e^{-s}\sin(1-w)\,s\,dw\,ds$$

$$=\frac{1}{2}\int_0^\infty \sin wx\left[\frac{e^{-s}(-\sin(1+w)s-(1+w)\cos(1+w)s)}{(-1)^2+(1+w)^2}-\frac{e^{-s}(-\sin(1-w)s-(1-w)\cos(1-w)s)}{(-1)^2+(1-w)^2}\right]_0^\infty dw$$

$$=\frac{1}{2}\int_0^\infty \sin wx\left[\frac{1+w}{1+(1+w)^2}-\frac{1-w}{1+(1-w)^2}\right]dw$$

$$=\frac{1}{2}\int_0^\infty \sin wx\left[\frac{1+w}{2+2w+w^2}-\frac{1-w}{2-2w+w^2}\right]dw$$

$$=\frac{1}{2}\int_0^\infty \sin wx\left[\frac{2-2w+w^2+2w-2w^2+w^3-2-2w-w^2+2w+2w^2+w^3}{(2+w^2)^2-(2w)^2}\right]dw$$

$$=\frac{1}{2}\int_0^\infty \sin wx\left[\frac{2w^3}{4+4w^2+w^4-4w^2}\right]dw$$

$$f(x)=\int_0^\infty \frac{w^3 sinwx}{w^4+4}dw$$

Since, $f(x) = \frac{\pi}{2} e^{-x} cosx$, we get

$$\frac{\pi}{2} e^{-x} cosx = \int_0^\infty \frac{w^3 sinwx}{w^4+4} dw$$

16. Find Fourier integral representation of $f(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} & x = 0 \\ e^{-x} & x > 0 \end{cases}$

[N15/Chem/7M][M16/Chem/6M]

Solution:

$$f(x) = \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(s) \cos w(s-x)\ dw\ ds$$
$$= \frac{1}{\pi} \int_0^\infty \int_0^\infty e^{-s}\ \cos w(s-x)\ dw\ ds$$
$$= \frac{1}{\pi} \int_0^\infty \left[\frac{e^{-s}}{(-1)^2+w^2} (-1.\cos(ws-wx) + wsin(ws-wx)) \right]_0^\infty dw$$
$$= \frac{1}{\pi} \int_0^\infty \left[0 - \frac{-coswx - wsinwx}{1+w^2} \right] dw$$
$$= \frac{1}{\pi} \int_0^\infty \left[\frac{coswx + wsinwx}{1+w^2} \right] dw$$

17. Express $f(x) = \begin{cases} sinx & 0 < x \leq \pi \\ 0 & x < 0, x > \pi \end{cases}$ as Fourier integral.

Hence deduce that $\int_0^\infty \frac{cos\frac{\omega\pi}{2}}{(1-\omega^2)} d\omega = \frac{\pi}{2}$

[M15/CompIT/8M]

Solution:

$$f(x) = \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(s) \cos w(s-x)\ dw\ ds$$
$$= \frac{1}{\pi} \int_0^\infty \int_0^\pi sins\ \cos w(s-x)\ dw\ ds$$
$$= \frac{1}{2\pi} \int_0^\infty \int_0^\pi 2\ sins\ \cos w(s-x)\ dw\ ds$$
$$= \frac{1}{2\pi} \int_0^\infty \int_0^\pi [\sin(s+ws-wx) + \sin(s-ws+wx)]\ dw\ ds$$
$$= \frac{1}{2\pi} \int_0^\infty \left[-\frac{\cos(s+ws-wx)}{1+w} - \frac{\cos(s-ws+wx)}{1-w} \right]_0^\pi dw$$
$$= \frac{1}{2\pi} \int_0^\infty \left[-\frac{\cos(\pi+\pi w-wx)}{1+w} - \frac{\cos(\pi-\pi w+wx)}{1-w} + \frac{\cos(-wx)}{1+w} + \frac{\cos(wx)}{1-w} \right] dw$$
$$= \frac{1}{2\pi} \int_0^\infty \left[\frac{\cos(\pi w-wx)}{1+w} + \frac{\cos(\pi w-wx)}{1-w} + \frac{\cos(wx)}{1+w} + \frac{\cos(wx)}{1-w} \right] dw$$
$$= \frac{1}{2\pi} \int_0^\infty \left[\frac{\cos(\pi w-wx)+coswx}{1+w} + \frac{\cos(\pi w-wx)+coswx}{1-w} \right] dw$$
$$= \frac{1}{2\pi} \int_0^\infty \left[\frac{2[\cos(\pi w-wx)+coswx]}{1-w^2} \right] dw$$
$$f(x) = \frac{1}{\pi} \int_0^\infty \frac{coswx + cosw(\pi-x)}{1-w^2} dw$$

Putting $x = \frac{\pi}{2}$

$$f\left(\frac{\pi}{2}\right) = \frac{1}{\pi}\int_0^\infty \frac{\cos\frac{\pi w}{2} + \cos\frac{\pi w}{2}}{1-w^2}\,dw$$

$$\sin\frac{\pi}{2} = \frac{2}{\pi}\int_0^\infty \frac{\cos\frac{\pi w}{2}}{1-w^2}\,dw$$

$$1 = \frac{2}{\pi}\int_0^\infty \frac{\cos\frac{\pi w}{2}}{1-w^2}\,dw$$

$$\frac{\pi}{2} = \int_0^\infty \frac{\cos\frac{\pi w}{2}}{1-w^2}\,dw$$

18. Express $f(x) = \begin{cases} sinx & 0 < x \le \pi \\ 0 & x < 0, x > \pi \end{cases}$ as Fourier sine integral

[M16/CompIT/8M]

and evaluate $\int_0^\infty \frac{sinwx\ .sin\pi w}{1-w^2}\,dw$

[M16/Biot/6M]

Solution:

By Fourier Sine Integral,

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin wx \int_0^\infty f(s)\sin ws\,dw\,ds$$

$$= \frac{2}{\pi}\int_0^\infty \sin wx \int_0^\pi sins\ sinws\,ds\,dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \int_0^\pi 2sins\ sinws\,ds\,dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \int_0^\pi (\cos(1-w)s - \cos(1+w)s)\,ds\,dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \left[\frac{\sin(1-w)s}{1-w} - \frac{\sin(1+w)s}{1+w}\right]_0^\pi dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \left[\frac{\sin(\pi-\pi w)}{1-w} - \frac{\sin(\pi+\pi w)}{1+w}\right] dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \left[\frac{sin\pi w}{1-w} + \frac{sin\pi w}{1+w}\right] dw$$

$$= \frac{1}{\pi}\int_0^\infty \sin wx \left[\frac{2sin\pi w}{1-w^2}\right] dw$$

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{sin\pi w}{1-w^2}\,sinwxdw$$